Introduction to Computational Biology

Complete notes with exam answer guides (2, 4, 5, 12 and 13 marks)

Course 26BTB1001T, Biotechnology, L2 T0 P0 C2. CLA-1 covers Units I and II; CLA-2 covers Units III to V.

Unit 1: Cell Basics and Computing Analogies

Unit 2: Biomolecules as Data Structures

Unit 3: Protein Structures and Prediction Algorithms

Unit 4: AI/ML Foundations in Biology

Unit 5: Deep Learning, Healthcare and Ethics

Unit 1

Cell Basics and Computing Analogies

The cell is the smallest working unit of life. Here we compare it to things you already know from computing.

S1 Cell theory and the differences between prokaryotic/eukaryotic organisms

2 MARKS Cell theory states that all living things are made of cells, the cell is the basic unit of life, and every cell arises from a pre-existing cell.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: 1.1 Biology as one of the fields of science; Testing the hypothesis rigorously by gathering new data; 1.2 Cell: basic unit of life; Multicellular organisms possess developing properties.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. 1.1 Biology as one of the fields of science
2. Testing the hypothesis rigorously by gathering new data
3. 1.2 Cell: basic unit of life
4. Multicellular organisms possess developing properties

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: State the three postulates of cell theory.
5 marks: Differentiate prokaryotic and eukaryotic cells (any five points).
12 marks: Explain cell theory and describe the structure of prokaryotic and eukaryotic cells with neat diagrams.

Complete notes

The Cellular Organization of a Living Organism Biology is a broad field of science focusing on the “study of life.” It covers a wide-range of subjects from cell biology that explains the workings of the cells to the study of ecosystems that govern the survival of the living organisms. Technological advances have made it possible to examine the contents of the cells, the organelles, in minute details to unraveling the contents of these organelles especially, the deoxyribonucleic acid (DNA) inside the nucleus. As long as there is an urge to lead a life without diseases and live longer with better quality, it is important to study biology because it provides clues to solving many of these problems. Understanding the nature of the cellular organelles and its functions provides basic information about the cell that can help us to devise strategies to correct the problems seen in this cellular machinery in diseases. Genetic information stored in the form of DNA in the chromosomes of the nucleus is not only important for propagation of species but also in the production of various proteins that is critical for the survival of an organism. Genes serve as blueprints for synthesis of proteins that have several functions in the body: provide structural support, regulate metabolism, and stimulate chemical reactions involved in anabolism and catabolism of biomolecules. Metabolism is a key process through which the living organism is able to generate energy for its sustenance. Plants get energy through photosynthesis using sunlight, water, and carbon dioxide while animals depend on external sources including plants. All the processes within the organism are carried out by maintaining equilibrium in order to adjust to the external environment. This process of homeostasis allows the organism to live under different climatic conditions and during healthy and diseased states. Reproduction is a key process for the propagation of living organisms and is done through both mitosis and meiosis involving cell division.

1.1 Biology as one of the fields of science

Science is an organized approach to a systematic study of the universe with the aim to increase our knowledge about the universe. Biology is one of the realms of science and its sole purpose is to

understand the living organisms. A living organism is defined as a complex entity of physicochemical components that is capable of surviving through self-regulatory mechanisms such as metabolism, reproducing, and coping with the external environment. It also possesses the capabilities to move, respond to external and internal stimuli, grow and develop to adjust to the environment in which they live in.

Biologists believe in the concept of evolution which states that the present day living organisms have originated from ancestral forms of life which had undergone continuous modifications over a period of time. These concepts have been developed by Jean Baptiste Lamarck (1801) and later modified by Charles Darwin (1859). It also helped in the classification of living organisms into groups in such a manner that they are linked chronologically and morphologically. The branch of science dealing with classification organisms is called as taxonomy. A kingdom is the broadest classification in taxonomy and all living organisms belong to one of the 5 kingdoms (Table 1.1).

The five kingdoms are Monera, Protista, Fungi, Plantae, and Animalia. Living organisms are further classified into prokaryote and eukaryote based on the presence or absence of nucleus. Monera consists of unicellular organisms that lack a nucleus and many of the specialized cell parts, called organelles. Such organisms are said to be prokaryotic (pro "'before"; karyotic "kernel," "nucleus'"') and consist of bacteria. All of the other kingdoms consist of eukaryotic (eu: "'true'') organisms, which have cells that contain a nucleus and a number of specialized organelles.

(Table 1.1 Classification of kingdoms with their characteristic 'features

S.No. | Ki ingdom __| Characteristics Examples 1 Monera | Single-celled, prokaryotic, cells do not have Bacteria _____| nucleus and specialized organelles z Protista Single-celled, eukaryotic, cells possess Protozoa _____| nucleus and specialized organelles 3 Plantae Multicellular, eukaryotic, produce their own | Trees _| food for survival and reproduction

Single or multicellular, eukaryotic, acquire Yeasts and molds food from their surroundings

5 Animalia | Multicellular, eukaryotic, obtain food from Fishes, Birds, Repthe environment and Process it through meta- | tiles, Cows, Human bolic pathways for survival and reproduction | beings

Biologists have been able to classify the living organisms by investigating their characteristic features. This was accomplished by adopting rigorous

_scientific methods as a model of research developed by Francis Bacon (1561—

1626). It has the following sequences:

Recognizing the problem ~ Gathering data through experiments (measurement, observations, etc)

Analyzing the data for meaningful connections between various parameters chosen for study and any deviations

Formulating a hypothesis (a generalization), which is an educated guess that explains the existing data and suggests further avenues of investigation

Testing the hypothesis rigorously by gathering new data

6. Confirming, modifying, or rejecting the hypothesis in light of the new

Scientists have strict guidelines of doing research as postulated Bacon. Any procedure or data that do not conform to the ri and go wrong are discarded and the whole process is repeated some instances, errors have lead to new discoveries. '

- Charles Goodyear accidently dropped a mixture of rubber and gultice

° Alexander Fleming found a contaminated mold growing in a baceea laboratory where he had a kept a chemical that later led to the. disco ety of X-rays. g ona hot stove and found the mixture to be flexible and Bees led to _ discovery of vulcanization process y culture on a petri dish. Any bacteriologist would have thrown aw: that petri dish because of contamination. But Fleming observed an area around the mold had no bacterial growth. He realized that this may have been due to the product produced by the mold. He isolated the chemical from the mold which happens to be penicillin, 'the first antibiotic that revolutionized the treatment options for various bacterial

1.2 Cell: basic unit of life

All living beings are made up‘of the basic unit of life- cells (Fig: 1.1). Itis the smallest unit of living matter and each cell is able to carry out the processes of life. Some structures transport materials, some make food and others

| release energy for the cell to use. All the cells are involved in metabolic Processes through cellular respiration, reproduction (meiosis), and growth (mitosis). They are also capable of transforming into different types of cells, a specialization called as cell differentiation. The cells have a few things in common such as having cell membrane, deoxyribonucleic acid (DNA), cytoplasm, and ribosome but they are different with regard to structure, shape, and functions.

The cells were first described by Robert Hooke in 1665. He examined thin slices of cork and found pores. But his examination did not indicate any nucleus, and other organelles. Later in 1674, Antony van Leeuwenhoek found moving organisms by observing them under lenses and microscopes. The observation of Hooke and others led to the development of cell theory. The cell theory states

- * All living things and organisms are made up of cells and their products. | Multicellular organisms (example: humans) are composed of many | cells while unicellular organisms (example: bacteria) are composed of only one cell. bee, ips s Cells are the basic building units of. life. |
- They are the smallest structures capable of surviving on their own.
- New cells arise from the pre-existing cells by division. i: 3° ee For example, new cells arise from cell division and ; a zygote (the very first cell formed when ; an organism is produced) arises from | the fusion of an egg cell and a sperm cell. ; i Evidence for the cell theory When scientists started to look at the structures of organisms under the microscope, they discovered that all living organisms where made up of these small units which they named them as cells. When these cells were taken from tissues they were able to survive for some period of time. Nothing smaller than the cell was able to live independently and so it was concluded that the cell was the smallest unit of life. For some time, scientists thought that cells must arise from non-living material but it was eventually proven that this was not the case, and instead proved to arise from preexisting cells.

A simple experiment to prove this can be done as follows:

- Take two containers and put food in both of these
- Sterilize both of the containers so that all living organisms are killed
- Leave one of the containers open and seal the other closed

PRe KA Rows -

~ The idea is that the whole is greater than ¢ re will be growth of mold in the open h in the sealed container. The er, cells are able to enter the

One will be able to observe that the container whereas there will be no such growl reason for this is because in the open contain container from the external environment and start to divide and grow.

- to the scaling on the other container no cells could enter and

However, du | that cells cannot arise from hence, no mold growth was observed proving non-living material.

Multicellular organisms possess developing properties

Multicellular organisms are the product of development from simpler_to complex structures. For example: cells form tissues, tissues form organs, organs form organ systems and organ systems form multicellular organisms. he composition of its parts. For example your lungs are made of many cells, However, the cells by themselves aren’t much use. It is the many cells working as a unit that allow the lungs to perform their function.

Every cell in multicellular organisms contains all the genes of that organism. However, the genes that are activated vary from cell to cell. The reason we have different types of cells in our body (the cells in your eyes are not the same as the ones that make up your hair) is because different genes are activated in different cells. For example, the gene that produces keratin will be active in hair and nail cells. Keratin is the protein which makes up hair and nails. Genes encode for proteins and the proteins affect the cell’s structure and function so that the cell can specialize. This means cells

develop in different ways. This is called differentiation. Differentiation depends on gene expression which is regulated mostly during transcription. It is an advantage for multicellular organisms as cells can differentiate to be more efficient unlike unicellular organisms that have to carry out all of the functions within that one cell.

These kinds of organisms do not have a nucleus, mitochondria or any other embrane-bound organelles. In other words, neither their DNA nor any other of their metabolic functions is collected together in a discrete_membrane enclosed area. Instead everything is openly accessible within the cell; though some bacteria have internal membranes as sites of metabolic activity, these membranes do not enclose a separate area of the cytoplasm.

The prokaryote cell is simpler and therefore smaller than a eukaryote cell. The prokaryotes do not have a nucleus and most other organelles of

| eukaryotes. There are two kinds of prokaryotes, bacteria. and archaea. Nuclear material of prokaryotic cell consists of a single chromosome. which is in direct contact with cytoplasm. Here the undefined nuclear region in the cytoplasm is called nucleiod. A prokaryotic cell has the following features:

. “On the outside, flagella and pili project from the cell’s surface. These are structures (not present in all prokaryotes) made of proteins that facilitate movement and communication between cells;

- Enclosing the cell is the cell envelope consisting of a cell wall covering a plasma membrane though some bacteria also have a further covering layer called a capsule. The envelope gives rigidity to the cell and separates the inferior of the cell from its environment.
- Prokaryotes can carry extra chromosomal DNA elements called

\ plasmids, which are usually circular. Plasmids enable additional

Ww X : flnctions, such as antibiotic resistance and therefore, survival.

Eukaryotic cells are about 15 times wider than a typical prokaryote and can be as much as 1000 times greater in volume. The major difference between prokaryotes and eukaryotes is that eukaryotic cells contain membrane-bound compartments in which specific metabolic activities take place. Most important among these isa cell nucleus, a membrane-delineated compartment that houses the eukaryotic c€Il’s DNA. This nucleus gives the eukaryote its name, which means “true nucleus.” Other differences include: :

- The plasma membrane resembles that of prokaryotes in function, with minor differences in the setup. Cell wall may or may not be present.
- The eukaryotic DNA is organized in one or more linear molecules, called chromosomes, which are associated with histone proteins. All chromosomal DNA is stored in the cell nucleus, separated from the cytoplasm by a membrane. Some eukaryotic organelles such as mitochondria also contain some DNA.
- Many eukaryotic cells are ciliated with primary cilia. Primary cilia play important roles in chemo-sensation, mechano-sensation, and thermo-
- Eukaryotes can move using motile cilia. The flagella are more complex than those of prokaryotes.

+ Plastids are a feature of most plant cells but are not found in the cells of animals. Vacuoles are quite prominent in plant cells, but are far less significant in or absent from animal cells. ~>

S2 Organelles as CPU components analogy

2 MARKS Organelles are specialised structures inside a cell, each with a job: the nucleus acts like the CPU, mitochondria like the power supply and ribosomes like assembly workers.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: 1.3 Cell structure and function; Cell/plasma membrane; Nucleus and Nucleolus; Cytoskeleton; Golgi apparatus/bodies.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. 1.3 Cell structure and function
2. Cell/plasma membrane
3. Nucleus and Nucleolus
4. Cytoskeleton
5. Golgi apparatus/bodies

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: State the function of mitochondria.
5 marks: Compare the cell with a CPU using any five organelles.
12 marks: Describe the structure and functions of the major cell organelles with a labelled diagram and computing analogies.

Complete notes

1.3 Cell structure and function

Earlier before the advent of electron microscope, cell was described as a unit possessing a limiting outer membrane and an inner central nucleus surrounded by large mass of cytoplasm (Fig. 1.1). Staining of cells facilitated to . aS . pemeatrnan, Me ei oc nonmaa identification of subcellular features such as chromosome and nucleolus. of the nucleus. Electron microscopic studies made it possible for the elucidation of the subcellular structures called as organelles that perform specialized functions. Most of the organelles are derived from membranes but some of them do_ not possess membrane structure such as tibosomes, microtubules, and microfilaments, flagella, cilia, and centrioles.

Cell/plasma membrane: The outermost layer of the living cell that gives structure and shape is the cell membrane. The chief function is to regulate the passage of materials into and out of the cell. Initial structural studies using electron microscope revealed that it consists of inner and outer dense protein layer enclosing a less dense phospholipid layer. Channels were also observed to traverse the through exterior surface of the membrane. Recent studies by Singer and Nicholson have demonstrated that the cell membrane has fluid

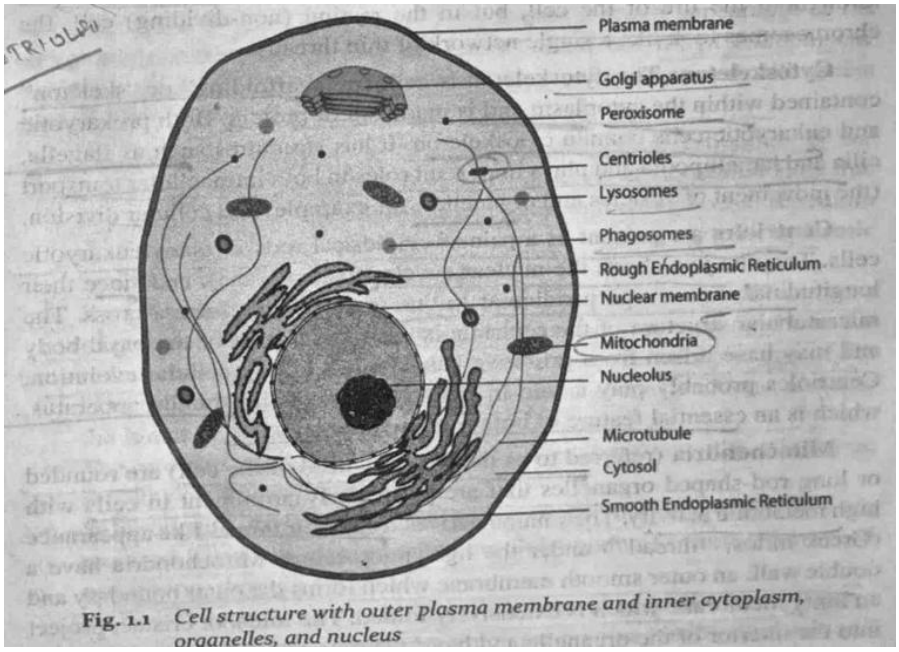


Fig. 1.1 *Cell structure with outer plasma membrane and inner cytoplasm, organelles, and nucleus*

mosaic model which better explains the dynamic nature of the proteins in the cell membrane. The only difference in the proposed model was the arrangement of phospholipids: their polar ends facing the inner_and outer surfaces and the hydrophobic, non-polar ends opposed at the center of the “Dilayer. nen

Nucleus and Nucleolus: The nucleus is a round or oval body lying in the center of the cell enclosed by a double membrane that is called as nuclear membrane or envelope. In certain regions of the nuclear envelope, ‘these membranes join together and there may be pores in these areas. The ‘pores serve 88 a direct passage route for the substances tg leave the nucleus. The outer membrane of the nuclear envelope is continuous with the endoplasmic reticulum and thus, may facilitate passage of materials from the nucleus directly into the channels of endoplasmic reticulum. Within the nucleus, one or more nucleoli may be seen. These are dense bodies containing the subunits for the ribosomes, the cytoplasmic organelles involved in the synthesis of j protein. The nucleolus is involved in the assembly and synthesis of ribosomes. “The nucleus is the storehouse of the genetic material called as chromosomes. Each chromosome exists as a tiny individual rod or string throughout the life of the cell, but in the resting (non-dividing) cell, the chromesemes took tike-a single-network of thin threads.

Cytoskeleton: The cytoskeleton is cellular “scaffolding” _or “skeleton” contained within the cytoplasm and is made out of protein. Both prokaryotic and eukaryotic cetis contain cytoskeleton. It has structures such as flagella, cilia 4nc+tamellipodia and plays important roles in both intracellular transport [! (thé movement of Vesicles and organelles, for example) and cellular division.

Centrioles are present as a pair of cylindrical rods in many eukaryotic cells. They Tie just above the nuclear envelope (membrane), and since their longitudinal axes are perpendicular to one another, they_ferm-a cross. The microtubular structure of the centriole is the same as that of the basal body anid may have arisen from primitive basal bodies during cellular evolution. Centrioles probably play a role in the formation of the spindle apparatus, which is an essential feature of both mitosis and meiosis.

Mitochondria (referred to as the ‘Powerhouse’ of the cell) are rounded or long rod-shaped organélles that are particularly prominent in cells with high metabolic activity. Their name derives from their thread-like appearance (Greek mitos, “thread’”’) under the light microscope. Mitochondria have a double wall: an outer smooth membrane which forms the outer boundary and aninner membrane which is extensively folded. The folds, or cristae, project into the interior of the organelle and have a variety of enzymes embedded in them. These cozymes are involved in the systematic degradation_of organic iifolecules to yield energy for the cell. They are responsible for the breakdown of sugar molecules to release ATP (adenosine triphosphate), which is used to transport energy within the cell

for n metabolism. (through citric acid cycle and eleciron transport chain as explained in chapter 3). Like the chloroplasts of plants, the mitochondria contain their own DNA and ribosomes; they replicate independently of the rest of the'cell and appear to control the synthesis of their membranes. __. "i y

Endoplasmic reticulum (ER) is a series of membranous channels that traverse the cytoplasm of most eukaryotic cells. It forms a continuous network of the cell, itmay appear as a series of flattened disks or sacs. The endoplasmic retivujum Serves many general functions, including the facilitation of protein folding and the transport of synthesized proteins in sacs called cisternae. Only properly-folded proteins are transported from the rough ER to the Golgi _ complex. Boer

\Un-many parts of the cell, the endoplasmic reticulum is associated with e small dense granules lying along the outer border of its membrane. These structures are known as ribosomes. They impart a rough appearance to the endoplasmic reticulum, so that the ER is called the rough endoplasmic reticulum (RER) in these regions, which are usually associated with active protein synthesis.

_ ¢ The primary function of the rough endoplasmic reticulum is the ~ production and processing of specific proteins at ribosomal sites that are later exported. The ribosomes in the rough endoplasmic reticulum synthesize proteins which are then sent in to the rough endoplasmic reticulum for advanced processing.

- Rough endoplasmic reticulum function involves creation of two types of proteins. One is the type which fortifies and gets embedded into the reticulum membrane. The other types are water-soluble proteins which after production at ribosomal sites, pass through the membrane and into the lumen.
- The proteins that enter are further folded inside by the chaperone proteins present in the lumen.
- Subsequently, these proteins are transported to the sites where they are required. They may also be sent to the Golgi bodies for further advanced processing, through vesicles. m1

The smooth endoplasmic reticulum (SER) does not contain any ribosomes and is associated with cellular regions which are involved in the synthesis and transport of lipids or the detoxification of a variety Seve 9 endoplasmic reticulum is found ina variety of cell types anc aoenireta functions in each, It consists of tubules and vesicles that branc! sail! "in network. The network of smooth endoplasmic reticulum Se . stirface area for the action or storage of key enzymes and the products o these enzymes, , | z

Ribosomes are the components of cells that make proteins from the amino acids. Ribosomes are the workhorses_of protein icauather*: process of translating messenger ribonucleic acid sey eee "central dogma" of biology is that DNA is used tomake RNA, whi¢h,in turn, is used to make protein (discussed in detail in Chapter yas aa Oe in genes is copied Ctranscription) into an mRNA. Ribosomes then read { le ifformation in this RNA and use it to create proteins. This process is known as! translation; i.e., the ribosome "translates" the genetic information From

'A into proteins. Ribosomes do this by binding to an mRNA and using it asa template for the correct sequence of amino acids in a particular protein. The amino acids are attached to transfer RNA (tRNA) molecules, which enter one part of the ribosome and bind to the messenger RNA sequence. The attached amino acids are then joined together by another part of the ribosome. The ribosome moves along the mRNA, "reading" its sequence and producing a chain of amino acids.

Ribosomes are made from complexes of RNAs and proteins. Ribosomes are divided into two subunits, one larger and another smaller subunit. The sifialler subunit binds to the mRNA, while the larger subunit binds to the tRNA and the amino acids. When a ribosome finishes reading an mRNA, these two subunits split apart. Ribosomes have been classified as ribozymes, since the ribosomal. RNA seems to be most important for the peptidyl transferase activity that links amino acids together.

Golgi apparatus/bodies: They exist as stacks of flattened sacs, or vesicles tat are continuous with the "channels of the smooth endoplasmic reticulum. Their major function is the storage, modification, and packing of materials for release outside the cell membrane. These organelles are particufar ty prominent in secretory cells such as those of the pancreas. The outer-portion Of The Golgi apparatus releases its secretory material within membrane enclosed globules (seqretory vesicles) that Mise of the cell. The Golgi apparatus may actually be part of ; a dynamic sysienrof membranous channels

within the cell in which all elements such as the nuclear envelope, the endoplasmic reticulum, the Golgi apparatus, and the cell meffibrane are connected to each other without sharp boui indaries. This pptereonmiceted network facilitates transport of materials across the cell. ~

Lysosomes are similar in shape to mitochondria_but_are_ smaller and_, consist of a single membrane covering the structure. They contain powerful enzymes that would digest the cellular contents if they were not contained svithin the impermeable Tysosomal membrane. Rupture of this membrane releases these enzymes. The lysosome plays a role in intracellular digestion and may also be important in the destruction OF certain structures during the PODS Or evelOpmen a eee “<¥iggables are discrete, clear regions within the cell that contain water and dissolved materials Materials, The vacuole may act as a reservoir for fluids and salts that might otherwise interfere with metabolic processes occurring in the cytoplasm The membrane enclosing the vacuole is called a tonoplast. Many pretozoans have a coniraétite-vacuole, which periodically contracts and forces fluid and salts out of the cell. The structure serves to prevent an accumulation of fluids in ‘organisms that live in fresh water. Vacuoles containing digestive enzymes may also be formed around ingested food particles in a variety of cells. In the cells of many plants, a large, central vacudle is a prominent feature; this vacuole may swell, press against the rigid ceil wall, and give the cell a high degree of rigidity or turgor, a Fe alt Organelles in plant cell Plant and animal cells differ in the number and structure of the above organelles, but the most fundamental difference in plant cell organelles isthe presence of the chloroplast and cell wall. Plants differ integrally from animals in their ability to prepare food within their cells by the process of

Chloroplasts are organelles found in plant cells and other eukaryotic organisms that conduct photosynthesis. The material within the chloroplast is called the stroma containing the stacks of thylakoids, the sub-organelles, which are the site of photosynthesis (described in detail in chapter 3).

Cell Wall is a structure made out of polysaccharide, peptidoglycan or glycoprotein that provides structural support and protection to the cell. In combination with the vacuole, which is large in plant cells, the other cell wall

4 function includes controlling the turgidity of the cell.

All cells, whether prokaryotic or eukaryotic, have a membrane that envelops the cell, separates its interior from its environment, regulates what moves in and out (selectively permeable), and maintains the electric potential of the cell, Inside the membrane, a salty cytoplasm takes up most of the cell volume. All cells possess DNA, the hereditary material of genes, and RNA, containing the information necessary to build various proteins such as enzymes, the cell’s primary machinery. There are also other kinds of biomolecules in cells. a :

Extra points and computing analogy

Organelles are tiny compartments, each with a job. Nucleus = CPU control unit (holds the instructions, DNA). Mitochondria = power supply (make ATP, the energy currency). Ribosomes = assembly workers that build proteins. Endoplasmic reticulum and Golgi = data bus and packaging/shipping. Lysosomes = garbage collector. Cell membrane = firewall that decides what enters.

In depth

- Nucleus: double envelope with pores, chromatin (DNA plus histones) and a nucleolus that makes rRNA. Mitochondria: double membrane with folded cristae; glycolysis, Krebs cycle and oxidative phosphorylation yield about 30-32 ATP per glucose.
- Rough ER carries ribosomes and makes secreted proteins; smooth ER makes lipids and detoxifies. Golgi modifies, sorts and packs proteins into vesicles.
- Lysosomes hold digestive enzymes at acidic pH; chloroplasts do photosynthesis; vacuoles store; the cytoskeleton gives shape and tracks for transport.
- Where the analogy breaks: cells have no central clock, work chemically and in parallel, and can repair and copy themselves.

Remember: Analogies help memory, but cells work in parallel and chemically, not by clock cycles.

S3 DNA replication compared to code duplication

2 MARKS Genetic information is in the chromosomes found in the nucleus.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: 1.4 Genetic information and protein synthesis; Pentosugar; Nitrogenous base; Phosphate.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. 1.4 Genetic information and protein synthesis
2. Pentosugar
3. Nitrogenous base
4. Phosphate

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: What is semi-conservative replication?
5 marks: List the enzymes in DNA replication and their roles.
12 marks: Explain how genetic information is stored and copied, and compare DNA replication with code duplication.

Complete notes

1.4 Genetic information and protein synthesis

Genetic information is in the chromosomes found in the nucleus. This is

; necessary for reproduction of species and therefore, its propagation on earth. It is coded along the length of a polymeric molecule composed of four types of monomeric units. This polymeric molecule is deoxyribonucleic acid (DNA). It is the chemical basis of heredity which is organised into genes, the fundamental units of genetic information. Genes control the synthesis of various types of ribonucleic acid (RNA). }

Nucleic acid is a polynucleotide consisting of nucleotides as the repeating subunits. Each nucleotide is made up of three components are, (i) pentosugar,

| (ii) nitrogenous base and (iii) phosphate. The nucleotides are joined to one another through phosphodiester linkages between the 3’ of one nucleotide and the S’ of the adjacent one. This linkages repeated many times to build up large structures containing hundreds to millions of nucleotides within a single giant molecule.

Pentosugar: It is a type of cyclic 5 carbon sugar, which connects two phosphate groups one at 3" and second at 5" carbon and also nitrogenous bases linked to the 1“ carbon. The type of sugar molecule in DNA is deoxyribose, where as in RNA is ribose.

Nitrogenous base: Nucleic acids contains 5 major heterocyclic bases, adenine (A), guanine (G), cytosine (C), thymine (T) and uracil (U). First four

"bases are common in DNA, in case of RNA thymine is replaced with uracil.

Further, adenine and guanine known as purine, other three cytosine, thymine and uracil known as pyrimidine.

Phosphate:(& phosphatg group is attached to the 5’ carbon of the sugar by a phosphodiester linkage. This phosphate group is solely responsible for the strong negative charge of the nucleic acids.

The structures of DNA and RNA, types of RNA, and protein synthesis are described in detail in Chapter 2.

Extra points and computing analogy

Before a cell divides it copies its DNA. Helicase unzips the double helix, primer-laying enzymes start the job, and DNA polymerase adds matching bases (A with T, G with C) to each strand. Each new molecule keeps one old strand and one new strand: this is called semi-conservative replication. Like

copying a file, it needs an original, a copier and error checking; polymerase proofreads, like a checksum.

In depth

- Steps: origin of replication opens, helicase unwinds, single-strand binding proteins keep strands apart, topoisomerase relieves supercoils, primase lays an RNA primer, DNA polymerase III extends 5' to 3'.
- Leading strand is made continuously; lagging strand is made as Okazaki fragments, primers are removed by polymerase I and gaps sealed by DNA ligase.
- Accuracy: polymerase proofreads with 3' to 5' exonuclease activity and mismatch repair fixes leftovers; final error rate is about 1 in 10^9 bases.
- Eukaryotes use many origins at once to finish fast; telomerase protects chromosome ends. Computing link: copy with verification, many forks in parallel.

Remember: Error rate is about 1 in a billion bases after proofreading and repair.

S4 Cell division as a concept of parallel computing

2 MARKS Mitosis is cell division that produces two genetically identical daughter cells from one parent cell, for growth and repair.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: 1.8 Cell growth and reproduction; Eukaryotic Cell Division; Interphase and the control of cell division; Reproductive signal; Replication of genetic material; Segregation of DNA.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. 1.8 Cell growth and reproduction
2. Eukaryotic Cell Division
3. Interphase and the control of cell division
4. Reproductive signal
5. Replication of genetic material
6. Segregation of DNA
7. Cytokinesis
8. Prophase
9. Metaphase
10. Anaphase

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: Define mitosis.
5 marks: Differentiate mitosis and meiosis.
12 marks: Explain the cell cycle and the phases of mitosis and meiosis; relate cell division to parallel computing.

Complete notes

1.8 Cell growth and reproduction

Cells, like an organism, have the ability to grow, reproduce, communicate, and carryout a multitude of reactions and are regarded as the basic units of life. In order to have an optimum functionality, a cell must maintain a higher surface area to volume ratio. As an object grows larger, its volume increases more rapidly than its surface area. Similarly as a living cell grows larger, its rate of waste production and its need for resources increases faster than its surface area creating an unhealthy condition. So a cell must divide to have a small volume in order to maintain a larger surface area to volume ratio.

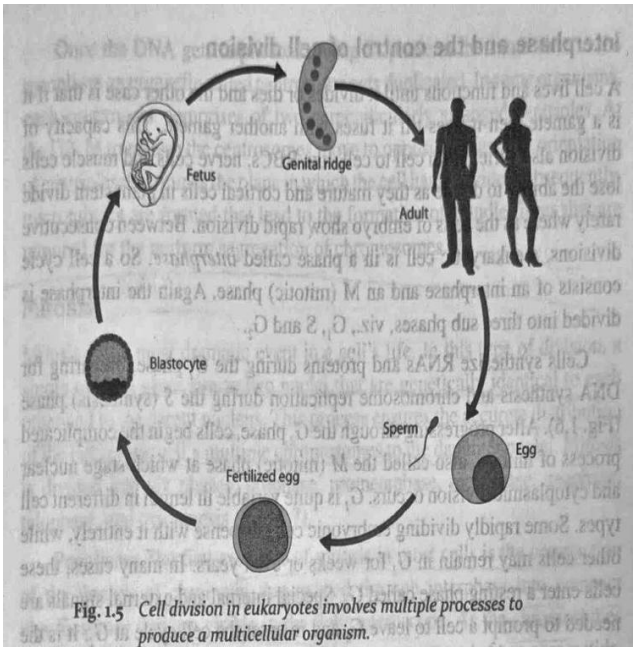


Fig. 1.5 *Cell division in eukaryotes involves multiple processes to produce a multicellular organism.*

Growth and reproduction are the major characteristics of life without which life would come to an end. Reproduction continues life and provides the basis for evolution. Cells reproduce by duplicating their contents and then dividing into two. In case of unicellular organisms, such as bacteria, each cell division produces a whole new organism. This controlled cycle of duplication and division is termed cell-cycle. An ordered series of events must occur for acell to divide uniformly which can be summed up as follows: - a) A reproductive signal that activates the cellular reproductive machinery. b) Duplication of the genetic material and other vital components so that each cell after division becomes complete and identical to each other. c) Segregation of the genetic material.

Prokaryotic Cell Division In prokaryotes, there is neither mitosis nor meiosis but rather a different type of cell division. The type of cell division is termed fission wherein the cel] grows, replicates its genetic material and then divides into two single-celled complete entities.

Reproductive signal: The reproductive mechanism of many prokaryotes depends upon the environmental stimuli. The cell division in case of bacterium, Escherichia coli, takes 40 minutes at 37°C where as in presence of adequate carbohydrates and salts, the cycle time gets reduced to 20 minutes, So this suggests that materials in the environment act as signals to control the division of a cell.

Replication of genetic material : Most of the prokaryotes contain a single circular DNA, which is packaged tightly to fit into a tiny cell that is a hundred times smaller than it. Functionally, the prokaryotic chromosome has

} two regions that are important for cell reproduction:

- The site where replication of the circle starts: the origin of replication, designated as “ori”
- The site where replication ends: the terminus of replication, “ter’’.

Several functional enzymes that are collectively termed as replisomes act sequentially to carry out replication starting from an unwound region of DNA called as replication fork.

Segregation of DNA : DNA replication is followed by the segregation of the replicated DNA molecules to the two new cells. The first region to be replicated is ori, which is attached to the plasma membrane. After replication, the two ori regions separate as the new chromosome forms and new plasma membrane forms between them as the cell grows longer. By the end of replication, there are two chromosomes, one at either end of the lengthened bacterial cell.

Cytokinesis : The division of cytoplasm starts once the cell has successfully replicated its genetic material. The process starts with the pinching in of the cell membrane. As the membrane pinches in the materials required for cell wall synthesis are produced and ultimately the cells are separated.

Eukaryotic Cell Division

Cell division in eukaryotes, like that in prokaryotes, involves reproductive signals, DNA replication, segregation, and cytokinesis, but are much more complicated (Fig. 1.5). First, cell division does not depend upon the environmental cues of a single cell rather than on the needs of the wholt

Interphase and the control of cell division

A cell lives and functions until it divides or dies and the other case is that if it is a gamete then it lives till it fuses with another gamete. This capacity of division also varies from cell to cell, like RBCs, nerve cells and muscle cells lose the ability to divide as they mature and cortical cells in plant stem divide rarely where as the cells of embryo show rapid division. Between consecutive divisions, a eukaryotic cell is in a phase called interphase. So a cell cycle consists of an interphase and an M (mitotic) phase. Again the interphase is divided into three sub phases, viz., G,, S and G,.

Cells synthesize RNAs and proteins during the G, phase, preparing for DNA synthesis and chromosome replication during the S (synthesis) phase (Fig. 1.6). After progressing through the G, phase, cells begin the complicated process of mitosis, also called the M (mitotic) phase at which stage nuclear and cytoplasmic division occurs. G, is quite variable in length in different cell types. Some rapidly dividing embryonic cells dispense with it entirely, while

} other cells may remain in G, for weeks or even years. In many cases, these cells enter a resting phase called G,. Special internal and external signals are needed to prompt a cell to leave G, and re-enter the cell cycle at G,. It is the preparing stage for S phase. During G, phase, the cell makes preparations for mitosis like for example by synthesizing microtubules that will help in segregation of chromosomes.

segregation of chromosomes.

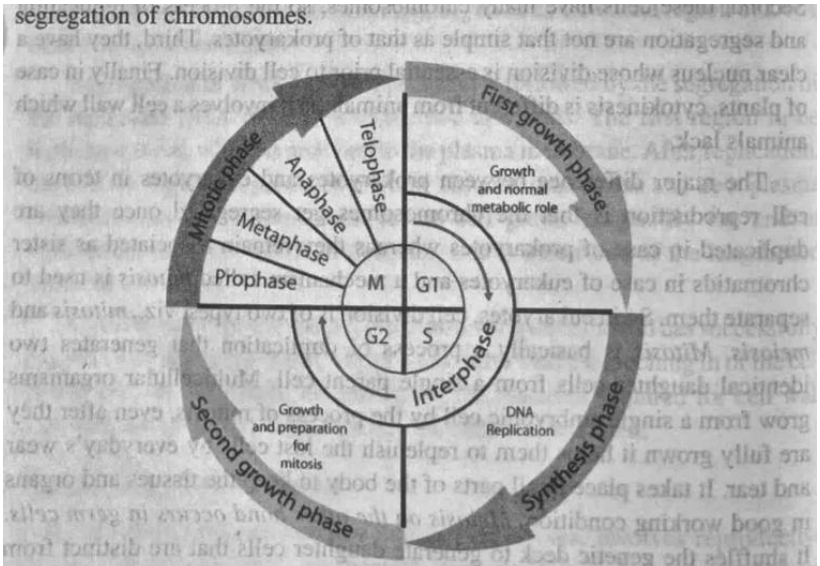


Fig. 1.6 *Major events in a cell cycle that consists of growth phases (1 and 2), DNA synthesis phase, and mitotic phase.*

Once the DNA gets duplicated during S- phase at the same time in the cytoplasm an organelle called centrosome gets duplicated. In some organisms, each centrosome comprises of two perpendicularly arranged centrioles. At the G2- M transition the centrosomes move to opposite poles. The orientation of centrioles determines the plane in which the cell has to divide. Subsequently, microtubules are formed that lead to the formation of spindle fibres that are required for the uniform segregation of chromosomes.

Mitosis is the most dramatic event in a cell's life. In this type of division, a single nucleus gives rise-to-two.nuclei.that are.genetically identical to each other and to the parent nucleus. This process ensures the accurate distribution of the eukaryotic cell's multiple chromosomes to the daughter nuclei. Mitosis is divided into six phases: prophase, prometaphase, metaphase, anaphase, telophase, and cytokinesis (Fig. 1.7).

Prophase: The first evidence of mitosis in most cells is the compaction of the threads of chromatin that-existed through interphase into compact chrofhosomeés that are visible in the light microscope,

AS the chromosomes cdmpact, each can be seen to be paired structures composed of two chromatids. This is the visible effectof the DNA molecules having been replicated in interphase. Chromosome condensation reduces the chance of long DNA molécules becoming tangled and broken. Each chromosome has a constriction | ye iF RSS Interphase Prophase Prometaphase ie OOD Faewes fee ni NN! Bat bat cn Metaphase Anaphase Telophase y

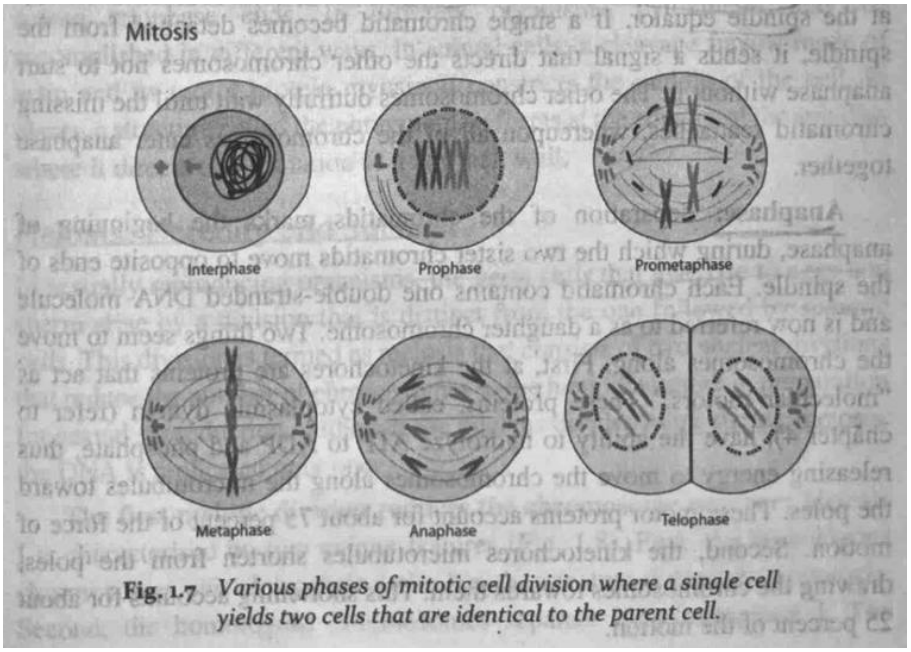


Fig. 1.7 *Various phases of mitotic cell division where a single cell yields two cells that are identical to the parent cell.*

Called the kinetochore, a structure that forms around a region rich in satellite DNA called the centromere. The kinetochore is the point of attachment of the chromosome to the spindle. At the same time as the chromosomes are condensing within the nucleus, the centrosomes, which lie on the cytoplasmic side of the nuclear envelope, begin to separate to establish the mitotic spindle. _®rometaphase: At the breakdown of the nuclear envelope, the chromosomes become free to interact With the forming spindle. Microtubule assembly from the centrosomes is random and dynamic. The growing ends of individual microtubules make chance contact with and are captured by the kinetochores. Because of the random nature of these events, the kinetochores of “chromatid pairs are initially associated with different numbers of microtubules, and the forces acting upon each chromosome are unbalanced: Initially, therefore, the spindle is highly unstable and chromosomes make frequent excursions toward and away from the poles. Gradually, a balance of forces is established and the chromosomes become aligned at the equator, with the kinetochores of each member of a chromatid pair oriented toward opposite poles.

Metaphase: Metaphase is the most stable period of mitosis. The system can be regarded as being at steady-state with the chromosomes lined up at the al eee ere spindle consists of two major groups of — microtubules: those connecting the chromatids to the poles and a second group extending from each pole ‘toward the other. The second group ove group Overlap at the spindle-equator. If a single chromatid becomes detached from the spindle, it sends a signal that directs the other chromosomes not to start anaphase without it. The other chromosomes dutifully wait until the missing chromatid reattaches, whereupon all of the chromosomes enter anaphase

Anaphase: Separation of the chromatids marks the beginning of anaphase, during which the two sister chromatids move to opposite ends of the spindle. Each chromatid contains one double-stranded DNA molecule and is now referred to as a daughter chromosome. Two things seem to move the chromosomes along. First, at the kinetochores are proteins that act as

“molecular motors.” These pfoteins, called cytoplasmic dynein (refer to chapter 4), have the ability to hydrolyze ATP to ADP and phosphate, thus releasing energy to move the chromosomes along the microtubules toward the poles. These motor proteins account for about 75 percent of the force of motion. Second, the kinetochores microtubules shorten from the poles, drawing the chromosomes towards them. This shortening accounts for about 25 percent of the motion. :

During anaphase the poles of the spindle are pushed farther apart, _—douibling the distance between them, The distance between poles increases pecause the overlapping polar microtubules extending from opposite ends of | the spindle contain motor proteins that cause them to slide past each other, in | much the same way that microtubules slide in cilia and flagella. This polar separation further separates one set of daughter chromosomes from the other. During this segregation process the chromosomes move very slowly at about | {pm per minute ensuring their proper segregation. Sheds, Gilt ’] ; AAlophase: This stage sees the reversal of many of the events of prophase; reforms, Golgi_apparatus and endoplasmic reticulum reform, and the ices ees Fh ge ea oi me ml of the genome from if€ Father and one copy from the mother. When these and othier changes are complete, telophase—and mitosis—is at an end, and each of the dalghT@F WUGL] enters another interphase.” PYOfoq Wala /-Mitosis is beautifully precise. It result in two nuclei that are identical to each other and to the parent nucleus in chromosomal makeup, and hence in genetic constitution. Next, the two nuclei must be isolated in separate cells, which require the division of the cytoplasm. into Me Cytokinesis: division of the Cytoplasm: Mitosis refers only to the division of the nucleus. The division of the cell’s cytoplasm, which follows mitosis, is accomplished by cytokinesis. This process may actually begin before’ telophase ends. In different organisms, cytokinesis may be accomplished in different ways. In animal cells, a cleavage furrow made of actin and its motor protein myosin II constricts the middle of the cell. In plants, a structure called the phragmoplast forms at the equator of the spindles where it directs the formation of a new cell wall. ys ia Meiosis: Heredity and survival r In sexually reproducing organisms, the germ cells that give rise to eggs and sperm arise by a division that is distinct from the one followed by somatic cells. This division is termed as meiosis that consists of two nuclear divisions that reduce the number of chromosomes to the haploid number in preparation ‘ for sexual reproduction. Although the nucleus divides twice during meiosis, the DNA is replicated only once. > The first meiotic division reduces the chromosome number: Meiosis Lis characterised by two unique features (Fig. 1.8). First, the homologous chromosomes pair along their entire length, a feature distinct from mitosis. Second, the homologous chromosomes separate after metaphase I. The

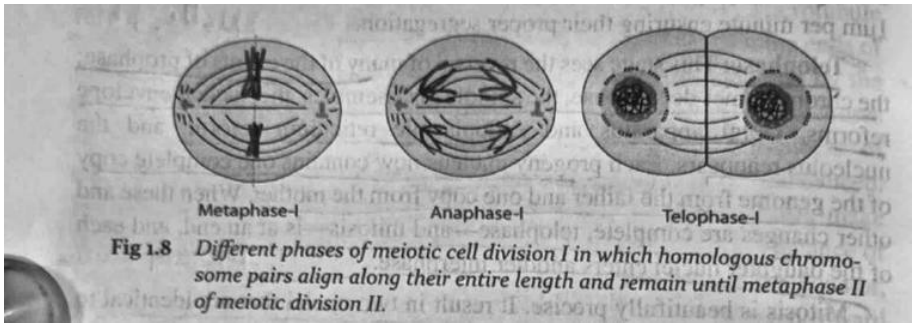


Fig 1.8 *Different phases of meiotic cell division I in which homologous chromosome pairs align along their entire length and remain until metaphase II of meiotic division II.*

’ of meiotic division II.

D... chromosomes, each consisting of two sister chromatids, remain

7 intact until the end of metaphase II in the second meiotic division.

: Like mitosis, meiosis I is preceded by an interphase with an S phase during which each chromosome is teplicated. As a result, each chromosome Consists of two sister chromatids, held together by cohesin proteins. Meiosis I begins with a long Prophase I during which the chromosomes change markedly. The homologous chromosomes pair by adhering along their lengths, a process called synapsis. This process lasts from prophase I to the end of metaphase I. By the time chromosomes can be clearly seen under light microscope, the two homologs are already tightly joined. This joining begins at the centromeres and is mediated by recognition of homologous DNA Sequences on homologous chromosomes. In addition, a’ special group of Proteins may form a scaffold called the synaptonemal complex, which runs lengthwise along the homologous chromosomes and appears to join them

The four chromatids of each pair of homologous chromosomes form what is called a tetrad, or bivalent. In other words, a tetrad consists of four chromatids, two each from two homologous chromosomes. For example, there are 46 chromosomes ina human diploid cell at the beginning of

meiosis, so there are 23 homologous pairs of chromosomes, each with two chromatids (that is, 23 tetrads), for a total of 92 chromatids during prophase I. Throughout prophase I and metaphase I, the chromatin continues to coil and compact, so that the chromosomes appear ever thicker. At a certain point, the homologous chromosomes seem to repel each other, especially near the centromeres, but they are held together by physical attachments mediated by cohesins. These cohesins are different from the ones holding the two sister chromatids together. Regions having these attachments take on an X-shaped appearance and are called chiasmata, in 5 f

A chiasma reflects an exchange of genetic material between nonsister chromatids on homologous chromosomes—what geneticists call crossing over: The chromosomes begin exchanging material shortly after synapsis begins, but chiasmata do not become visible until later, when the homologs are repelling each other. Crossing over increases genetic variation among the products of meiosis by reshuffling genetic information among the homologous pairs. There seems to be plenty of time for the complicated events of prophase I to occur. Whereas mitotic prophase is usually measured in minutes, and all of mitosis seldom takes more than an hour or two, meiosis can take much longer. In human males, the cells in the testes that undergo meiosis take about a week for prophase I and about.a month for the entire meiotic cycle. In the cells that will become eggs, prophase I begins long before a woman’s birth, during her early fetal development, and ends as much as decades later, during the monthly ovarian cycle. Prophase I is followed by prometaphase I, during which the nuclear envelope and the nucleoli disaggregate.

A spindle forms, and microtubules become attached to the kinetochores of the chromosomes. In meiosis I, the kinetochores of both chromatids in each chromosome become attached to the same half-spindle. Thus the entire chromosome, consisting of two chromatids, will migrate to one pole. There is a randomness in the pairing of homologous chromosome pair to each halfspindle, and the migration to the pole. By metaphase I, all the chromosomes have moved to the equatorial plate. Up to this point, homologous pairs are held together by chiasmata. The homologous chromosomes separate in anaphase I, when the individual chromosomes, each still consisting of two chromatids, are pulled to the poles, with one homolog of a pair going to one pole and the other homolog going to the opposite pole. Each of the two daughter nuclei from this division thus contains only one set of chromosomes, not the two sets that were present in the original diploid nucleus. However, because they consist of two chromatids rather than just one, each of these chromosomes has twice the mass that a chromosome at the end of a mitotic

In some organisms, there is a telophase I, with the reappearance of the nuclear envelopes. When there is a telophase I, itis followed by an interphase, called interkinesis, similar to the mitotic interphase. During interkinesis the chromatin is partially uncoiled; however, there is no replication of the genetic material, because each chromosome already consists of two chromatids. | Furthermore, the sister chromatids in interkinesis are generally not genetically identical, because crossing over in prophase I has reshuffled genetic material | between the maternal and paternal chromosomes. In other organisms, the chromosomes move directly into the second meiotic division. | The second meiotic division separates the chromatids: Meiosis II is similar to mitosis in many ways (Fig. 1.9). In each nucleus produced by | meiosis I, the chromosomes line up at equatorial plate at metaphase IL. The centromeres of the sister chromatids separate because of cohesin breakdown | and the daughter chromosomes move to the poles in anaphase I. r The result of meiosis is four nuclei; each nucleus is haploid and has a single set of unreplicated chromosomes that differs from other such sets in its exact genetic composition. The differences among the haploid nuclei result from crossing over during prophase I and from the random segregation of homologous chromosomes during anaphase I. MEIOSIS-II Prophase-| Metaphase-II ANS, BOs wie ele Anaphase-Il Telophase-N

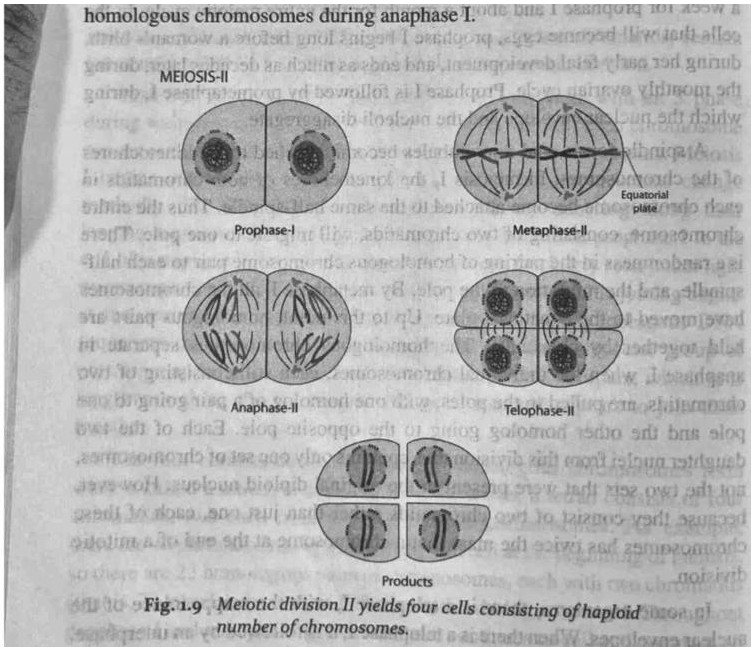


Fig. 1.9 *Meiotic division II yields four cells consisting of haploid number of chromosomes.*

Extra points and computing analogy

Mitosis makes two identical cells (growth, repair); meiosis makes four cells with half the chromosomes (gametes). Mitosis phases: prophase, metaphase, anaphase, telophase, then cytokinesis. One task (a cell) is split into two identical workers that run side by side, like forking a process; billions of cells dividing at once is massive parallelism.

In depth

- Cell cycle: G1 (growth), S (DNA synthesis), G2 (preparation), M (mitosis). Checkpoints at G1/S, G2/M and the spindle checkpoint are controlled by cyclins and CDKs.
- Mitosis: prophase (chromosomes condense), metaphase (line up at the plate), anaphase (sister chromatids pulled apart), telophase (two nuclei form), then cytokinesis.
- Meiosis has two divisions, with crossing over in prophase I and independent assortment, so gametes are genetically different and haploid (n).
- Computing view: dividing is like fork(); many cells dividing is a multi-threaded system. Cancer is like a loop that lost its exit condition.

Remember: Checkpoints in the cell cycle act like assertions: if DNA is damaged, division stops.

S5 Genetic algorithms for biological optimization

2 MARKS A genetic algorithm searches for good solutions by evolving a population of candidates through fitness-based selection, crossover and mutation.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Definition; Key features or structure; Process or mechanism; Examples; Applications; Limitations.

12 TO 13 MARKS: ANSWER IN THIS ORDER

1. Definition
2. Key features or structure
3. Process or mechanism
4. Examples
5. Applications
6. Limitations
7. Conclusion

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS

2 marks: Define fitness function in a genetic algorithm.
5 marks: Explain selection, crossover and mutation.
12 marks: Explain a genetic algorithm with a flowchart and a biological optimisation example.

S6 Evolutionary computing in cell/tissue modelling

2 MARKS Evolutionary computing is a family of search methods, such as genetic algorithms, that imitate natural evolution to solve optimisation problems.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Definition; Key features or structure; Process or mechanism; Examples; Applications; Limitations.

12 TO 13 MARKS: ANSWER IN THIS ORDER

1. Definition
2. Key features or structure
3. Process or mechanism
4. Examples
5. Applications
6. Limitations
7. Conclusion

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS

2 marks: What is evolutionary computing?

5 marks: Give three uses of evolutionary computing in cell and tissue modelling.

12 marks: Discuss evolutionary computing and agent-based modelling of cells and tissues, with limitations.

Extra 1 Protein structure

2 MARKS Proteins are the most abundant organic molecules of the living system.
4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: 1.5 Protein structgre; Forces influencing protein structure; Hydrogen bonds; Hydrophobic interactions; Van der Waals interactions; B-helix.
12 TO 13 MARKS: ANSWER IN THIS ORDER 1. 1.5 Protein structgre 2. Forces influencing protein structure 3. Hydrogen bonds 4. Hydrophobic interactions 5. Van der Waals interactions 6. B-helix 7. Dynamic functions *Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.*

Complete notes

1.5 Protein structgre

Proteins are the most abundant organic molecules of the living system. They occur in every part of cefl and constitute about SO%|of cellular dry weight. They form the“Tundamental basis of structure and function of life. The functional properties of proteins depend upon their three-dimensional structures. They are the molecules that actually do almost all the work in the cell And are largely responsible TOF expressing the information present ; in the DNA molecule. They range in size from a few kilodaltons to hundreds of kilodaltons. All proteins are built up from 20 amino acids, which constitute 7 the monomers. The structure and function of proteins are directly derived from the chemical nature of aminoacids.

Forces influencing protein structure

Several different kinds of non-covalent interactions are of vital importance in protein structure. Hydrogen bonds, hydrophobic interactions, electrostatic bonds, and Van der Waals forces are all non covalent in nature, yet are extremely important influences on protein conformations.

Hydrogen bonds: Hydrogen (H) bonds are generally made wherever possible within a given protein structure. In most protein structures that have been examined to date, component atoms of the peptide backbone tend to form hydrogen bonds with one another. Furthermore, side chains capable of forming H bonds are usually located on the protein surface and form such bonds primarily with the water solvent. Although each hydrogen bond may contribute an average of only about 12 kJ/mol in stabilization energy for the protein structure, the number of H bonds formed in the typical protein is very large, For example, in a-helices, the CO and NH groups of every residue participate in H bonds. The importance of H bonds in protein structure cannot be overstated.

Hydrophobic interactions: Hydrophobic “bonds,” or, more accurately, interactions, form because non-polar side chains of amino acids and other non-polar solutes prefer to cluster in a non-polar environment rather than to

/ intercalate in a polar solvent such as water. The : forming of hydrophobic bonds minimizes the interaction of non-polar residues with. water and Is therefore highly favourable. Such clustering is entropically driven. The side chains of the amino acids in the interior or core of the protein Structure. are almost exclusively hydrophobic. Polar amino acids are almost never found in ihe interior of a protein, but the protein surface may consist of both polar and | non polar residues. 7 Electrostatic interactions: Ionic interactions arise either as electrostatic Attractions between opposite charges or repulsions b between like charges. Amino acid side chains can carry positive charges, as in the case of lysine, arginine, and histidine, or negative charges, as in aspartate and glutamate. In addition, the NH,-terminal and COOH-terminal residues of @ protein or peptide chain usually exist in ionized states

and carry positive BENG J charges, respectively. All of these may experience electrostatic Waa a protein structure. Charged residues are normally located on the_protein | surface, where they may interact optimally with the water solvent. It is | energetically unfavourable for an ionized residue to be located in the hydrophobic core of the protein. Electrostatic interactions between charged groups on a protein surface are often complicated by the presence of salts in > the solution. For example, the ability of a positively charged lysine to attract a nearby negative glutamate may be weakened by dissolved NaCl. . The Nat and Cl- ions are highly mobile, compact units of charge, compared d to the amino acid side chains and thus, compete effectively for charged sites on the protein. In this manner, electrostatic interactions among amino acid residues on protein surfaces may be damped out by high concentrations of salts. Nevertheless, these interactions are important for protein stability.

Van der Waals interactions: Both attractive and repulsive forces are included in Van der Waals interactions. The attractive forces are due primarily to instantaneous dipole-induced dipole interactions that arise because of fluctuations in the electron charge distributions of adjacent non bonded atoms. Individual Van der Waals interactions are weak ones (with stabilization energies of 4.0 to 1.2 kJ/mol), but many such interactions occur ina typical protein, and, by sheer force of numbers, they can represent a significant contribution to the stability of a protein. Van der Waals interactions between tightly packed groups in the interior of the protein are a major contribution to protein stability. Structure of proteins The amino dod Sequence of a protein's polypeptide chain is called its primary structure (Fig. 1.2). Different regions of the sequence form local regular eit, secondary structure tertiary structure quaternary structure

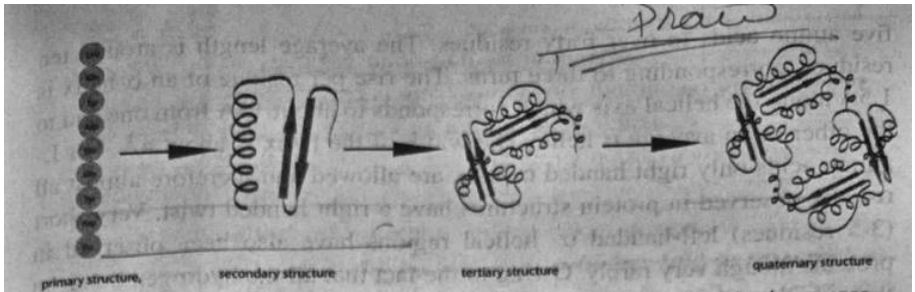


Fig. 1.2 *The sequence of amino acids forms different protein structures: primary, secondary, tertiary, and quaternary.*

secondary structures, such as alpha («) helices or beta (B) strands. The tertiary structure is formed by packing Such structural elements into one or several compact globular units called domains. The final protein may contain several polypeptide chains arranged in a quaternary structure. By formation of such tertiary and quaternary structure amino acids far apart in the sequence are brought close together in three dimensions to form a functional region, an active site

All the 20 amino acids have in common a central carbon atom (C,) to which are attached a hydrogen atom, an amino group (NH,) and a carboxyl group (COOH). Amino acids are joined end-to-end during protein synthesis by the formation of peptide bonds when the carboxyl group of one amino acid condenses with the amino group of the next to eliminate water (figure 3). This process is repeated as the chain elongates. One consequence is that the amino group of the first amino acid of a polypeptide chain and the carboxyl group of the last amino acid remain intact and the chain is said to extend from its amino terminus to its carboxy terminus.

Secondary structure The conformation of polypeptide chain by twisting or folding is referred to as secondary structure. The amino acids are located close to each other in their

B-helix: The a-helix has 3.6 residues per turn with hydrogen bonds between C=O of residue n and NH of residue n+4. Thus all NH and CO groups are joined with hydrogen bonds except the first NH groups and the last CO groups at the ends of the a-helix. As a consequence, the ends of a helices are polar and are almost always at the surface of protein molecules. In globular proteins © helices vary considerably in length ranging from four or five amino acids to over forty residues. The average length is around ten residues, corresponding to three turns. The rise per residue of an o.-helix is 1.5A along the helical axis which corresponds to about 15A from one end to the other of an average © helix. The width of the helix is about 4A. For Lamino acids only right handed o.-helix are allowed and

therefore almost al} o-helix observed in protein structures have a right handed twist. Very short (3-5 residues) left-handed ©- helical regions have also been observed in proteins though very rarely. Owing to the fact that all the hydrogen bonds in the @-helix point in the same direction, the helix as a whole has a dipole moment such that the amino end or the N- terminal has a net partial positive charge and the carboxy end or the C- terminal has a net partial negative charge. Alpha helices are preferentially made up of certain of the 20 amino acid residues. Among the strongest helix- formers are alanine, glutamine, leucine and methionine. Conversely proline, glycine, tyrosine and serine occur in helices only rarely. An o-helix often occurs on the surface of a protein. Then it is usually made up of hydrophobic residues on the side facing the interior of the protein and hydrophilic residues on the other.

Beta (f) sheets usually have their B strands either parallel or antiparallel: The second major structural element found in globular proteins is the B sheet. This structure is built up from a combination of several regions of the polypeptide chain, in contrast to the o-helix which is built up from one continuous region.

These regions, B strands, are usually from 5 to 10 residues long and are in an almost fully extended conformation with y, y angles within the broad structurally allowed region in the upper left quadrant of the Ramachandran plot. Beta strands can interact in two ways to form a pleated sheet. Either the amino acids in the aligned f strands can all run in the same biochemical direction, amino terminal to carboxy terminal in which case the sheet is described as parallel or the amino acids in successive strands can have alternating directions, amino terminal to carboxy terminal followed by carboxy terminal to amino terminal, followed by amino terminal to carboxy terminal and so on in which case the sheet is called antiparallel. Each of the two forms has a distinctive pattern of hydrogen- bonding. The antiparallel B sheet has narrowly spaced hydrogen bond pairs that alternate with widely spaced pairs.

Within both types of f sheets all possible main chain hydrogen bonds are formed except for the two flanking strands of the B sheet that have only one neighbouring f strand. Beta strands can also combine into mixed B sheets with some f strand pairs parallel and some antiparallel. There is a strong bias against mixed B sheets; only about 20% of the strands inside the f sheets of known protein structures have parallel bonding on one side and antiparallel bonding on the other. As they occur in known protein structures, almost all B sheets-parallel, antiparallel and mixed have twisted strands.

Tertiary structure Various segments of a protein chain fold into various secondary structural units such as a-helices or B sheets. These units further fold up into compact super secondary structural units or domains. This is called the tertiary structure of the proteins and involves interactions between amino acid residues that are not necessarily close together in the primary sequence. One of the most important of such tertiary interactions is the disulphide bond. The ‘disulphide bond is formed between the sulphur atoms of the cysteine residues. Such bonds could occur within a single polypeptide chain or between two different chains. The chief function of the disulphide bridge is to provide mechanical stability to the protein structure. They may also determine chemical properties by stabilising the correct active conformation. Soa In some proteins, the disulphide bond may play a catalytic role. Another major stabilising factor in the tertiary structure of proteins is the so-called ~ entropic or hydrophobic forces. Since the protein is made up of both polar and non-polar side chains in an aqueous solution the protein molecule behaves like a drop of oil, with polar groups on the surface, in contact with the solution and the non-polar groups in the interior of the molecule. The solvent molecules in the near of the protein also assume a certain amount of ordering around polar groups. For non-polar groups, however, the solvent molecules make a large negative contribution to the entropic energy, leading to the folded state being favoured over the denatured state. Other factors which stabilize the tertiary structure of proteins include salt bridges or strong columbic interactions between oppositely charged amino acids such as glutamic acid and aspartic acid. Hydrogen bonds in the interior of the protein are another important stabilizing factor not only of the secondary structural motifs but also of the tertiary structure /The secondary structural units often form clearly identifiable domains or super-secondary structures. Quaternary structure __Maniy proteins in their active or functional forms exist as aggregates of more than one folded polypeptide chain. The macromolecular structure so built up is called the quaternary structure of proteins. Usually

the subunits of quaternary structure associate by non-covalent interactions, though disulphide linkages may exist between different subunits. The monomers which form the quaternary structure may be identical or may be different. F or example, haemoglobin is a tetrameric protein consisting of two chains each of two different polypeptides called the chain and the number of subunits Tanges " from 2 to about lee with a majority of possessing 2 or 4 subunits. The exceptions are large enzyme complexes and viruses, which may have even | 100 or more monomers. Most protein multimers possess point. group symmetry consisting of one or.more interacting rotation axes. Some of the rotation axes found in proteins include 2-fold, 3-fold, 4-fold, 5-fold, 6-fold and 8-fold axes.

A tetrahedral symmetry is also seen where the monomers are placed at the vertices of an imaginary tetrahedron. Hexameric proteins may form trigonal prisms, hexagons or octahedral. Octameric proteins may form cubes or square prisms. In order to attain a definite function there are several reasons why an assembly of subunits is evolutionary favourable, rather than giant covalently linked molecules. One of the reasons is that less DNA is needed to code such a protein, since the monomers also contain the information to self assemble into the multimer. Again, it is easier to make smaller units in an error free fashion than large units. Similarly defective units can be more economically discarded if they are small. A very important reason is probably that subunits with a particular function may be combined with different subunits to make up different proteins. This modular approach to building up a protein exists at the level of domains too. A particularly striking example is the structure of functions, such as _ glyceraldehydes-3-phosphate dehydrogenase, phosphoglycerate kinase and phosphorylase. Functions of proteins Proteins perform a great variety of specialized and essential functions in the living cells. These functions may be broadly grouped as structural and dynamic. Structural functions: Certain proteins perform brick and mortar roles and are primarily responsible for structure and strength of body. These include collagen and elastin found in bone matrix, vascular system and other organs and o1-keratin present in epidermal tissues.

Dynamic functions: The dynamic functions of proteins are more diversified in nature. These include proteins acting as enzymes, hormones, blood clotting factors, immunoglobulins, membrane receptors, storage proteins, besides their function in genetic control, muscle contraction, respiration, (isy Proteins performing dynamic functions are appropriately regarded as “the working horses” of cell. y al, SOUS VIER eg Sica cea Bonin cee Beate me ra (date AI, Saka deat avetyN cae esse as i

Extra 2 Cell metabolism

2 MARKS There are two broad categories based on energy sources mainly, phototrophs (‘“trophs” means nourishment) trap and use sunlight, and chemotrophs derive their energy from oxidation of nutrients.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: 1.6 Cell metabolism; An overview of Fatty Acid Metabolism; Amino acid metabolism; Amino acid biosynthesis; Fatty Acid Metabolism.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. 1.6 Cell metabolism
2. An overview of Fatty Acid Metabolism
3. Amino acid metabolism
4. Amino acid biosynthesis
5. Fatty Acid Metabolism

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

Complete notes

1.6 Cell metabolism

Living organisms can be classified according to how they obtain the energy and carbon they need for synthesizing cellular material from their cellular environment. There are two broad categories based on energy sources mainly, phototrophs (‘“trophs” means nourishment) trap and use sunlight, and chemotrophs derive their energy from oxidation of nutrients. All chemotrophs require a source of organic: nutrients; (they cannot fix CO, into organic compounds). The phototrophs can be further divided into those that can obtain all needed carbon from CO, (autotrophs) and those that require organic chemical nutrients (heterotrophs). Chemotrophs may be further classified according to a different criterion: whether the fuels they oxidize are inorganic ; (lithotrophs) or organic (organotrophs). ay.

Living cells and organisms must perform work to stay alive that requires ; the input of energy and is given to cell through cell metabolism. Metabolism is a highly coordinated cellular activity in which many multienzyme systems _ (metabolic pathways) cooperate to (1) obtain chemical energy by capturing solar energy (autotrophs) or degrading energy-rtich nutrients from the environment (heterotrophs) ; (2) convert nutrient molecules into the cell’s own characteristic molecules, including precursors of macromolecules; (3) polymerize monomeric precursors into macromolecules: proteins, nucleic acids, and polysaccharides; and (4) synthesize and degrade biomolecules required for specialized cellular functions, such as membrane lipids, intracellular messengers, cell signalling. ef

Cell metabolism is thousands of enzyme-catalyzed chemical reactions in cells. These are functionally organized into many sequences of consecutive reactions, called pathways, in which the product of one reaction becomes the reactant in the next. Some pathways degrade complex organic nutrients into simple end products in order to extract chemical energy and convert it into a

= useful to the cell; together these degradative, free-energy-yielding ie =~ catabolism. Other pathways start with small ei ae

| precursor molecules and convert them to progressively larger and more . complex molecules, including proteins and nucleic acids. Such synthetic pathways, which invariably require the input of energy, are collectively designated anabolism. ‘

The overall network of enzyme-catalyzed pathways constitutes cellular metabolism. ATP is the major connecting link (the shared intermediate) between the catabolic and anabolic components of this network also reducing power of NADH, NADPH, and FADH. The pathways of enzyme-catalyzed reactions that act on the main constituents of cells like proteins, fats, sugars, and nucleic acids that are virtually identical in all living organisms.

ATP plays important role in cellular metabolism: ATP is an energy currency of cell. It plays special role that links catabolism and anabolism. Heterotrophic cells obtain free energy in a chemical form by the catabolism of nutrient molecules, and they use that energy to make ATP from ADP and Pi. ATP then donates some of its chemical energy to endergonic processes such as the synthesis of metabolic intermediates and macromolecules such as carbohydrates, fats, lipids from smaller precursors. This donation of energy from ATP generally involves the covalent participation of ATP in the reaction that is to be driven, with the eventual result that ATP is converted to ADP and Pi or, in some reactions, to AMP and 2 Pi.

ATP provides energy through Phosphoryl group transfer and by not mean of simple hydrolysis: Most of the reactions of cell metabolism are not single step though it looks like simple but it is not simple conversion of ATP to ADP and Pi (or, in some cases, of ATP to AMP and pyrophosphate, PPi). When written this way, these reactions of ATP appear to be simple hydrolysis reactions in which water displaces Pi (or PPi). This is not the case. ATP hydrolysis accomplishes nothing but the liberation of heat. A single reaction arrow such as that in almost invariably represents a two-step process in the one of the reaction firstly, a phosphoryl or pyrophosphoryl group or the adenylate moiety (AMP), is transferred to a substrate molecule or to an amino acid residue in an enzyme, becoming covalently attached to the substrate or the enzyme and raising its free-energy content. Then, in a second step, the phosphate-containing moiety transferred in the first step is displaced, generating Pi, PPi, or AMP. Thus ATP participates covalently in the enzyme-

Catalyzed reaction to which it contributes free energy. ATP plays an important role in metabolism (catabolism and anabolism). sion of complex nutrients (carbohydrates, fats, proteins) into erates ATP which is used by our body to carry out : / fe is excess of monomers incell, itcan be converted to complex nutrients (carbohydrates, fats, proteins) with utilisation of ATP (anabolism). Thus, ATP plays important role in cell

Amino acid metabolism: Amino acids are amphipathic molecules with amino and carboxyl group at each of its terminal. They play several physiological roles in our body: a) These are building blocks of protein, which is an important constituent of biological or plasma membrane. b) Many proteins are modified by the covalent attachment of fatty acids (lipoprotein) or carbohydrate moieties (glycoprotein) which targets them, to membrane locations (such as membrane-bound receptor). c) Amino acids may serve as fuel molecules when carbohydrates are either unavailable or not properly utilised. d) Amino acid serves as cofactors for various enzymes, hormones and intracellular messengers of cellular activities like cell signalling, cell adhesion etc.

Amino acid biosynthesis: All amino acids are derived mainly through the intermediates of glycolysis, the citric acid cycle, or the pentose phosphate pathway. Nitrogen enters these pathways with the help of two amino acids namely glutamate and glutamine. Ten of the amino acids are just one or several steps removed from the common metabolite from which they are derived. The biosynthetic pathways for others, such as the aromatic amino acids, are more complex.

Organisms vary greatly in their ability to synthesize the 20 common amino acids. Whereas most bacteria and plants can synthesize all 20, : mammals can synthesize only about half of them generally those with simple pathways. These are called as nonessential amino acids, not needed in the diet. The remainder, the essential amino acids, must be obtained from food. The 10 essential amino acids are phenylalanine, valine, threonine, tryptophan, isoleucine, methionine, histidine, arginine, leucine, and lysine.

Fatty Acid Metabolism: Fatty acid is a long hydrocarbon chain with a terminal carboxylate group. Fatty acids play several major physiological roles in our body. a) They are building blocks of phospholipids and glycolipids, which are components of biological membranes or plasma membrane. b) Many proteins are modified by the covalent attachment of fatty acids

(lipoprotein), which targets them, to membrane locations (such as membrane bound receptor). c) Fatty acids are fuel molecules. They are stored in our body a, triacylglycerols (which are uncharged esters of fatty acids with glycero}) also called neutral fats or triglycerides. Fatty acids .mobilized from, triacylglycerols are oxidized to meet the energy needs of a cell or organism, d) Fatty acid derivatives serve as hormones (sex hormones) and intracellular messengers of cellular activities like

cell signaling, cell adhesion etc.

An overview of Fatty Acid Metabolism

Fatty acid degradation and synthesis are relatively simple processes which are reverse of each other (Fig. 1.3). The process of degradation converts a fatty acid (an aliphatic compound) into a set of activated acetyl units (most of the time acetyl CoA moieties) that can be processed by the citric acid cycle. An activated fatty acid is oxidized to introduce a double bond; which on hydration introduce oxygen; then four carbon fragment is cleaved by coenzyme A to yield acetyl CoA and a fatty acid chain with two carbons shorter as in parent fatty acid. If the fatty acid has an even number of carbon atoms and is saturated, the process is simply repeated until the fatty acid is completely converted into acetyl CoA units, however fatty acids with odd number of carbon atoms and unsaturated fatty lipid has separate enzymatic machinery which subsequently give rise to intermediates of citric acid cycle.

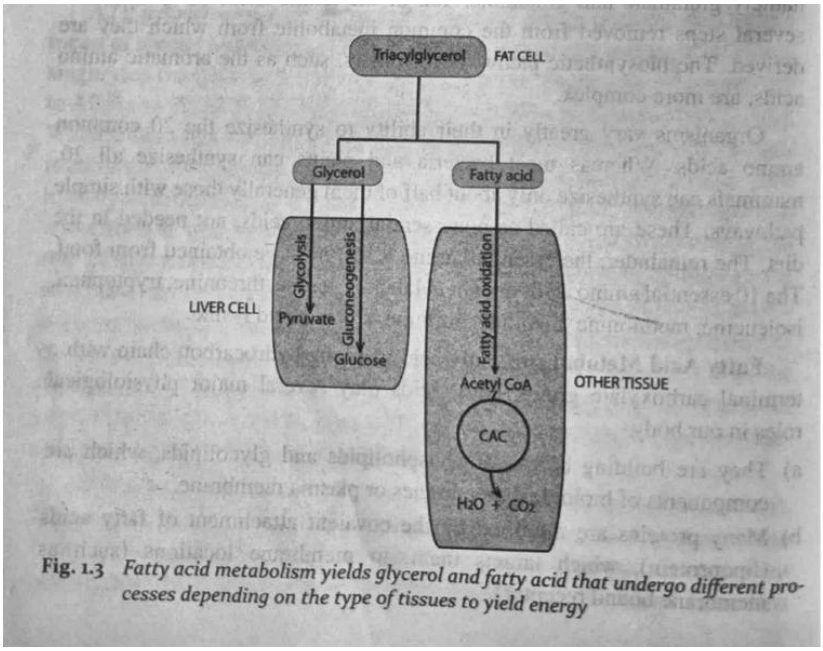


Fig. 1.3 *Fatty acid metabolism yields glycerol and fatty acid that undergo different processes depending on the type of tissues to yield energy*

Fatty acid synthesis is essentially the reverse of fatty acid breakdown. Because the result is the formation of polymer, the process starts with monomers, the activated acyl group (most simply, an acetyl unit) and malonyl units. The malonyl unit is condensed with the acetyl unit to form a fourcarbon fragment. To produce the required hydrocarbon chain, the carbonyl must be reduced. The fragment is reduced, dehydrated, and reduced again (exactly the opposite of degradation) to bring the carbonyl group to the level of a methylene group with the formation of butyryl CoA. Another activated malonyl group condenses with the butyryl unit and the process is repeated until a C16 fatty acid is synthesized.

The oxidation of fatty acid takes place in the mitochondrial matrix. The acyl CoA molecules are transported across the inner mitochondrial membrane by carnitine, a zwitter ionic compound formed from lysine. Acetyl CoA, NADH and FADH2 are generated in each round of fatty acid oxidation. A saturated fatty acid (1-acyl CoA) is degraded by a recurring sequence of four reactions: Oxidation by FAD, hydration, oxidation by NAD+, and thiolysis by CoA. As a result, the fatty acyl chain is shortened by two carbon atoms, FADH and NADH and acetyl CoA are generated. These reactions constitute the B oxidation pathway. mi Carbohydrate Metabolism ore Carbohydrates play several physiological roles in our body. F a) Itis an important constituent of biological or plasma membrane. b) Many carbohydrates are modified by the covalent attachment of fatty acids (glycolipids) or protein moieties (glycoprotein) which targets them, to membrane locations (such as membrane bound receptor). c) Carbohydrates serve as fuel molecules. d) Carbohydrates provide energy for various enzymes, hormones and intracellular messengers of cellular activities like cell signaling, cell adhesion etc. .

Carbohydrate metabolism involves several metabolic pathways for the generation of energy for cellular respiration. Some of them are glycolysis, citric acid cycle (tricarboxylic acid cycle or Kreb’s

cycle) and electron transport chain that utilize glucose to generate ATPs.

Glycolysis is a Greek word which means that splitting of sweet, in this a molecule of glucose is degraded in a series of enzyme-catalyzed reactions to yield two molecules of the three-carbon compound pyruvate and is occurs in ten steps. During the sequential reactions of glycolysis, some of the free energy released from glucose is conserved mainly in the form of ATP ang

Glycolysis is an almost universal central pathway of glucose catabolism, the pathway with the largest flux of carbon in most cells. The glycolytic breakdown of glucose is the sole source of metabolic energy in some mammalian tissues and cell types such as erythrocytes, renal medulla, brain, and sperm. Some plant tissues that are modified to store starch (such as potato tubers) and some aquatic plants (watercress, for example) derive most of their energy from glycolysis; many anaerobic microorganisms are entirely dependent on glycolysis.

Citric acid cycle (Tricarboxylic acid cycle or Kreb’s cycle) occurs in mitochondria. It is the common pathway to completely oxidize fuel molecules which mostly is acetyl CoA, the product of the oxidative decarboxylation of pyruvate. It enters the cycle and passes ten steps of reactions that yield energy

Electron Transport Chain (ETC) and Oxidative Phosphorylation: Main source of NADH is from TCA cycle this can be used for reductive biosynthesis; however, the reducing potential of mitochondrial NADH is most often used to supply the energy for ATP synthesis via oxidative phosphorylation. Oxidation of NADH with phosphorylation of ADP to form ATP is processes supported by the mitochondrial electron transport assembly and ATP synthase, which are integral protein complexes of the inner mitochondrial membrane. The electron transport assembly is comprised of a series of protein complexes that catalyze sequential oxidation reduction reactions; some of these reactions are thermodynamically competent to

Metabolic engineering deals with optimizing regulatory factors such as enzymes and processes within the cell to augment the production of novel substances of industrial and medical importance. Mathematical and computational modeling is widely used to delete constraints or overexpress the genes in a metabolic pathway that can ultimately enhance the production of desired substances. es

>) | © Increased lysine, a amino acid, production was achieved by removing the constraints in metabolic network of the bacteria, Corynebacterium glutamicum é Pens ote

- Metabolism of fatty acids in Escherichia coli, a bacteria, was reversed

_ by genetic manipulation to achieve efficient production of alcohols and

| _carboxylic acids useful in the manufacture of biofuels and chemicals _ support ATP production via ATP synthase provided a coupling mechanism, such as a common intermediate, is available. Proton translocation and the development of a transmembrane proton gradient provide the required coupling mechanism.

These topics are further discussed in detail in Chapter 3 under photosynthesis, metabolism and bioenergetics.

Extra 3 Homeostasis

2 MARKS Maintaining a constant internal enyironment with all that the cells need to survive (O,, glucose, minerals, ions, and waste removyal) is necessary for individual cells.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: 1.7 Homeostasis; Extrinsic homeostatic system; Intrinsic homeostatic system; Nervous System; Endocrine System.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. 1.7 Homeostasis
2. Extrinsic homeostatic system
3. Intrinsic homeostatic system
4. Nervous System
5. Endocrine System

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

Complete notes

1.7 Homeostasis.

Human beings have trillions of cells all working together for the maintenance ~ : of the entire body. While Certain cells may perform very different functions, all the cells are quite similar in their metabolic requirements and perform certain basic functions essential for their own survival such as + Obtaining nutrients and oxygen from the environment surrounding the 2 Nutrients and oxygen to provide energy. : + Eliminating CO, and other wastes produced during chemical reactions —~ py the cells into the surrounding environment. se Synthesizing proteins and other components needed for cell structure. + Controlling the exchange of materials between the cell and its . surrounding environment.

Maintaining a constant internal enyironment with all that the cells need to survive (O,, glucose, minerals, ions, and waste removyal) is necessary for individual cells. The processes by which the body regulates its internal environment are referred to as homeostasis. ae The concept of homeostasis was first articulated by the French scientist Claude Bernard. He said “all the vital mechanisms, varied as they are have only one object that of preserving constant the conditions of life in the internal environment”. The term, homeostasis, was coined by American physiologist Walter cannon, author of the “Wisdom of the Body” (1932). The word comes from the Greek homoios (same) and stasis (to stand). For e.g., after cut or abrasion, blood comes out of the wound site. In order to prevent further loss of blood and promote survival of the organism, following process need to 7 take place:

+ Primary homeostasis: This is defined as the formation of the platelet sen triage ; _*_ Secondary homeostasis: This is defined as the formation of fibrin ~ through the coagulation cascade.

/ Tertiary homeostasis: This. is defined as the formation’ of plastpa.for | reakdown lot. f we | re st “vm The adjusting of Pi aie the fyi oro snain stl tera evionment and which eas \/constant monitoring and adjustments as conditions change is..¢ ed y homevvtaic repuarion, Homeostatic regulation, involves thr€® parts (or % ecapbie’ The receptor receives information that mometeny changes in || the efivironment. ee \\ SO Gaiithel eeneer:Teroeetvee and processes information from the receptor. \ Effectors: It responds to the commands ¢ of the control center by either ‘Tn regulating body temperature there are temperature receptors in the skin which communicate information to the brain, which is the control center, and the effectors aré our blood vessels and sweat glands in our skin. Following , factors homeostatically regulated:

1. Concentration of nutrients: Cells need a constant supply of nutrients molecules Tor energy production. Energy is needed to support life 4 sustaining and specialized cell activities. 2. Concentrations of O, and CO,: Oxygen used to carry out the energy) yielding chemical reactions and the CO, produced during thesé reactions must be removed so that acid forming CO, does not increase the acidity of the internal environment, AR ae eee ae 3. Concentrations of waste products: Some of the chemical reaction produce that may have a toxic effect on the cells if these wastes are

accumulate. 4. Concentrations of water, salt and other electrolytes: The concentration of salt (NaCl) and water in extracellular fluid influence how much water enters or leaves the cells, these concentrations are carefully regulated to maintain the proper volume of cells.

5. Volume and pressure: The circulating component of the internal environment, the plasma, is maintained by volume and Pressure. Pathways that alter homeostasis A. variety of homeostatic mechanisms maintain the internal environment within tolerable limits. Homeostasis is maintained through a series of control mechanisms or the body suffers various illness or disease. When the cells in the body begin to malfunction, the homeostatic balance becomes disrupted. Disease and cellular malfunction can be caused in two basic ways, either

. deficiency (cells not getting all they need) or toxicity (cells being poisoned by things they do not need),

Extrinsic homeostatic system: Most homeostatic systems are extrinsic and regulated by nervous system and endocrine system in higher animals.

|, The nervous system depends on sensors in the skin or sensory organs to

7 receive stimuli and transmit a message to the spinal cord or brain. Sensory input is processed and a signal is sent to an effector system, such as muscles or glands that affects the response to the stimulus.

2, The endocrine system is the second type of extrinsic control, and involves a chemical component to the reflex. Sensors detect a change within the body and send a message to an endocrine effector (parathyroid), which makes parathyroid hormone(PTH). PTH is released into the blood when blood calcium levels are low. PTH causes bone to release calcium into the

A bloodstream, raising the blood calcium levels and ultimately, shutting vy down further production of PTH. ei

7 ¥ Some reflexes have a combination of neryous and endocrine responses.

Ry The thyroid gland secretes thyroxine (which controls the metabolic rate) into

NY fe the bloodstream. Falling levels of thyroxine stimulate receptors in the brain yA to signal the hypothalamus to release a hormone that acts on the pituitary

"gland to release thyroid-stimulating hormone (TSH) into the blood, TSH acts on the thyroid, causing it to increase production of thyroxine. c “)

Intrinsic homeostatic system: The control systems usually involve only one organ or tissue. When muscles use more oxygen, and also produce more carbon dioxide, intrinsic controls cause dilation of the blood vessels allowing more blood into those active areas of the muscles. Eventually the vessels will return to “normal”.

Each organ system contributes to the homeostasis of other systems and of the entire organism, No system of the body works in isolation, and the wellbeing of the person depends upon the well-being of all the interacting body systems. A disruption within one system generally has consequences for several other body systems. Nervous system and endocrine system (discussed in detail in Chapter 5) are critical to the maintenance of homeostasis (Fig. 1.4).

Nervous System: The nervous system receives a continuous supply of

| nutrients from the bleod. Any interruption to the flow of blood may bring brain damage or death. The nervous system maintains homeostasis. by controlling and regulating the other parts of the body. A deviation from a normal set point acts as a stimulus to a receptor, which sends nerve impulses to a regulating center in the brain. The brain directs an effector to act in such

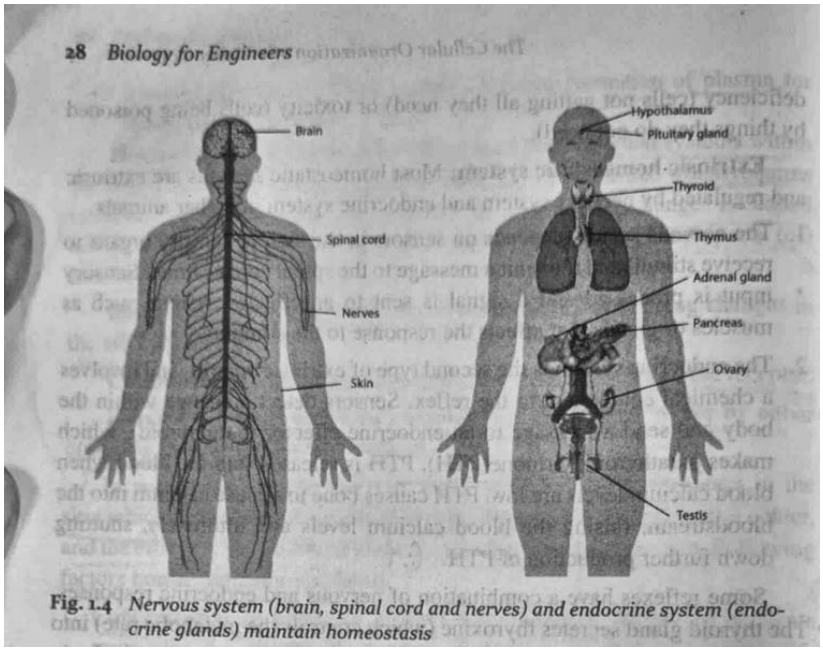


Fig. 1.4 *Nervous system (brain, spinal cord and nerves) and endocrine system (endocrine glands) maintain homeostasis*

i a way that an adaptive response takes place. For e.g., if there is abnormal j lowering of body temperature, the effector acts to increase body temperature. i The adaptive response returns the body toa state of normalcy and the receptor, | the regulating center, and the effector temporarily cease their activities. | The nervous system has two major portions: the central nervous system and the peripheral nervous system. The peripheral nervous system consists of , the spinal nerves. The autonomic nervous system is a part of peripheral nervous system and contains motor neurons that control internal organs. It operates at the subconscious level and has two divisions, the sympathetic and parasympathetic systems. In general, the sympathetic system brings about those results we associate with emergency situations, often called fight or flight reactions, and the Parasympathetic system produces those effects necessary to our everyday existence.

Regulating centers are located in the central nervous system, consisting of the brain and spinal cord. The hypothalamus is a portion of the brain particularly concerned with homeostasis; it influences the action of the medulla oblongata, a lower part of the brain, the autonomic nervous system, and the pituitary gland.

Endocrine System: The endocrine system consists of glands which secrete special compounds called hormones into the bloodstream Each hormone has an effect on one or more target tissues, In this way the endocrine system regulates the metabolism and development of most body cells and body systems. For e.g. the endocrine system has sex hormones that can activate sebaceous glands, development of mammary glands, alter dermal blood flow, and release lipids from adipocytes etc besides governing

Bone growth is regulated by several hormones, and the endocrine system helps with the mobilization of minerals like calcium by the hormone, calcitonin. In the muscular system, hormones adjust muscle metabolism, energy production, and growth. In the nervous system, hormones affect neural metabolism, regulate fluid/electrolyte balance and help with reproductive hormones that influence CNS (central nervous system), development and behaviours. In the cardiovascular system, hormones regulate heart rate and blood pressure. Hormones also have anti-inflammatory effects and control the lymphatic system. a

Extra 4 Cell differentiation and chapter highlights

2 MARKS] Differentiation is the biological process whereby an unspecialized cell acquires the properties of a specialized cell (e,g., stem cells as described in chapter 2).
4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: 1.9 CELL DIFFERENTIATION; Chapter highlights; Genomic equivalence.
12 TO 13 MARKS: ANSWER IN THIS ORDER 1. 1.9 CELL DIFFERENTIATION 2. Chapter highlights 3. Genomic equivalence *Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.*

Complete notes

1.9 CELL DIFFERENTIATION

] Differentiation is the biological process whereby an unspecialized cell acquires the properties of a specialized cell (e,g., stem cells as described in chapter 2). It defines the specific structures and functions of a cell (Fig. 1.10). Mitosis produces daughter cells that are genetically identical but still the cells of multicellular organism are obviously not all identical in structures or functions. This is because of the regulation of the gene expression of various parts of the genome. The regulation of gene expression starts from the embryo level. When the embryo consists of only a few cells, each cell has the potential to develop in many different ways. As development proceeds, the possibilities available to individual cells gradually narrow, until each cells fate is fully determined and the cell has differentiated.

Differentiated cells are different from one another starting from their protein products to morphology. So the main cause is differential gene expression for the cellular differentiation process to happen. The fertilized | egg or zygote has the ability to give rise to every type of cell in the adult bod ee tt Oe oy a Gitar) eit tht and hence, referred to as totipotent. The genomes of these cells carry instructions for all of the structures and the functions that will arise throughout eT ee Pospe, ater Y sagem it one io "T hea agin wad ei pea @— Endoderm Mesoderm © Etoderm 1 |e | Epidermis.skin, Pe wrsines Pancreas irae aa Liver brea pobre — +e | Vessels ob soe ; Outer covering Lining of thoracic Waite ef Eat: of * of internal organs: ‘and abdominal cavities 4

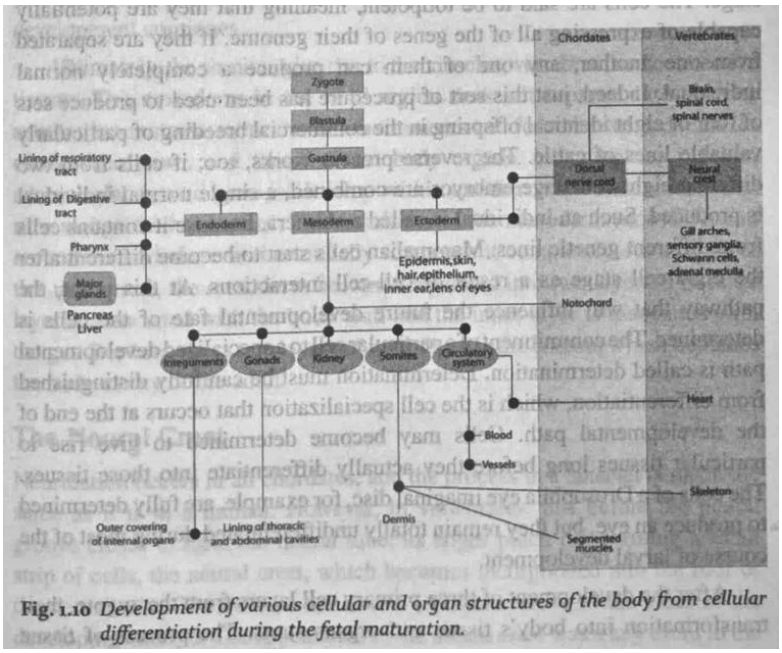


Fig. 1.10 *Development of various cellular and organ structures of the body from cellular differentiation during the fetal maturation.*

the life cycle. Later in the development of animals, the cellular descendents of the zygote lose” their totipotency and Become determined cel lis, then differentiate into specific types Of cells” ~ :

~ Changes in the genome during differentiation: Changes are of two types, either reversible or irreversible. Differentiation is irreversible in certain types of cells like mammalian RBC. It loses its nucleus during development and become specialized for oxygen uptake and transportation.

Genomic equivalence: No information is lost from the nucleus of cells as they pass through the early stages of embryonic developments.

The zygotic cells are regarded to be totipotent as they hold the ability to generate the whole organism. As the cell divides by responding to internal and external stimuli, its potency decreases. The cells at the blastula stage are then referred to as pluripotent, as they can generate all the cell types except the trophoectoderm. Further upon division, they become more restricted and give rise to multipotent stem cells like hematopoietic stem cell that can generate different blood cell types.

The mammalian egg is symmetrical in its contents as well as in shape, so that all of the cells of an early blastoderm are equivalent up to the eight-cell if" stage. The cells are said to be totipotent, meaning that they are potentially } capable of expressing all of the genes of their genome. If they are separated from one another, any one of them can produce a completely normal individual. Indeed, just this sort of procedure has been used to produce sets of four or eight identical offspring in the commercial breeding of particularly valuable lines of cattle. The reverse process works, too; if cells from two different eight-cell-stage embryos are combined, a single normal individual is produced. Such an individual is called a chimera, because it contains cells from different genetic lines. Mammalian cells start to become different after the eight-cell stage as a result of cell-cell interactions. At this point, the pathway that will influence the future developmental fate of the cells is determined. The commitmentofa particularcell toa specialized developmental path is called determination. Determination must be carefully distinguished from differentiation, which is the cell specialization that occurs at the end of the developmental path. Cells may become determined to give rise to particular tissues long before they actually differentiate into those tissues. The cells of a Drosophila eye imaginal disc, for example, are fully determined to produce an eye, but they remain totally undifferentiated during most of the course of larval development.

| After the development of three primary cell layers from the zygote, their transformation into body’s tissue and organs starts. The process of tissue

[* The Cellular Organization ofa Living Organism 41 differentiation begins with the formation of two morphological features found only in chordates, the notochord and the hollow dorsal nerve cord. This development of the dorsal nerve cord is called neurulation.

The notochord is first visible soon after gastrulation is complete, forming from mesoderm along the dorsal midline of the embryo. It is a flexible rod Jocated along the dorsal midline in the embryos of all chordates, although its function is replaced by the vertebral column when it develops from mesoderm in the vertebrates. After the notochord has been laid down, a layer of ectodermal cells situated above the notochord invaginates, forming a long crease, the neural groove, down the long axis of the embryo. The edges of the neural groove then move toward each other and fuse, creating a long hollow cylinder, the neural tube, which runs beneath the surface of the embryo’s back. The neural tube later differentiates into the spinal cord and brain. The dorsal lip of the blastopore induces the formation of a notochord, and the presence of the notochord induces the overlying ectoderm to differentiate into the neural tube. While the neural tube is forming from ectoderm, the rest of the basic architecture of the body is being determined rapidly by changes in the mesoderm. On either side of the developing notochord, segmented blocks of mesoderm tissue called somites form; more somites are added as development continues.

Ultimately, the somites give rise to the muscles, vertebrae, and connective tissues. The mesoderm in the head region does not separate into discrete somites but remains connected as somitomeres and form the striated muscles of the face, jaws, and throat. Some body organs, including the kidneys, adrenal glands, and gonads, develop within another strip of mesoderm that runs alongside the somites. The remainder of the mesoderm moves out and around the endoderm and eventually surrounds it completely. Asa result of this movement, the mesoderm becomes separated into two layers. The outer layer is associated with the body wall and the inner layer is associated with the gut. Between these two layers of mesoderm is the coelom which becomes the body cavity of the adult. '

The Neural Crest Neurulation occurs in all chordates, and the process in a lancelet is much the same as it is in a human. However, in vertebrates, just before the neural groove closes to form the neural tube, its edges pinch off, forming a small strip of cells, the neural crest, which becomes incorporated into the roof of the neural tube. The cells of the neural crest later move to the sides of the developing embryo. The appearance of the neural crest was a key event in the

: evolution of the vertebrates because neural crest cells, after migrating to different parts of the embryo, ultimately develop into the structures characteristic of (though not necessarily unique to) the vertebrate body. The differentiation of neural crest cells depends on their location. At the anterior end of the embryo, they merge with the anterior portion of the brain, the forebrain. Nearby clusters of ectodermal cells associated with the neuraj crest cells thicken into placodes, which are distinct from neural crest ce|]x although they arise from similar cellular interactions. Placodes subsequently develop into parts of the sense organs in the head. The neural crest ang associated placodes exist in two lateral strips, which is why the vertebrate sense organs that develop from them are paired. Neural crest cells located in more posterior positions have very different developmental fates. These cells migrate away from the neural tube to other locations in the head and trunk, where they form connections between the neural tube and the surrounding tissues. The migration of neural crest cells is unique in that it is not simply a change in the relative positions of cells, such as that seen in gastrulation, Instead, neural crest cells actually pass through other tissues. Some neural crest cells migrate ventrally toward the notochord and form sensory neurons in the dorsal root ganglia. Others become specialized as “ Schwann cells, which insulate nerve fibres and permit the rapid conduction of nerve impulses. Still others form the autonomic ganglia and the adrenal medulla. Cells in the adrenal medulla secrete epinephrine when stimulated by the sympathetic division of the autonomic nervous system during the fightor-flight reaction. The similarity in the chemical nature of the hormone epinephrine and the neurotransmitter norepinephrine, released by sympathetic neurons, is understandable— both adrenal medullary cells and sympathetic neurons are derived from the neural crest.

A variety of sense organs develop from the placodes. Included among them are the olfactory (smell) and lateral line (primitive hearing) organs. Neural crest cells contribute to tooth development and to some of the facial and cranial bones of the skull. Thus, differentiation of cells lead to the formation of the whole living organism.

Chapter highlights

- Living organisms are classified into 5 kingdoms, Monera, Protista,

Fungi, Plantae, and Animalia. Prokaryotes do not have nucleus while eukaryotic cells have a proper nucleus.

- Cell theory states that all living organisms are made up of cells which are the fundamental units of life and the new cells arise by cell division.

Unit 2

Biomolecules as Data Structures

Life is built from four main molecule classes. We treat them as storage, structure and information formats.

S1 Carbohydrates/lipids as energy and membrane storage

2 MARKS Carbohydrates are naturally occurring organic compounds composed primarily of

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Introduction; Why Does a Cell Need Energy?; Carbohydrates; Biological Importance- Carbohydrates; Classification; Journey of a Carbohydrate in the human body.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. Introduction
2. Why Does a Cell Need Energy?
3. Carbohydrates
4. Biological Importance- Carbohydrates
5. Classification
6. Journey of a Carbohydrate in the human body
7. ATP—The Universal Energy Currency
8. Glycogen: Nature's Cache Memory
9. Lipids
10. Why Are Lipids Better Energy Stores?
11. Characteristics of Lipids
12. Types of Lipids- 4 types

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: Define carbohydrates. What is a glycosidic bond?
5 marks: Classify carbohydrates with examples.
12 marks: Explain carbohydrates and lipids as energy and membrane storage molecules, with classification, types and functions.

Complete notes

Introduction

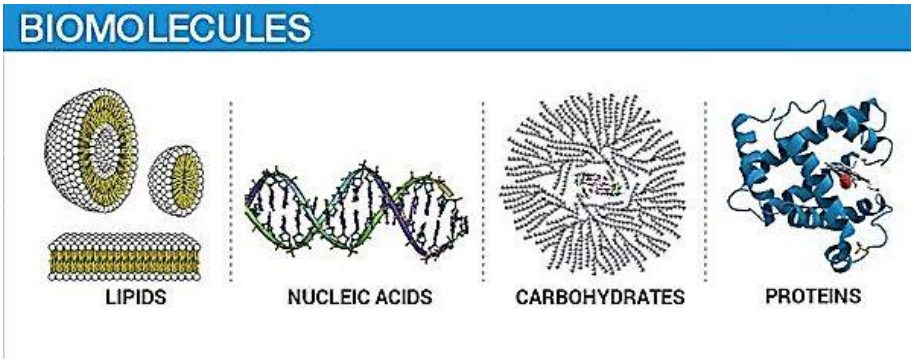
Every engineering system requires three things:

- Energy
- Storage
- Protection

- Similarly, every living cell requires
  - Energy
  - Information
  - Structure
  - Communication
- Nature solved these engineering challenges nearly **3.8 billion years** before the invention of computers.
- Cells use **biomolecules** to accomplish these tasks.

Figure

Biomolecule	Function	Engineering Analogy
Carbohydrates	Immediate energy	Laptop battery
Lipids	Long-term storage	Power bank
Proteins	Functional machines	CPU/Software
DNA/RNA	Information storage	Hard disk/Cloud database



Why Does a Cell Need Energy?

- Imagine your laptop. Without electricity,
  - CPU stops
  - RAM clears
  - Programs terminate
  - Communication ceases

Similarly, without ATP,

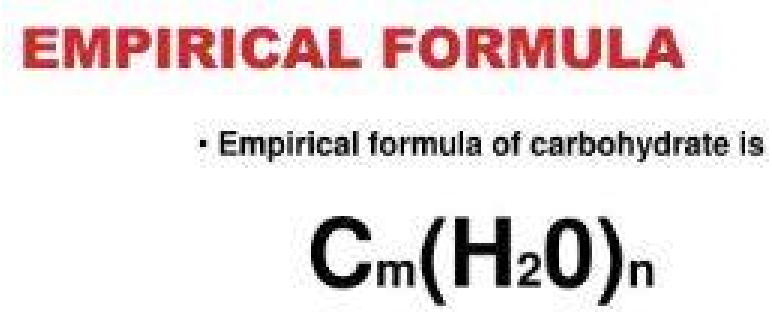
- muscles stop contracting
- DNA cannot replicate
- proteins cannot be synthesized
- neurons cannot transmit signals

Life requires continuous energy production.

Carbohydrates

Definition

- Carbohydrates are naturally occurring organic compounds composed primarily of
  - Carbon
  - Hydrogen
  - Oxygen
- m represents the number of carbon atoms. N represents the number of water molecules
- Historically, their empirical formula resembles Carbon and water hence called Carbohydrates
- Eg:- Glucose $C_6H_{12}O_6$ which approximates ($C(H_2O))_6$ where m=6 and n=6

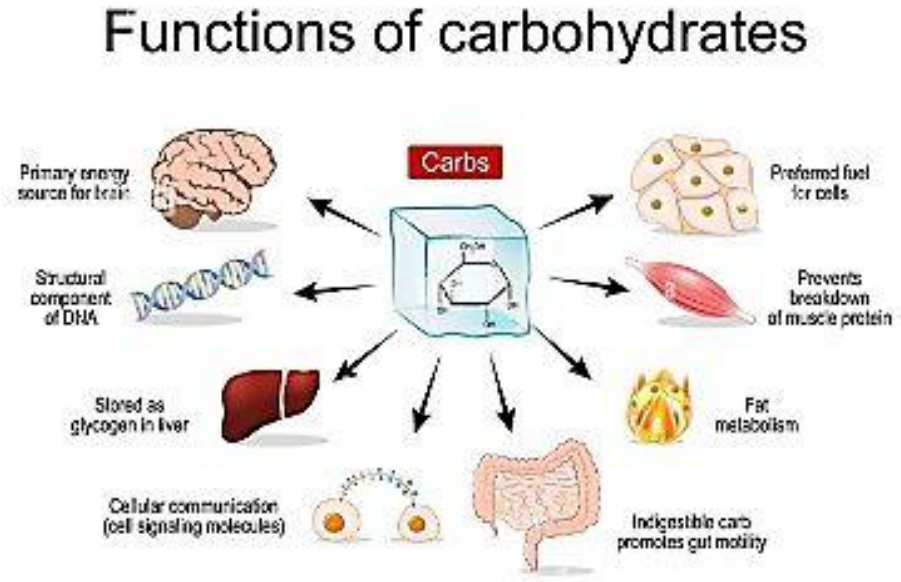


Biological Importance- Carbohydrates

Carbohydrates

- supply energy
- store energy
- provide structural support
- assist in cell recognition
- participate in signaling pathways

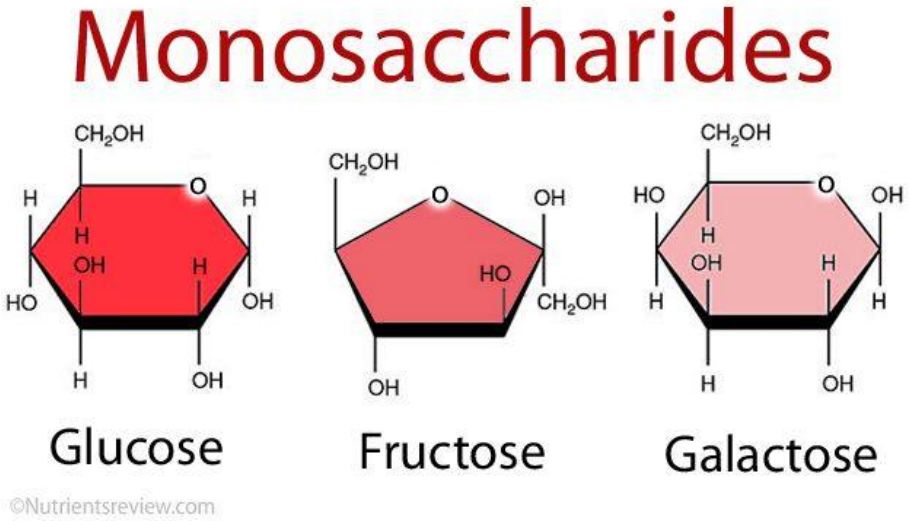
• Approximately **60–70% of daily human energy** comes from carbohydrates in a balanced diet.

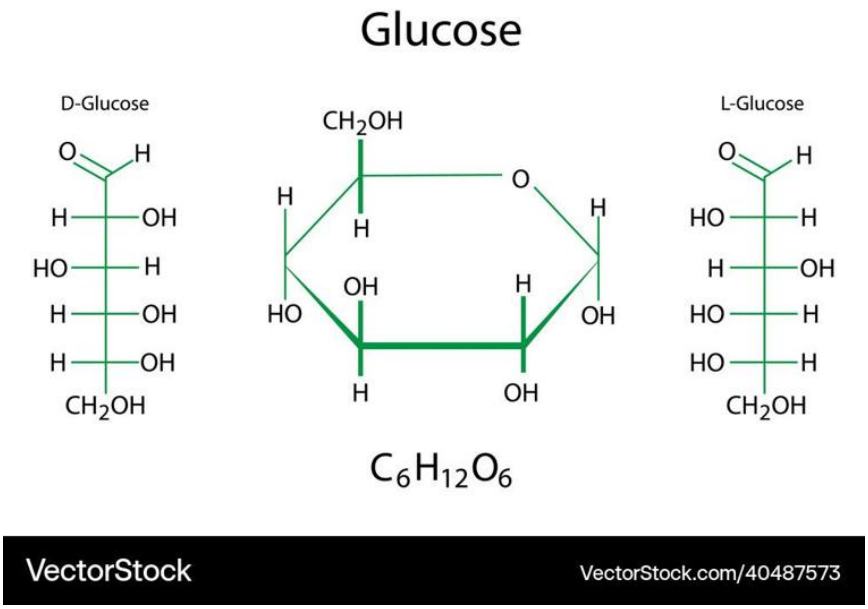


Classification

1. Monosaccharides

• Definition -Single sugar molecules.
• Examples –Glucose, Fructose, Galactose
• Characteristics- Sweet, water-soluble, Rapidly absorbed

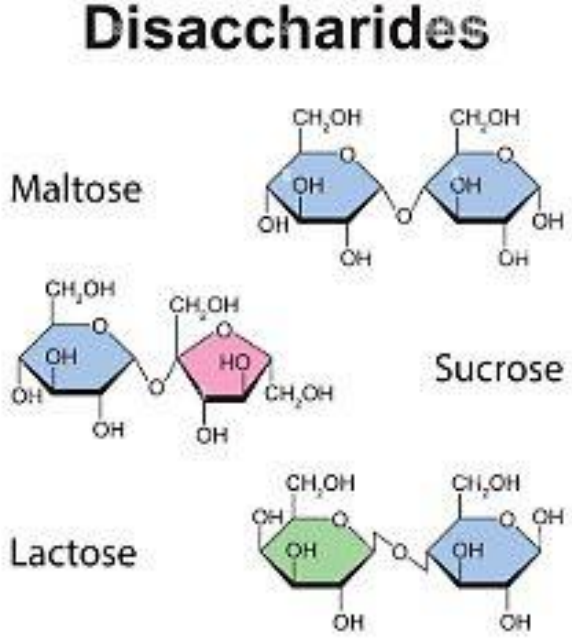


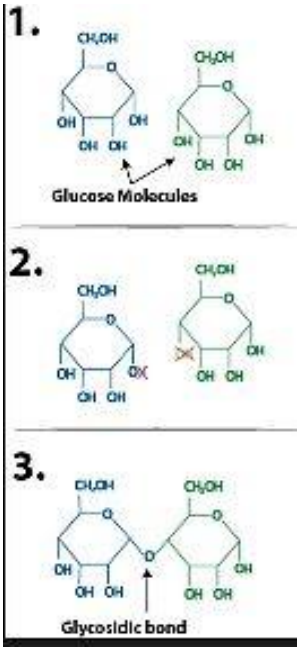


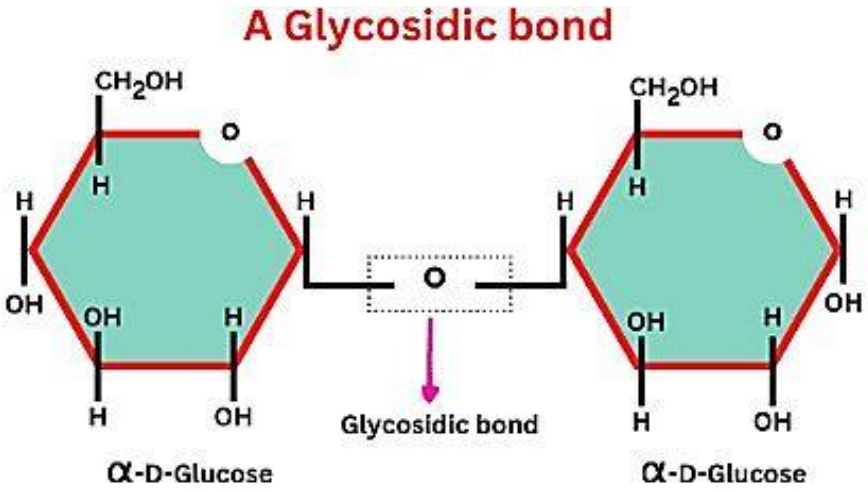
Figure

DiSaccharide	Composition
Sucrose	Glucose + Fructose
Lactose	Glucose + Galactose
Maltose	Glucose + Glucose

2. Disaccharides
Composed of two monosaccharides joined by a glycosidic bond.
The creation of a glycosidic bond occurs through condensation, also known as dehydration synthesis. This reaction involves joining two monosaccharide molecules and releasing a single water molecule.







Journey of a Carbohydrate in the human body

Consider eating a slice of bread.

Bread

↓

Starch

↓

Digestive enzymes

↓

Glucose

↓

Blood

↓

Cells

↓

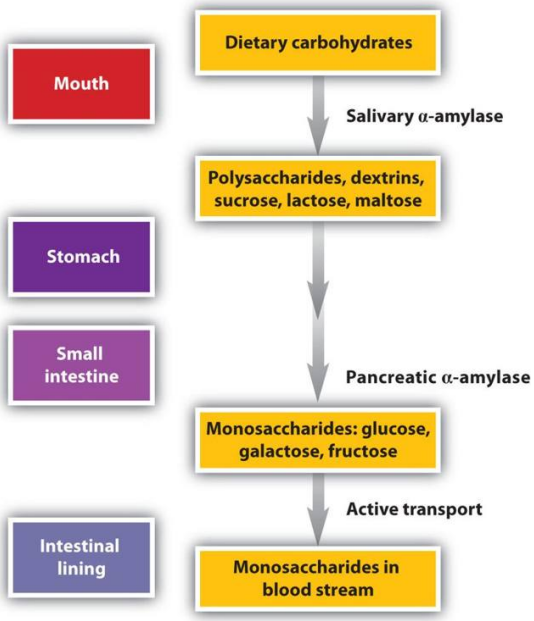
Mitochondria

↓

ATP

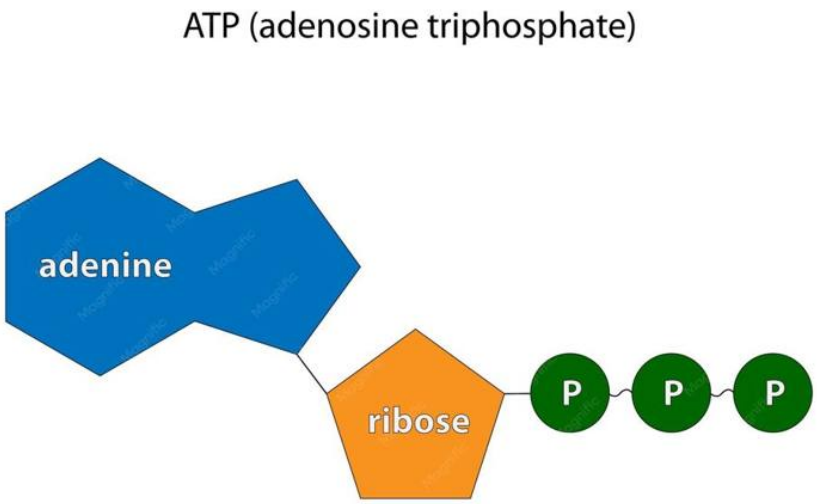
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Cellular work



ATP—The Universal Energy Currency

- ATP stands for **Adenosine Triphosphate**
- It consists of
  - Adenine
  - ribose sugar,
  - three phosphate groups
- Energy is stored in phosphate bonds.
- When ATP loses one phosphate,
- ATP → ADP + Pi + Energy
- This released energy powers virtually every cellular process.



Glycogen: Nature's Cache Memory

- Humans cannot store unlimited glucose in the bloodstream.
- The liver converts excess glucose into glycogen.
- When energy is required,

glycogen

↓

glucose

↓

ATP

- This resembles

CPU Cache

↓

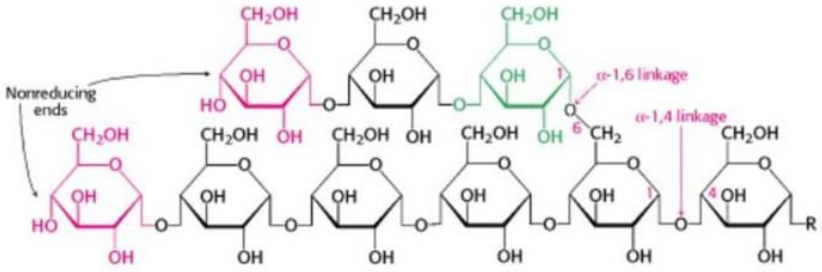
Immediate access

rather than

Hard Disk (Slow retrieval)

- After a meal, excess glucose is converted into glycogen and stored primarily in the liver and skeletal muscles. During fasting or prolonged exercise, glycogen is broken down back into glucose, thereby maintaining a steady supply of energy to vital organs.
- This sophisticated regulation prevents dangerous fluctuations in blood glucose levels and ensures that cells receive energy whenever required.

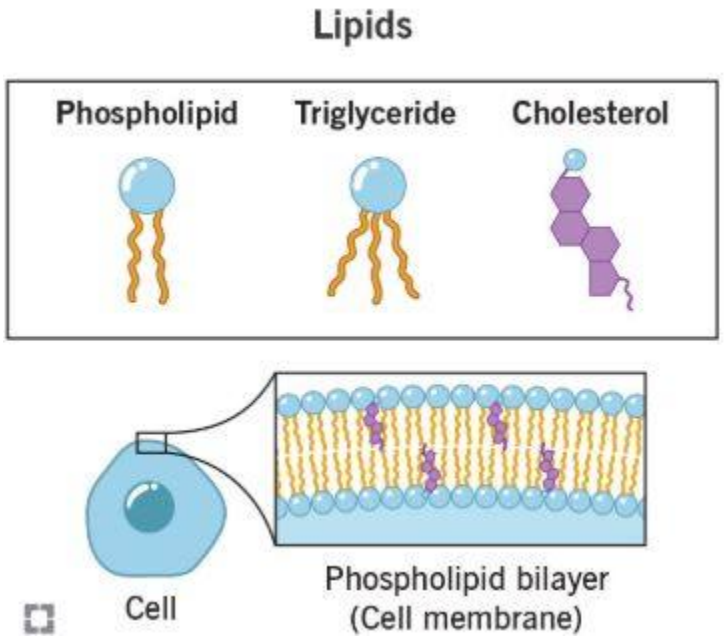
Glycogen Structure



Lipids

Definition

- Lipids are hydrophobic organic molecules composed mainly of
  - Carbon
  - Hydrogen
  - Oxygen
- Some lipids also contain
  - Phosphorus
  - Nitrogen
- They are insoluble in water but soluble in organic solvents.
- Soluble in organic solvents such as ether, chloroform, and benzene
- Lipids are broken down slowly and provide sustained energy over longer periods.
- This is why the human body stores excess dietary energy as fat rather than glucose.



Why Are Lipids Better Energy Stores?

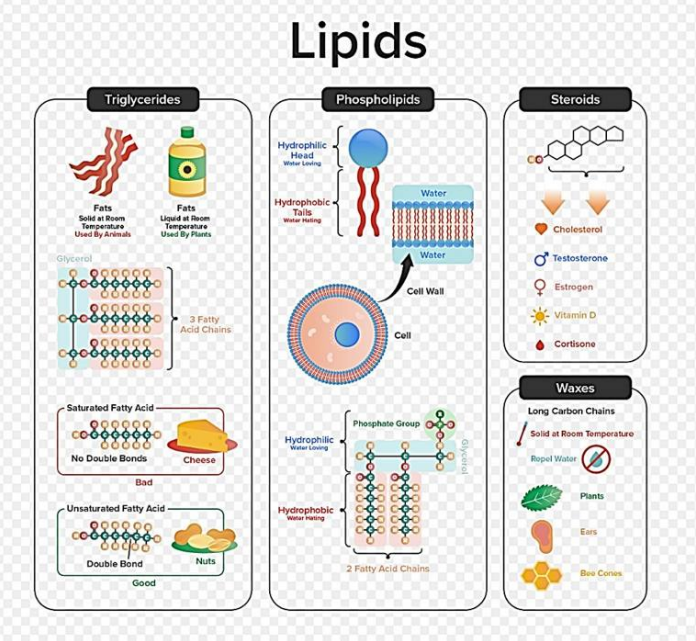
- Lipids contain more carbon-carbon bonds.
- Breaking these bonds releases more electrons, which ultimately generate more ATP.
- Thus, one gram of fat stores more than twice the energy of carbohydrates.

Molecule	Energy
Carbohydrates/ Proteins	4 kcal/g
Lipids	9 kcal/g

Characteristics of Lipids

- Although lipids include many different molecules, they share several common characteristics.
- They are insoluble in water but dissolve readily in non-polar organic solvents.
- They contain a high proportion of carbon and hydrogen atoms.
- They provide approximately **9 kcal of energy per gram**, compared to **4 kcal per gram** for carbohydrates.
- They serve both structural and functional roles in living organisms.
- They are stored mainly in adipose (fat) tissue in animals.

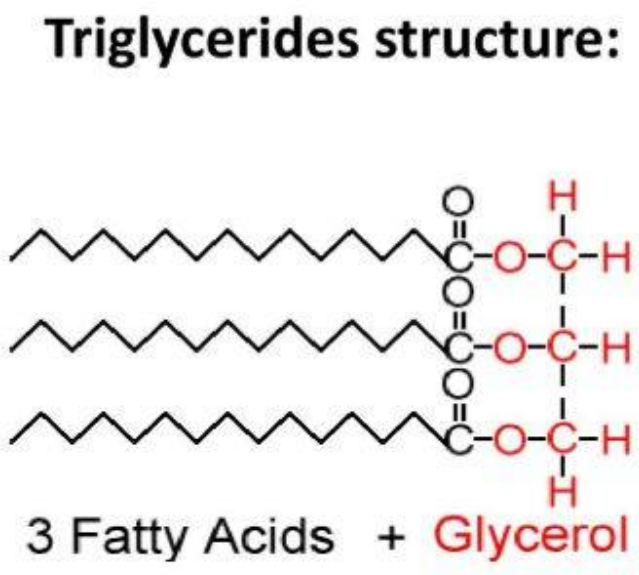
Figure



Types of Lipids- 4 types

1. Triglycerides

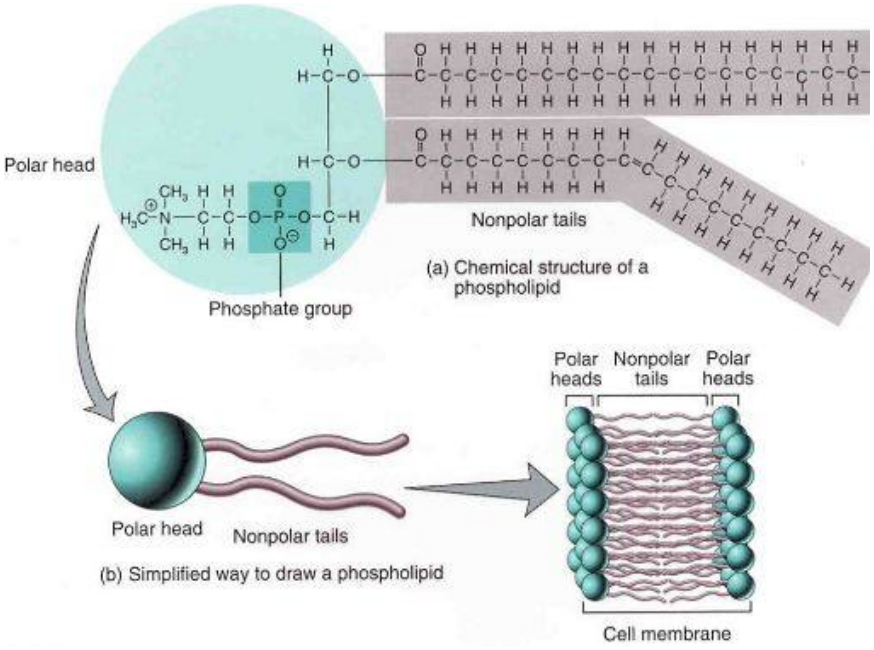
- Triglycerides are the most abundant lipids in the human body and represent the primary form of stored fat.
- They consist of **one glycerol molecule attached to three fatty acid molecules** through ester bonds.
- Whenever we consume more calories than our body requires, the excess glucose is converted into triglycerides and stored in adipose tissue. During fasting or prolonged exercise, these triglycerides are broken down to release fatty acids, which are then oxidized to produce ATP.
- **Functions of Triglycerides**
  - Store excess energy
  - Provide insulation against cold
  - Protect internal organs from mechanical injury
  - Supply energy during starvation and prolonged exercise



Types of Lipids- 4 types

2. Phospholipids

- Phospholipids are among the most important lipids because they form the **structural foundation of all cell membranes**. Unlike triglycerides, phospholipids contain:
  - One glycerol molecule
  - Two fatty acid chains
  - One phosphate-containing group
- This unique structure gives phospholipids two distinct regions:
- **Hydrophilic (water-loving) head**
- **Hydrophobic (water-fearing) tails**
- Because of this dual nature, phospholipids are described as **amphipathic molecules**.
- When placed in water, phospholipids spontaneously arrange themselves into a **bilayer**, with the hydrophilic heads facing outward toward water and the hydrophobic tails pointing



Types of Lipids- 4 types

3. Steroids

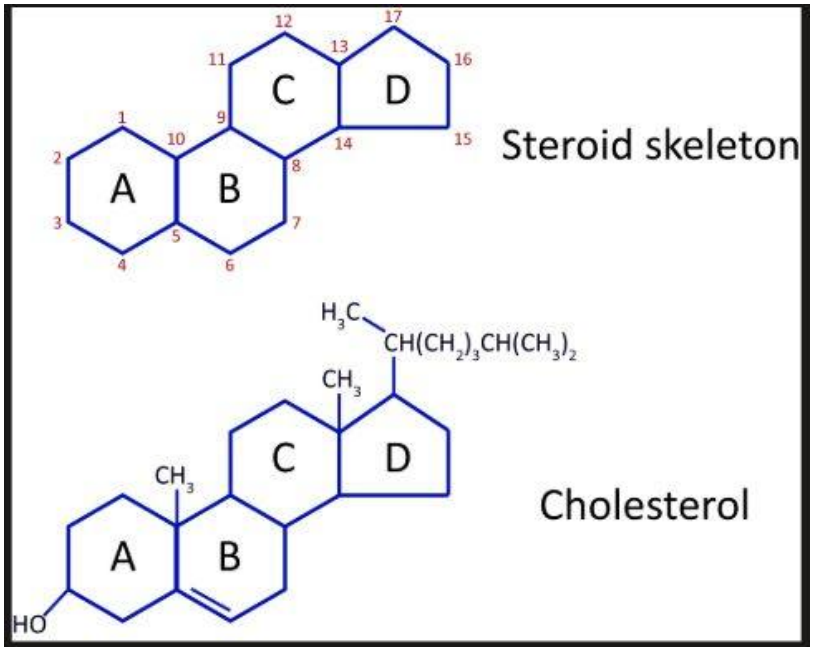
- Steroids differ structurally from triglycerides and phospholipids because they contain **four interconnected carbon rings** rather than long fatty acid chains.
- One of the best-known steroids is **cholesterol**, which is often misunderstood. Although excessive cholesterol levels are associated with cardiovascular disease, cholesterol is also an essential molecule that performs several important biological functions.

Biological Functions of Cholesterol

- Maintains cell membrane stability
- Serves as a precursor for steroid hormones
- Required for the synthesis of vitamin D
- Essential for the production of bile salts that aid fat digestion

Examples of steroid hormones include:

- Testosterone
- Estrogen
- Cortisol
- Progesterone



Types of Lipids- 4 types

4. Waxes

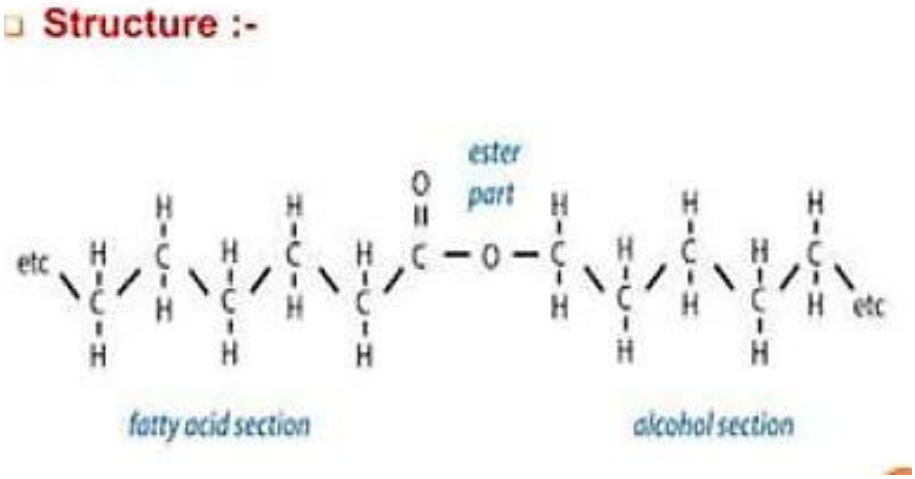
- Waxes are complex lipids formed by the combination of long-chain fatty acids and long-chain alcohols. They primarily serve protective functions rather than energy storage.

Examples

- Ear wax in humans
- Wax coating on plant leaves
- Beeswax produced by honeybees

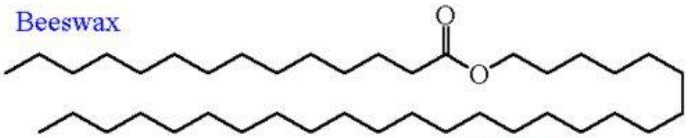
Functions

- Prevent water loss
- Protect against microbial infection
- Reduce mechanical damage
- Waterproof surfaces
- Plants growing in hot environments possess a thick waxy coating that minimizes water loss through evaporation.



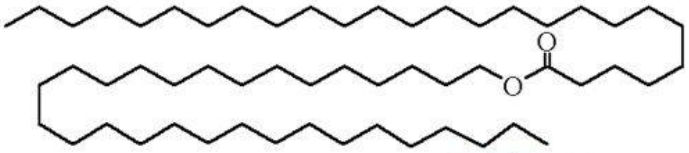
Waxes

Beeswax



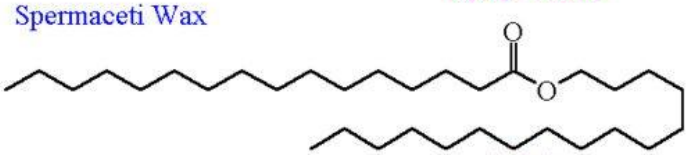
Ceryl Myristate

Carnauba Wax



Myricyl Cerotate

Spermaceti Wax



Cetyl Palmitate

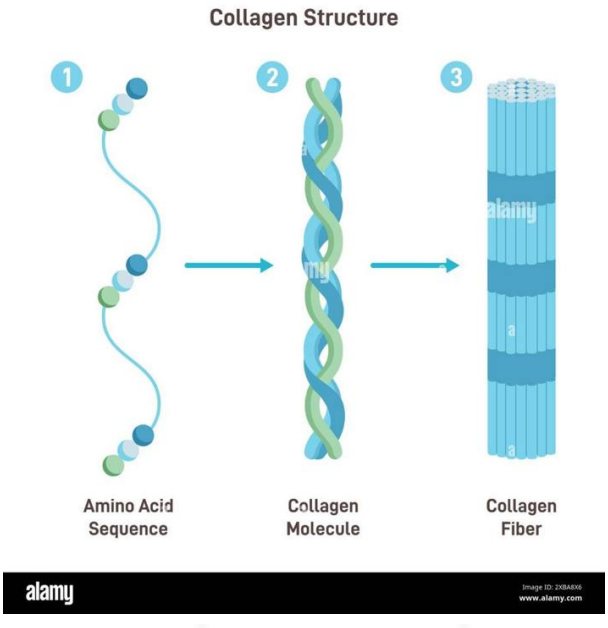
S2 Proteins/nucleic acids as functional data structures

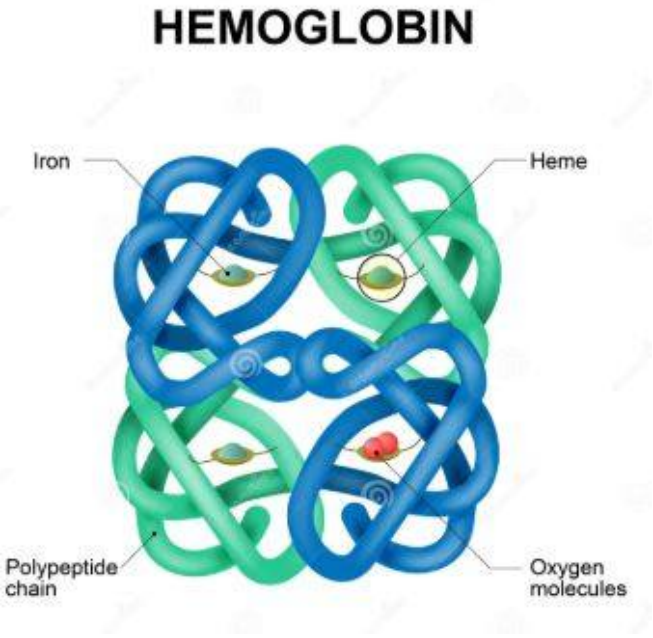
2 MARKS Proteins are the most versatile biomolecules found in living organisms.
4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Proteins – The Functional Machines of the Cell; Characteristics of Proteins; Functions of Proteins; 2. Structural Support; 4. Communication; 5. Defense.
12 TO 13 MARKS: ANSWER IN THIS ORDER 1. Proteins – The Functional Machines of the Cell 2. Characteristics of Proteins 3. Functions of Proteins 4. 2. Structural Support 5. 4. Communication 6. 5. Defense 7. Nucleic Acids – The Information Storage Molecules 8. DNA – The Biological Database 9. Each strand contains four nitrogenous bases: 10. The Central Dogma of Molecular Biology 11. RNA – The Information Messenger 12. Functions of Nucleic Acids 13. Nucleic Acids as Functional Data Structures *Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.*
PRACTICE QUESTIONS 2 marks: State the central dogma. 5 marks: Write short notes on DNA structure and the types of RNA. 12 marks: Explain proteins and nucleic acids as functional data structures, with structure, functions and engineering analogies.

Complete notes

Proteins – The Functional Machines of the Cell

- Proteins are the most versatile biomolecules found in living organisms. Nearly every biological activity depends on proteins.
- They act as enzymes, hormones, transport molecules, structural components, antibodies, and molecular motors.
- Every protein is built from smaller molecules called **amino acids**.
- Twenty different amino acids combine in various sequences to produce millions of unique proteins,
- The sequence of amino acids determines the three-dimensional structure of the protein, and its structure ultimately determines its function.
- For example**, hemoglobin** transports oxygen because of its unique three-dimensional structure, whereas **collagen** provides strength to skin and bones because its structure is long, fibrous, and highly resistant to stretching.



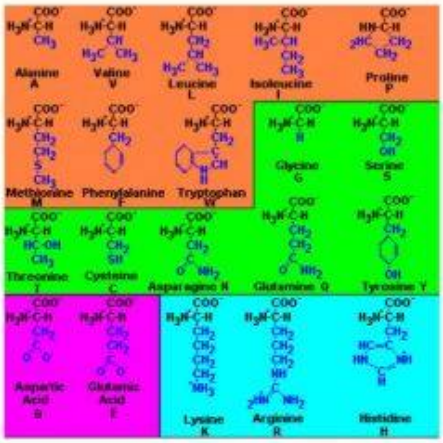


Characteristics of Proteins

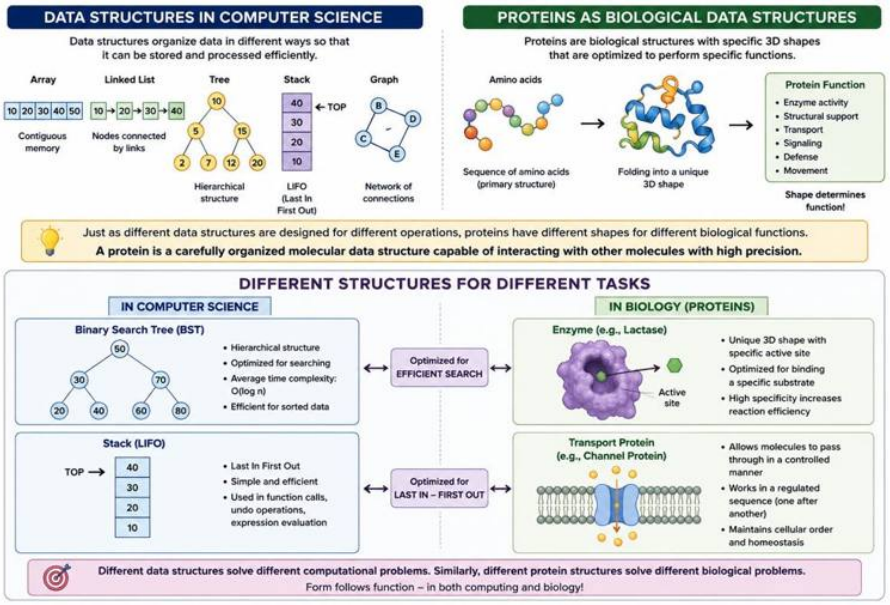
- They are polymers composed of amino acids.
- They contain carbon, hydrogen, oxygen, nitrogen, and sometimes sulfur.
- Their structure determines their biological function.
- They perform most of the work inside the cell.
- They are continuously synthesized and degraded according to the cell's needs.

Protein characteristics

- Unbranched polymer
- Folds into an accurate three dimensional structure (globular structure)
- Correct folding is essential for the protein to exert its functions- tight **structure-function relationship**



Figure



Functions of Proteins

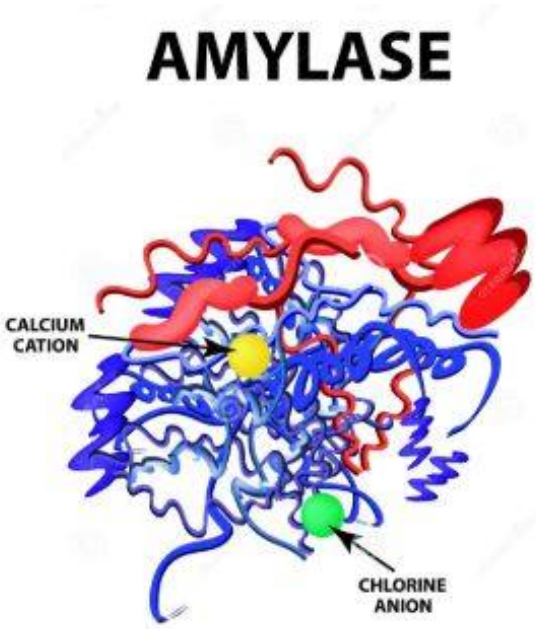
- Proteins perform an astonishing variety of functions within living organisms.

1. Catalysis

- Many proteins function as enzymes that accelerate biochemical reactions.

Examples include:

- Amylase – digests starch
- Pepsin – digests proteins
- DNA polymerase – synthesizes DNA
- Without enzymes, most biochemical reactions would occur too slowly to support life.

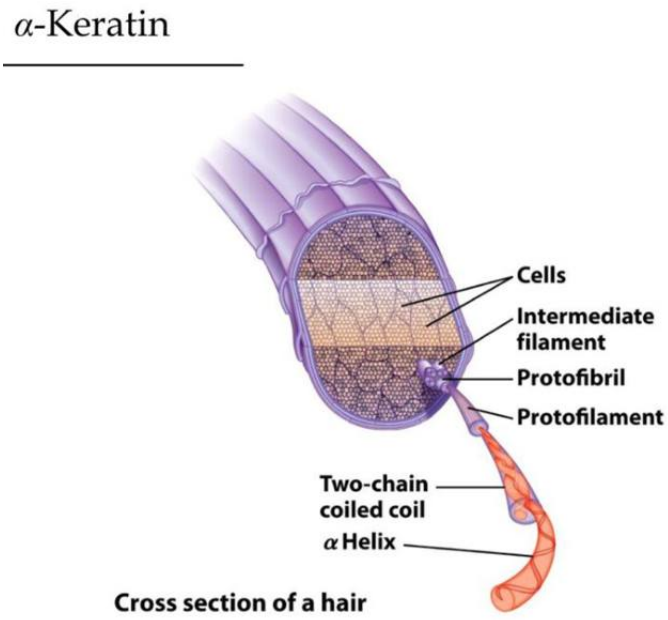


2. Structural Support

- Certain proteins provide mechanical strength to cells and tissues.

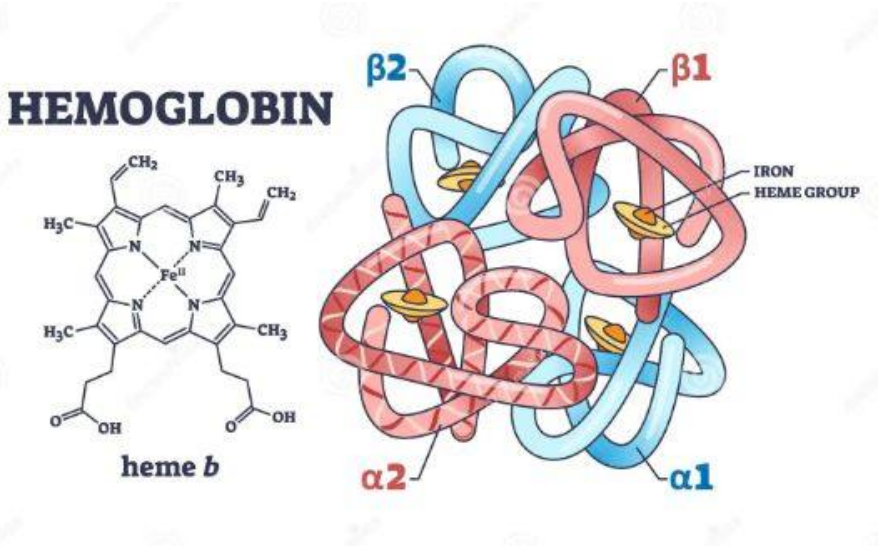
Examples:

- Collagen – connective tissue
- Keratin – hair and nails
- Elastin – skin elasticity



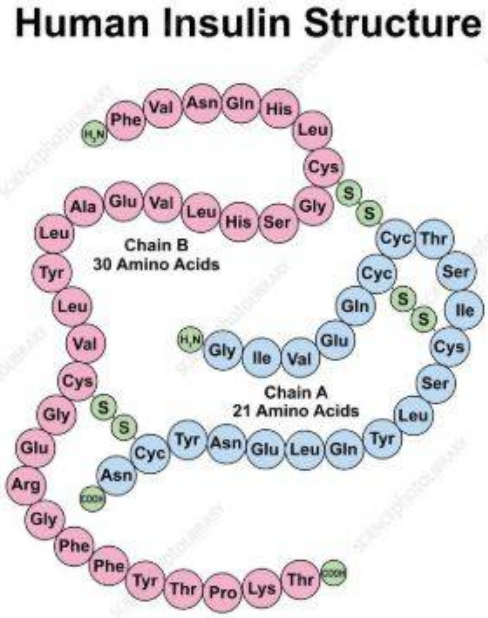
Figure

- **3. Transport**
- Some proteins transport molecules throughout the body.
- Example:
- Hemoglobin carries oxygen from the lungs to body tissues.



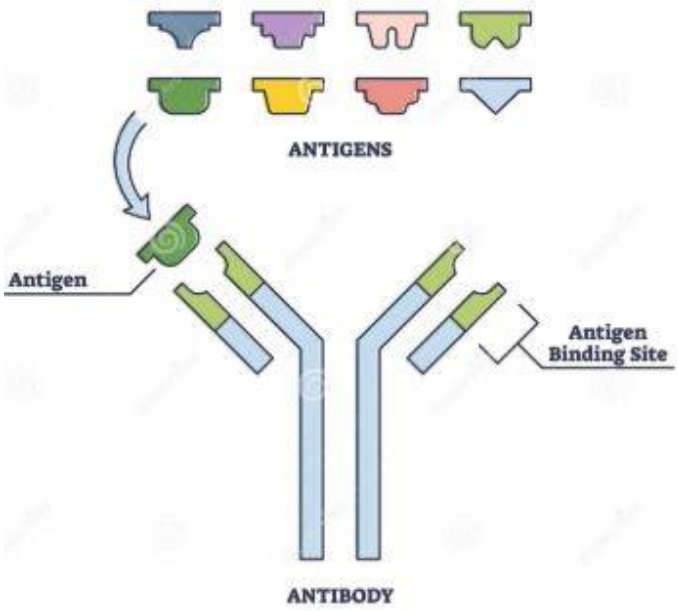
4. Communication

- Protein hormones coordinate activities between organs.
- Examples:
- Insulin
- Glucagon
- Growth hormone



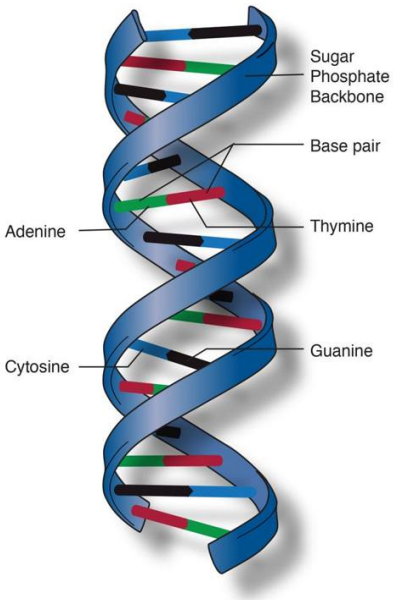
5. Defense

- Antibodies recognize and neutralize harmful microorganisms.
- Without antibodies, the immune system would be unable to protect the body against infections.



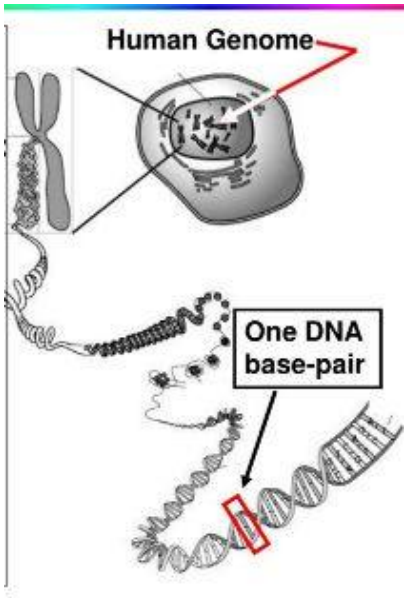
Nucleic Acids – The Information Storage Molecules

- While proteins perform cellular work, **nucleic acids store and transmit biological information**.
- Every living organism contains two major nucleic acids:
  - **DNA (Deoxyribonucleic Acid)**
  - **RNA (Ribonucleic Acid)**
- These molecules ensure that biological information is accurately preserved and transmitted from one generation to the next.
- Without nucleic acids, cells would have no instructions for producing proteins.
- DNA (Deoxyribonucleic Acid) is the primary hereditary material in almost all living organisms. It carries the complete set of genetic instructions required for growth, development, reproduction, and maintenance of life.
- The DNA molecule consists of **two long polynucleotide chains** twisted around each other to form the famous **double helix**



DNA – The Biological Database

- DNA is often described as the blueprint of life because it contains the complete genetic information required to build and maintain an organism.
- A single human cell contains approximately **3.2 billion base pairs** of DNA, organized into 23 pairs of chromosomes.
- This enormous amount of information is compressed into a nucleus only a few micrometers in diameter.
- The DNA molecule consists of two complementary strands twisted into a **double helix**.
- DNA molecules are extremely large, and they are constructed from small repeating units known as nucleotides.



Each strand contains four nitrogenous bases:

- Adenine (A)
- Thymine (T)
- Guanine (G)
- Cytosine (C)

Every nucleotide consists of three components:

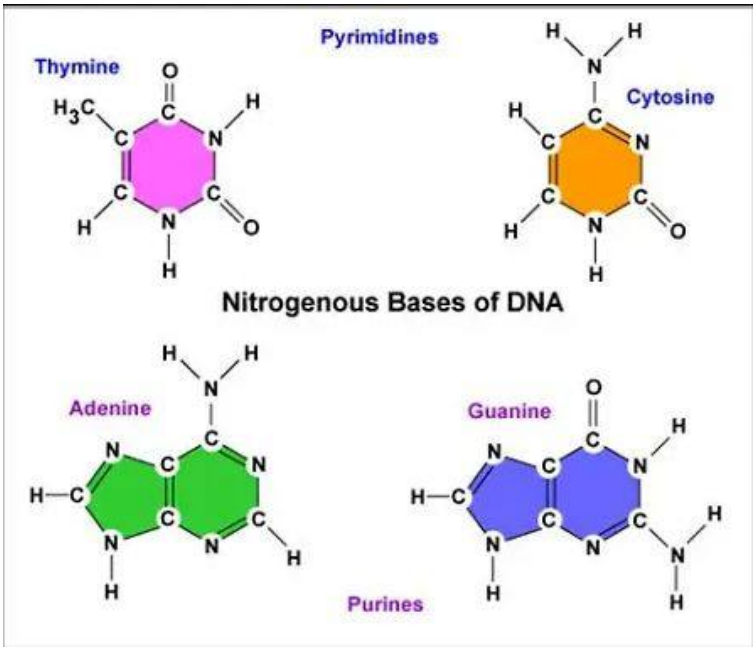
A nitrogenous base

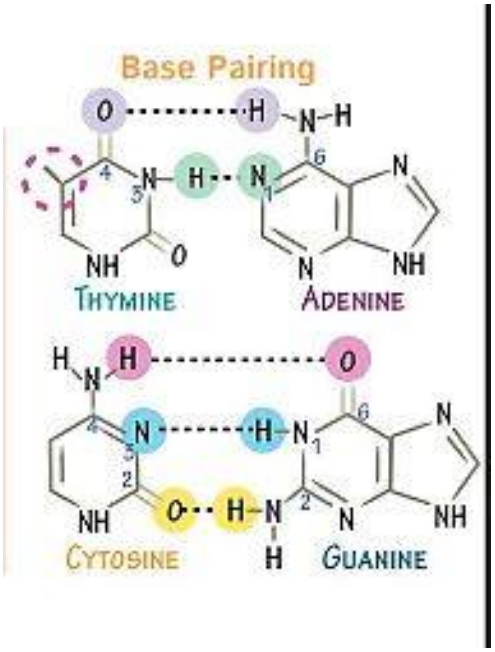
A five-carbon sugar

A phosphate group

These bases follow strict pairing rules:

- A pairs with T
- G pairs with C
- This complementary pairing ensures accurate replication of genetic information.





The Central Dogma of Molecular Biology

- The flow of genetic information follows a well-defined pathway known as the **Central Dogma**.

DNA

↓ Transcription

RNA

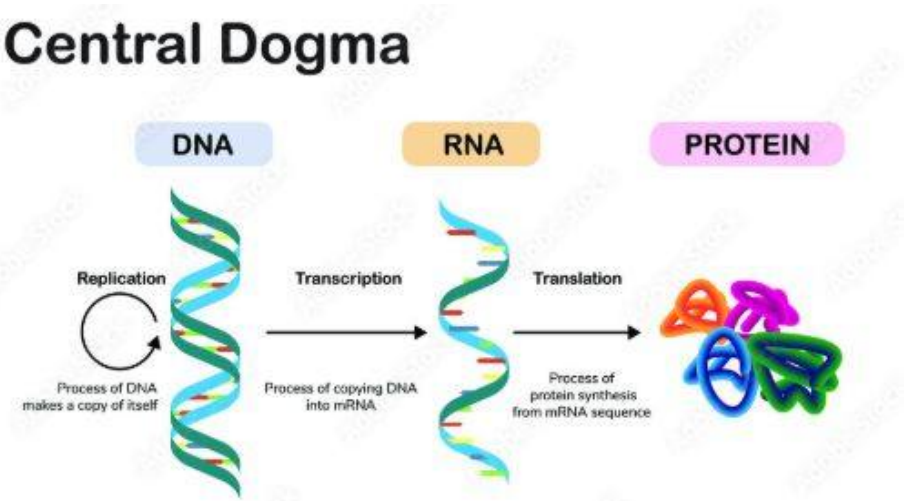
↓ Translation

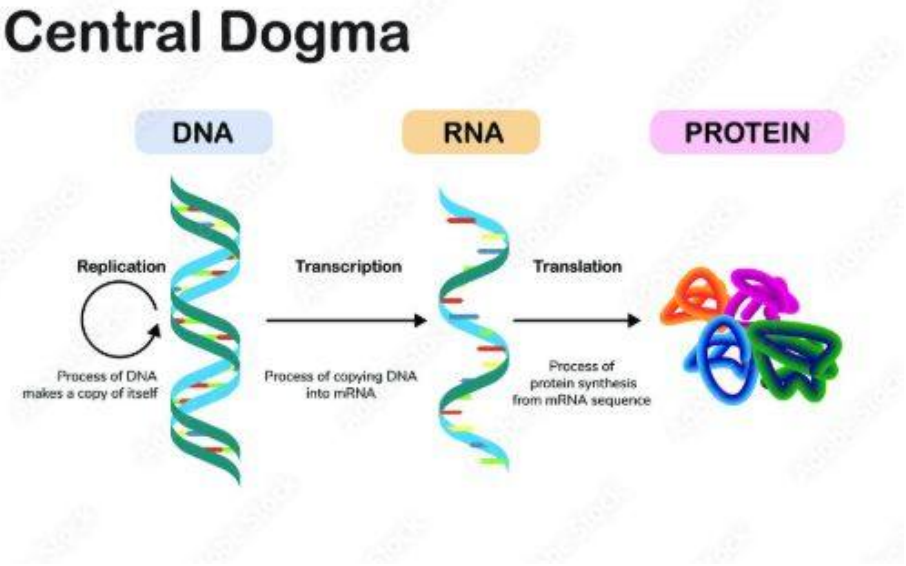
Protein

- This sequence explains how stored genetic information is converted into functional molecules.

From a computing perspective:

- **DNA** is the source code.
- **Transcription** copies the required code.
- **RNA** acts as the temporary instruction file.
- **Translation** converts the instructions into executable form

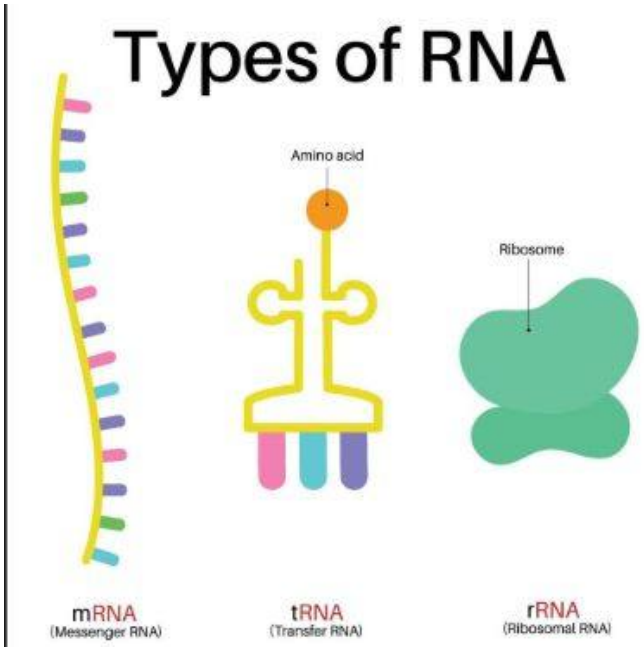




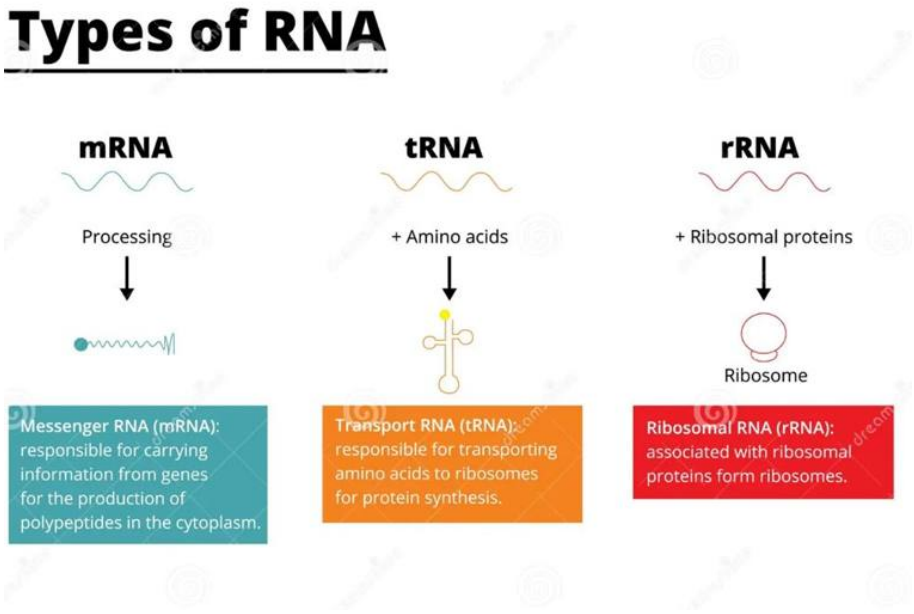
RNA – The Information Messenger

- Unlike DNA, RNA is usually single-stranded and functions as the intermediary between DNA and proteins.
- RNA carries genetic instructions from the nucleus to ribosomes, where proteins are synthesized.
- There are three major types of RNA: mRNA, rRNA, tRNA
- RNA is temporary.
- Once its job is complete, it is degraded,

allowing cells to rapidly adjust protein production according to changing conditions.



Figure



Functions of Nucleic Acids

- Nucleic acids perform several essential functions that ensure the continuity of life.
- They store hereditary information.
- They transmit genetic information from one generation to the next.
- They direct the synthesis of proteins.
- They regulate cellular activities by controlling gene expression.
- They contribute to growth, development, and reproduction.
- They enable organisms to adapt and evolve through genetic variation.

Nucleic Acids as Functional Data Structures

- A data structure must satisfy three fundamental requirements:
- It must store information efficiently.
- It must allow rapid retrieval of information.
- It must maintain data integrity during storage and transmission.

- DNA fulfills all three requirements exceptionally well.
- The sequence of nitrogenous bases—adenine (A), thymine (T), guanine (G), and cytosine (C)—forms a biological coding language capable of storing billions of pieces of information. These bases are arranged in a precise sequence, just as characters are arranged in a computer program. Even a small change in this sequence can alter the biological instructions, similar to how changing a single line of code may modify the behavior of a software application.
- Unlike electronic storage devices, DNA also possesses the remarkable ability to **self-replicate**, allowing identical copies of genetic information to be produced during cell division with extremely high accuracy

S3 Enzymes as biological catalysts

2 MARKS Enzymes are biological molecules, primarily proteins, that catalyze biochemical reactions occurring within living organisms.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Characteristics of Enzymes; Why are Enzymes Called Biological Catalysts?; Structure of an Enzyme; Mechanism of Enzyme Action; Enzyme Classes; Applications of Enzymes.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. Characteristics of Enzymes
2. Why are Enzymes Called Biological Catalysts?
3. Structure of an Enzyme
4. Mechanism of Enzyme Action
5. Enzyme Classes
6. Applications of Enzymes

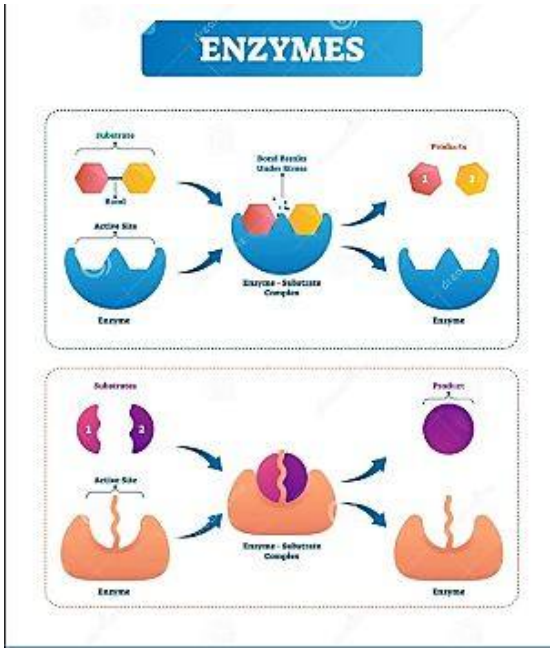
Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: Define an enzyme.
5 marks: Explain the mechanism of enzyme action.
12 marks: Describe enzyme structure, characteristics, mechanism of action, classes and applications.

Complete notes

Figure

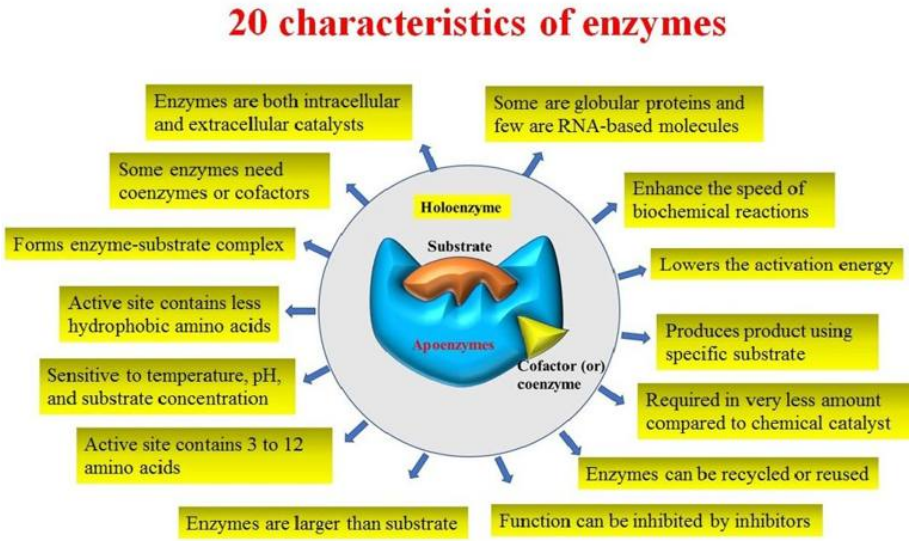
- Enzymes are biological molecules, primarily proteins, that catalyze biochemical reactions occurring within living organisms.
- Almost every metabolic process in the body depends on enzymes.
- Digestion, respiration, DNA replication, muscle contraction, blood clotting, and nerve impulse transmission all require specific enzymes.
- Without enzymes, life as we know it would not exist.



Characteristics of Enzymes

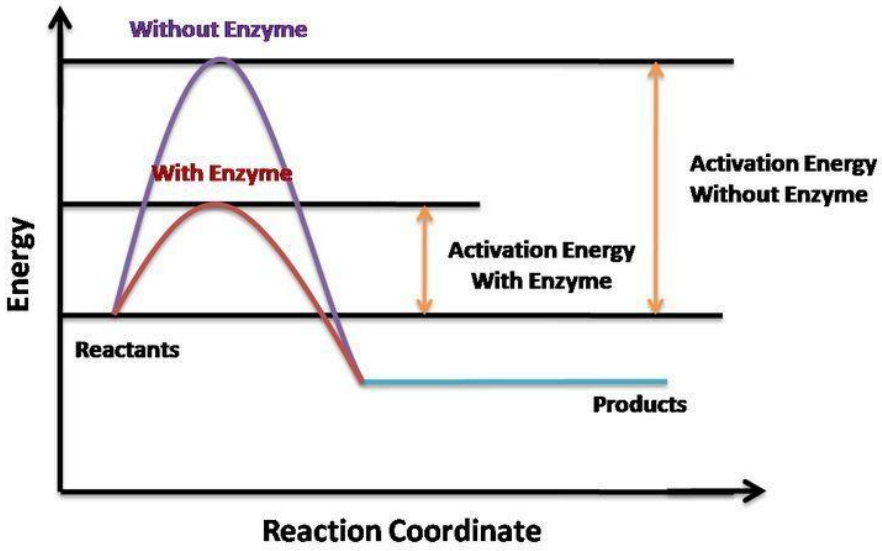
- Most enzymes are proteins, although a few RNA molecules (ribozymes) also exhibit catalytic activity.

- They are highly specific, with each enzyme acting on a particular substrate.
- They remain chemically unchanged after the reaction and can be reused.
- They function efficiently under mild conditions of temperature and pH.
- Their activity can be regulated according to the needs of the cell.



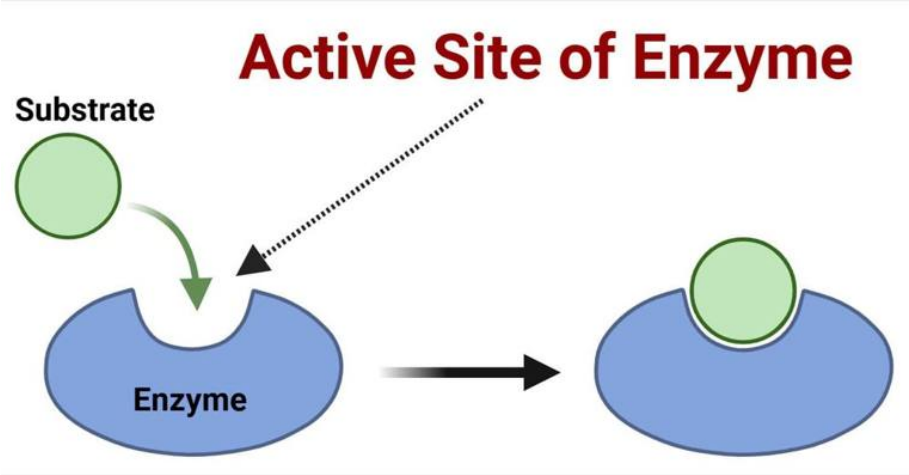
Why are Enzymes Called Biological Catalysts?

- A catalyst is a substance that increases the rate of a chemical reaction without being consumed during the process.
- Chemical reactions require a certain amount of energy to begin. This initial energy barrier is known as the activation energy.
- Enzymes reduce this activation energy, making it easier for reactant molecules to be converted into products.

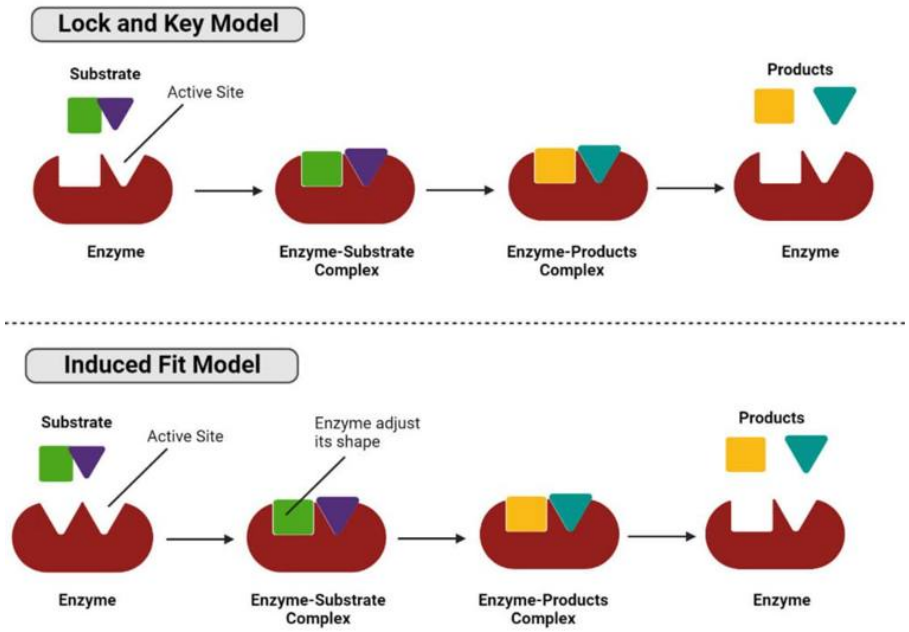


Structure of an Enzyme

- Enzymes possess a highly specific three-dimensional structure. This structure enables them to recognize and bind only to particular molecules.
- Every enzyme contains a region known as the active site. The active site is a small pocket or groove where the substrate binds and undergoes a chemical transformation.
- Because the active site's shape is highly specific, only molecules with the correct structure can bind successfully.



Figure



Mechanism of Enzyme Action

- **Lock-and-Key Model**
- According to this model, the enzyme's active site has a rigid shape that perfectly matches the substrate, just as a specific key fits only one lock.
- Only the correct substrate can bind to the enzyme and undergo the reaction.
- Although this model explains enzyme specificity, it does not account for the flexibility observed in many enzymes.

2. Induced-Fit Model

- According to this model, the enzyme is flexible rather than rigid. As the substrate approaches, the active site undergoes slight structural changes to fit the substrate more closely.
- This conformational adjustment improves binding efficiency and increases the reaction rate.
- The induced-fit model is widely accepted because it explains the dynamic nature of enzyme-substrate interactions.

Enzyme Classes

Enzyme Class	Function	Example
Oxidoreductases	Oxidation–reduction reactions	Catalase

Enzyme Class	Function	Example
Transferases	Transfer functional groups	Kinases
Hydrolases	Hydrolysis reactions	Amylase
Lyases	Addition or removal of groups	Decarboxylase
Isomerases	Rearrangement of molecules	Phosphoglucose isomerase
Ligases	Join molecules together	DNA ligase

Applications of Enzymes

Medicine

- Diagnostic tests (e.g., blood glucose estimation)
- Drug development
- Treatment of enzyme deficiency disorders

Food Industry

- Cheese production
- Bread making
- Fruit juice clarification
- Brewing

Environmental Applications

- Wastewater treatment
- Bioremediation
- Biodegradation of pollutants

More notes

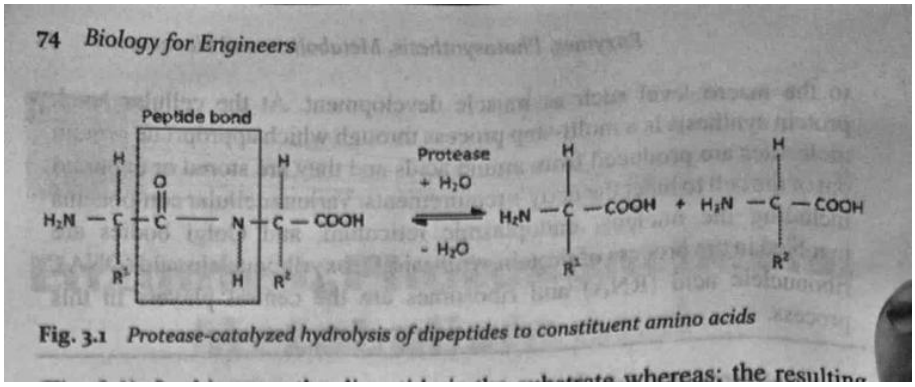
3.1 Enzymes as biological catalysts

Consider a biochemical reaction under physiological temperature, pH, and aqueous conditions, which left to itself is likely to take a relatively long duration, say hours or even days to reach equilibrium. Such a reaction might not be useful to a microorganism that needs to conduct metabolism fast for survival and reproduction. Also, considering the diversity of biochemicals and pathways involved in organisms, many wasteful side reactions are possible making cellular metabolism an energy inefficient process. Nature has evolved Biological catalysts termed as “enzymes” to_regulate_the biochemical reactions of cells and organisms. As the definition of catalyst suggests, enzymes facilitate biochemical reactions, but are not consumed in the reaction and emerge unmodified, thereby allowing them to be recycled for the next round of reaction. In these biochemical reactions, the reactants on which these enzymes act are termed as “substrates”. The enzyme facilitation involves regulating (increasing) the rate or velocity of the biochemical

} reactions as well as channelizing the substrates through useful pathways and preventing unwanted side reactions. This helps the cells and organisms to prioritize their needs for survival and reproduction as well as conserve energy and precious biochemicals.

Most enzymes are proteins, with the exception of non-proteinaceous enzymes such as ribozymes. Proteins comprising of amino acid residues linked through peptide bonds have complex tertiary and quaternary structures and this provides them with the structural features necessary to act as a catalyst. They work in aqueous solutions under mild pH and temperature conditions and are capable of increasing the reaction rates when compared to uncatalyzed reactions. They work in tandem with each other to catalyze and tegulate key biochemical processes.

A classic example of enzymes are proteases, which catalyze the reverse of peptide synthesis. An example is shown below which depicts the reversible hydrolysis of peptide bond between two amino acid residues of a dipeptide



(Fig. 3.1). In this case, the dipeptide is the substrate whereas; the resulting free amino acids are the products. The equilibrium of the reaction exists far to the right, with the products dominating the reaction mixture.

Naming and classification of enzymes: Most enzymes as ee after their substrate they act on and have their names ending with “ase’ For e.g. urease catalyzes the hydrolysis of urea. Some enzymes have their names based on either their broad functionality or the source of the enzyme. For ais ft proteases, papain and bromelain are named after their source, papaya (Carica t papaya) and pineapple (bromeliaceae family) respectively.

To prevent any ambiguities, the Enzyme Commission of the TUBMB (International Union of Biochemistry and Molecular Biology) has adopted a classification system. In this system, each enzyme is provided with a systematic name and divided into 6 classes: oxidoreductases, transferases, hydrolases, lyases, isomerases and ligases. They are further sub-classified and provided with a unique 4 number Enzyme Commission number (E.C. number) that indicates the class, sub-class efc. that go onto uniquely identify each enzyme in the database. For e.g., the E.C. number for papain is “3.4.22.2” with first number, “3” denoting “hydrolyases” class and the second number, “4” sub-classifying it as a hydrolyases that specifically acts on peptide bonds (peptidases). The third number, “22” denotes a peptidases that act with a cysteine endopeptidase mechanism and the last number, “2” denotes the specific cysteine endopeptidase, papain. Enzymes are located in cell cytoplasm, mitochondria, tissues and body fluids. Enzymes can be also classified in many ways: intra and extracellular, soluble and insoluble etc.

3.2 Significance of enzymes

(i) Catalytic power (reaction rate): An important question in chemical reactions is: What is the driving force for a reaction to proceed from reactants to products? From a chemical thermodynamics point of view, as long as the free energy of products is lower than that of the reactants, i.e., the change in rr ae Enzymes, Photosynthesis, Metabolism and Bioenergetics 75 eo free eneTBY: AG (= Gyre ~ Grexcans) i8 negative, the reaction is feasible. AG | isa thermodynamic concept that defines the usable energy available froma = chemical process and has the units of joules/mole. However, thermodynamics i! goes not take into account the transition states the reactants have to pass | ¢nrough before they form the products, The above equation shows that substrate, S in the presence of enzyme, EB has to £0 thorough an activated high energy unstable transition state, Pes before it forms the products.

A better example is the splitting of disaccharide lactose (substrate) into its constituent monosaccharides, glucose and galactose (products) by the enzyme lactase (Fig. 3.2a). The reaction coordinate is a measure of the progress of the reaction from the reactants to the product through various transition states. Figure 3.2a shows that the conversion of reactant to products proceeds with the formation of at least one high energy transition state which imposes an “energy barrier” for the reactants. To surmount this barrier, the reactants have to acquire a critical energy, called “activation energy” and form an energy rich transition state before forming products. Activation energy (joules per mole) can be defined as the minimum energy required for initiating a chemical reaction.

Fig. 3.2. A schematic representation of the free energy vs. reaction coordinate plot for hydro-

lysis of lactose to glucose and galactose in the absence (a) and presence (b) of the enzyme, lactase. The plots show that the enzymatic-catalysis lowers the activation energy and facilitates the formation of the transition complex needed to drive the reaction to product formation.

At any given instant, only a fraction of the reactant molecules in a chemical reaction are likely to have sufficient activation energy to surmount the barrier. Hence, the reaction rate would be relatively slow. In the case of

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Enzymes, Photosynthesis, Metabolism and Bioenergetics 77 Active ite sieeve | | Active cysteine | | t-domain —/ GOP. f \ Jay f-domain oe Paw-=

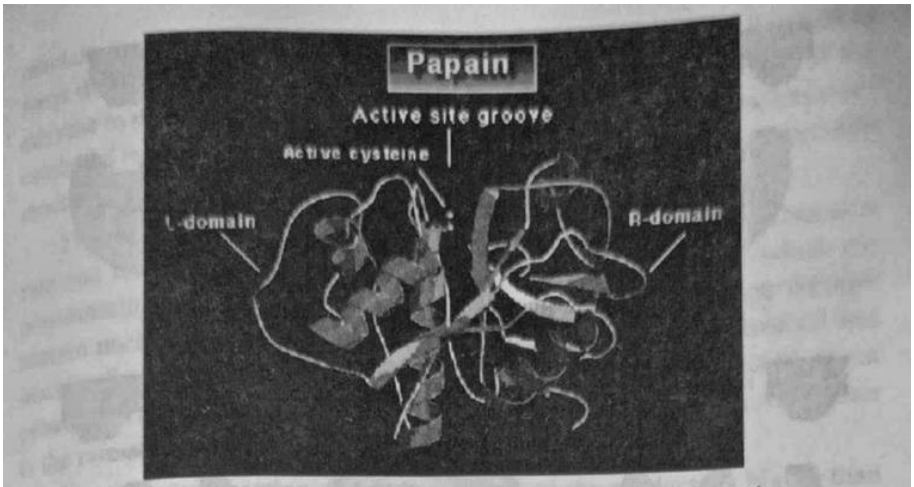


Fig. 3.3 *Structure of protease, papain with catalytic cysteine residue in the active site groove (reproduced with permission*

from Prof. Thierry Moreau, Inserm U618, France) regio-specificity also. This specificity also prevents unnecessary sidereactions, formation of by-products and waste of energy. Hence, a combination of steric complementarity, covalent and non-covalent interactions serve to provide aconcerted effect that optimizes the orientation of specific substrates and lowers the activation energy to produce high reaction rates. The above discussion suggests that a key requirement for enzyme catalysis is ability of the substrate molecule to complement the shape of the cleft on the active site. This led to the postulation of the “lock-and-key” hypothesis in 1894 (figure 3.4A). According to this hypothesis, enzyme specificity arises most probably by the ability of the enzyme to accommodate the substrate in the same way that a lock accommodates the key specific for it, This hypothesis explained the specificity part, but did not explain much about how this aids the formation of the intermediate transition state complex. It was until 1958 that the “induced-fit model” was used to explain this phenomenon (Fig. 3.4B). This model postulated that the merely fitting the right rigid key in the lock by itself was not sufficient and that the substrate needs to induce a structural reconfiguration of itself as well as the enzyme so as to form an enzyme-substrate complex. This enzyme-substrate complex would distort or orient the substrate to closely approximate the structure of the transition state complex The specificity of enzymes varies. Some enzymes have absolute specificity and can catalyze only one substrate such as protease Lys-C which cleaves peptides at the carboxy terminal of lysine residues. Some enzymes

Enzymes, Pho; | synthesis, Metabolism and Bioenergetics 79 | ts can either move back to f eo jucts. The prod orm reactants again or move forward to —] fo the products. Products can also utilize the catalyti ar ot the ! to surmount the activation barrier lytic pow i a7) ed reaction rate enhancement i and form reactants. Thus, enzyme- | cotaly ent is valid for both directions in a reversible it ts equilibri Portant concept of difference between reaction i || ae and reat a ee rium. Equilibrium is the steady state at which the | | concentrations rae eis and Product species in a reaction mixture 't | email unchanged with time. This occurs when the rate of the forward and \ | packward saat a eet An uncatalyzed reaction left to itself, when | provided with enough time will eventually attain equilibrium. Reaction rate } isthe rate of (dis)appearance of a species in a reaction mixture. ical reaction, i Pan Ie chem ‘ n, if the free energy of the products is higher than ipat of the reactants, the

net change in free energy, AG is positive making it thermodynamically unfavorable (Fig. 3.5). Enzymes cannot make such reactions favorable. On the contrary, if the AG is negative for a reaction, enzymes cannot convert this thermodynamically favourable reaction to an unfavorable one. Enzymes cannot alter the thermodynamically dictated feasibility and the final equilibrium established in biochemical reactions. Enzymes can lower the activation energy and enhance the reaction kinetics, ie reaction rate at which the equilibrium is attained. Reaction Kinetics Vs. Thermodynamics

- / | c, \, Reaction Kinetics = Activation Energy (E,) = How Fast ? Thermodynamics = Net Change in Free Energy (AG) = Spontaneous ? Reaction Coordinate

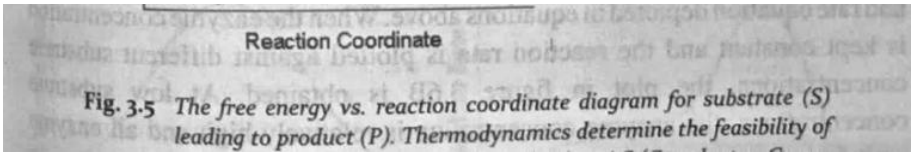


Fig. 3.5 *The free energy vs. reaction coordinate diagram for substrate (S) leading to product (P). Thermodynamics determine the feasibility of*

areaction. Reactions are feasible only when AG (Gproducts - Greactants) is negative. In the absence of an activation barrier, this reaction will be spontaneous. Catalysts (enzymes) lower the activation barrier and determine the reaction rate.

4 zymes, Photosynthesis, Metabolism and Bioenergetics 81 ay Pee" TORIES eer pas of Eneyme Kinetics Enzyme Concentration, [E] SRR Wersrey seme ne } f ‘Substrate Concentration, [S] Hl fig. 3° Effect of Enzyme (A) and Substrate (B) concentrations of reaction rate 1 | a cernin point, increase in substrate con concentration leads to saturation of the i . ayuihe enzyme active sites with no further increase in reaction rate reat (iii) Effect of temperature on enzyme activity: When the temperature ipreases inasystem, it Tead s to the increase in the kinetic energy and internal sjergy of the molecules (Fig, 3.7), In an enzyme-catalyzed system, this leads we more energetic collisions as well as increased frequency of collisions petween Substrates and enzyme molecules. The energy from such collisions cairin turn be converted to chemical potential energy and used to offset the activation energy needed to attain the transition states. Hence a moderate incTeaSe in temperature Teads to increase in ‘the Teactionc ratestaeeel ream figure 3.7A until a maximum is reached. ha This increase in temperatures also leads to increase in the internal energy of the molecules. At higher temperatures, the high internal energy can be converted to chemical potential energy and used to break the fragile noncovalent bonds in the tertiary and quaternary structure of enzymes. This would destabilize the enzyme structure leading to thermal “denaturation” i.e. loss of the enzymatic activity.

(iv) Effect of pH on enzyme activity: pH of a solution is a measure of the hydrogen ion concentration, [H*] in a system. This affects the ionization Temperature, °C pH r Rig. 3-7 Effect of Temperature (A) and pH (B) on reaction rate f state of charged residues in the amino acid residues of proteins. The aming _ functional groups of basic amino acids (lysine, arginine, and histidine) ang the carboxylic acids of acidic amino acids (aspartic acid and glutamic acid) can be protonated and deprotonated depending on the pH of the solution. Fo, e.g., at pH 7, the carboxylic acid side chains of both aspartic acid and glutami, acid are deprotonated, whereas at pH 4, aspartic acid is deprotonated ang glutamic acid is protonated.

This change in ionization status leads to alterations in the ability of the protein to form ionic bonds necessary for its structural integrity. This leag, to altered specificity, poor binding with its natural substrate or eye, complete denaturation of the protein. The pH can also similarly affect the shape and/or ionization status of the substrate leading to its poor binding with the enzyme. Each set of enzyme and substrates have a unique py optimum and moving away from this pH values results in loss of activity as comm ngaeo TS: ie aye

- (v) Effect of cofactors: Cofactors are non-proteinaceous compounds Le that are"needed by enzymes for their activity. Cofactors could be either ions (inorganic) or small organic molecules. Ions used as cofactors are usually heavy metals Mg*, Fe, Mn, Zn* etc. Small organic molecules used as cofactors are called coenzymes. Most coenzymes are derived from vitamins, Examples of coenzymes are thiamine pyrophosphate (TPP), flavin adenine | dinucleotide (FAD), nicotinamide adenine dinucleotide (NAD), pyridoxyl phosphate, coenzyme A (CoA) etc. Coenzymes_ actively participate in the catalysis and play a role in the active site of enzymes. They are used as carrierSof electrons or functional groups during catalysis (Fig. 3.8). ee reactive Fj Btls, Samiar 'ig.

3.8 Coenzyme thiamine pyrophosphate ME is ; bound to an enzyme surface. Copyfi * Peas a ie right 1994 from Molecular Biology of Oho', © the Cell, 3rd edition, Bruce Alberts |

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si Enzymes, Pho 4 synthesis, Metabolism and Bioenergetics 83 ect of inhibito : id * di oS on enzyme activity: Inhibitors are molecules or hat hinder or terminate the catalytie-q ig. 3.9 jon jon can be classified into two toe mney SE eee (Fig. 3.). orm ple inhibiti types: irreversible and reversible. ible nbition can be fy patel —— overs ssive, and. non-competitive | tther subdivided into competitive, aco , mpetitive inhibitors, Competitive inhibitor Noncompetitive inhibitor

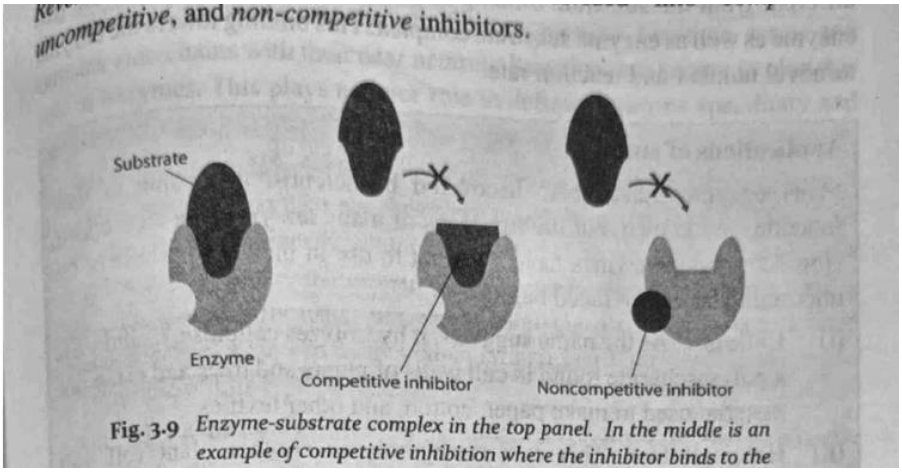


Fig. 3.9 *Enzyme-substrate complex in the top panel. In the middle is an example of competitive inhibition where the inhibitor binds to the*

active ra preventing access to the substrate and in the bottom panel Bee illustration of a noncompetitive inhibition where the inhibitor can bind to the enzyme and prevents the substrate from accessing the active site.

Irreversible inhibition: Inhibitors in this category resemble the substrate gmeturally with similar molecular geometry providing them the ability to intéract with the enzyme active site. The interaction can be covalent or noncovalent and is strong enough to bind irreversibly with ihe enzyme and destroy its catalytic properties. An interesting example is the action' of ailibiotic penicillin. Penicillin is an irreversible inhibitor of transpeptidase enzyme involved in bacterial cell wall synthesis. Penicillin mimics the natural Dalanine-D-alanine dipeptide substrate binding covalently to the active site of transpeptidase. This prevents synthesis of rigid cell walls and promotes bacterial cell lysis. Penicillin and its analogs are therefore important antibacterial drugs.

Reversible inhibition: Reversible "competitive" inhibitors have similar geometry as substrates and are able to compete with them for formation of theéizyme-substrate complex. The formation of the enzyme-inhibitor complex is reversible but it lowers the catalytic rate as the proportion of Substrates bound to enzymes is reduced.

Inreversible "uncompetitive" inhibition, the inhibitor is unable to bind to free enzyme by itself, but can bind once the enzyme-substrate complex is

— *uorm raze 7 eithl) Seanaagen ue '% latest Use tiereatone 7* 3.4 MECHANISMS OF ENZYME ACTION a The binding of the substrate to the active site is aided by covalent and noncovalent interactions. The covalent interactions involve cleavage™and~ formation of bonds between the substrate and the amino acid residues in the active site cleft. The chemistry behind this can be divided into the following 4 types: / Ae oy hie aie eae (i) General acid-base catalysis: The intermediates formed in chemical reactions may be charged making them unstable and revert back to fom

' Enzymes, Photosynthesis, Metabolism and Bioenergetics 4 bs | i gat shai ete heed Intermediate by donating or wits i | on from molecules other than water) will facilitate their transformation it | Pn In enzyme-catalyzed reactions, this role is played by precisely oa, oP jctive site amino acid residues which can donate and accept protons. a | seri. gn, cysteine, histidine, serine and 1] i | ine side-chains wi! their hear neutral pKa values are known to play this ie & Py enzymes. This plays a major role in defining enzyme specificity and | mr yt activity: 4 | on @ Covalent catalysis: This involves the formation of a temporary_but dl | able covalent t between the enzyme and the active site residues.

This i | wise jhe activation energy leading to further reactions which releases the pee TEE the enzyme. Amino acid residues that can form a jeopile are usually used in enzyme catalysis. Papain which utilizes a 4 jeophilic thiolate ion to form a covalent bond with the peptide backbone mesa OF this typé of mechanism.

- ii) Metal ion catalyst: Metals attached to substrates or active. site sides 2 1 play. a role in enzyme catalysis. With ionic interactions they abilize charged fransition States. They can also mediate oxidation-reduction reactions by changing the oxidation states of the metal ions. Some of the ions Enzymes usually use a combination of the mechanisms described above for catalysis. These are explained further with 4 specific examples described | (i) Proteases: As explained earlier, proteases are enzymes that catalyze the hydrolytic cleavage of peptide bonds in proteins and peptides. Although, peptide bond hydrolysis is thermodynamically favored, the reaction is too slow for quick turnover. Turnover refers to the rate at which unwanted proteins are degraded to constituent amino acids and recycled to form new Proteins. A quick turnover is necessary to match the fast metabolic Tequirements. Proteases are also needed to break down proteins and peptides in food before they are absorbed in the intestines. They also play a role in Tegulating the activity of many other enzymes. For e.g., inactive precursor — is processed by proteases to form mature insulin. Proteases are a ensure all these by lowering the activation energy to catalyze the tion product peptides.

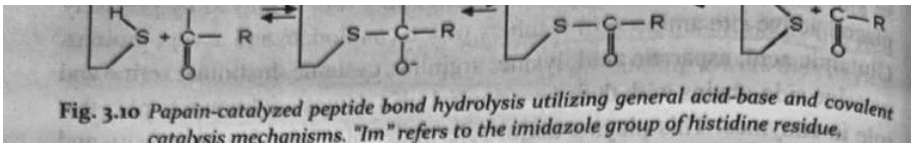


Fig. 3.10 *Papain-catalyzed peptide bond hydrolysis utilizing general acid-base and covalent catalysis mechanisms. "Im" refers to the imidazole group of histidine residue.*

catalysis mechanisms, “Im” refers to the imidazole group of histidine residue,

In papain, during peptide bond hydrolysis, a proton is donated from the sulfahydrl group of a catalytically active cysteine residue to an adjacent imidazole group of a histidine residue in the active site (general acid-base is followed by a nucleophilic attack by the thiolate anion on the carbonoyl

K group carbon of the peptide bond \ (covalent catalysis) to form an unstable tetrahedral intermediate.

The intermediate breaks down followed by hydrolysis to release the Fig. 3.11 Structure of papain. Image from the 4 peptides and donate the proton back RCSB PDB (www.pdb.org) of PDB ID to the cysteine and regenerating the IPPN, R. W. Pickersgill, R.W. Harenzyme. Hence, protease papain uses ris: Ke Gamnag (1992), Suctury fh é i Monoclinic Papain at 1.6 Angstroms two mechanisms: (i) acid-base and Resolution, Acta Crsytallogr., Sect. (ii) covalent catalysis. This example B 48: 59-67. also shows how enzyme-substrate H complex leads to the intermediate formation that can lower activation energy to drive the reaction forward.

(ii) Carbonic anhydrase: In mammals, aerobic metabolism in tissues results in the formation of carbon dioxide, a major end product. This carbon dioxide needs to be removed rapidly from the system to prevent harmful accumulation. This is accomplished by transferring the carbon dioxide to the blood, where it is converted to a soluble form and transported to the lungs. In the Jungs it is exhaled in exchange for oxygen. The solubilization of carbon dioxide in blood occurs due to its conversion to carbonic acid. Carbonic acid isa elatively weak acid and is formed by the reversible hydration of carbon dioxide as shown below:

Enzymes, Photosynthesis, Metabolism and Bioenergetics 87 it | | ever, the carbonic acid can be reversibly deprotonated to form | Hl | pe jon. These steps are coupled to facilitate the transport in blood tm | Peal ation of carbon dioxide as the blood passes through the lungs. This ad sd oe reaction 1S catalyzed rapidly to rates as high as million times per ii | “om pet enzyme molecule by carbonic anhydrase. Carbonic anhydrase hin | enn inc cofactor coordinated to imidazole side chains of 3 histidines in the | ive Sl : bias tigiesd : av? zinc atom coordinated with imidazole side groups of 3 histidine 1 a se in carbonic anhydrase is able to bind with a water molecule (Figs. sid 4 3:13)- This leads to polarization of the water molecule and 42 os ment of @ slight negative charge on the oxygen atom. This facilitates ger yonation of the water molecule and generation of a negatively spe oe? by droxyl ion coordinated to the positively charged zinc atom. The

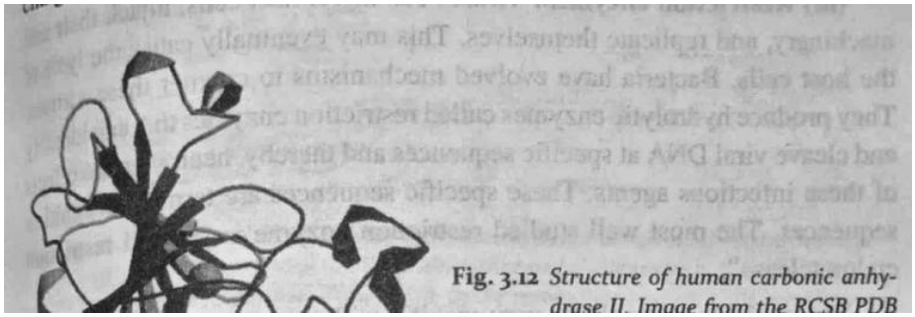


Fig. 3.12 *Structure of human carbonic anhydrase II. Image from the RCSB PDB*

(www. pdb.org) of PDB ID 1CAz, A.E.

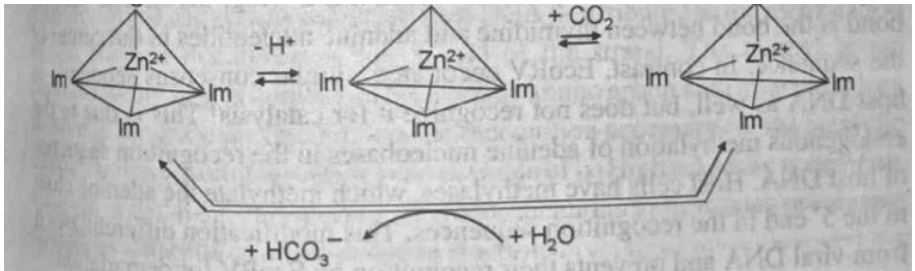


Fig. 3.13 *Role of zinc atom in carbonic acid catalyzed conversion of carbon dioxide to carbonic acid. "Im" refers to the imidazole group of histidine residues. This*

carbonic acid. “Im” refers to the imidazole group of histidine residues. This catalysis utilizes 2 mechanisms: metal-ion and general acid-base.

S4 Nucleic acid structure-function relationships

2 MARKS This concept is known as the structure–function relationship.
4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: What is a Structure–Function Relationship?; Comparison.
12 TO 13 MARKS: ANSWER IN THIS ORDER 1. What is a Structure–Function Relationship? 2. Comparison *Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.*
PRACTICE QUESTIONS 2 marks: State one structure-function relationship of DNA. 5 marks: Compare DNA and RNA. 12 marks: Explain how the structure of DNA and RNA determines their functions.

Complete notes

What is a Structure–Function Relationship?

- The shape, chemical composition, and organization of a biological molecule determine how it behaves inside a living cell. This concept is known as the **structure–function relationship**.
- DNA and RNA illustrate this principle exceptionally well. Although both are nucleic acids composed of nucleotides, small differences in their structures enable them to perform entirely different biological functions.

Comparison

Feature	DNA	RNA
Full Name	Deoxyribonucleic Acid	Ribonucleic Acid
Sugar	Deoxyribose	Ribose
Strands	Double	Usually single
Nitrogenous Bases	A, T, G, C	A, U, G, C
Stability	Highly stable	Less stable
Function	Long-term genetic storage	Protein synthesis and regulation
Cellular Location	Nucleus (primarily)	Nucleus and cytoplasm

Figure

- The structural differences between DNA and RNA explain why they perform different biological roles.
- DNA is chemically more stable because it contains deoxyribose sugar and is double-stranded. This stability makes DNA ideal for storing genetic information throughout an organism's lifetime.
- RNA contains ribose sugar and is generally single-stranded, making it less stable but much more flexible. As a result, RNA is ideal for short-term information transfer and protein synthesis.

S5 BLAST tool for sequence database queries

2 MARKS BLAST stands for Basic Local Alignment Search Tool.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: BLAST; Significance of BLAST; BLAST workflow; BLAST Metrics; Types of Blast Program; Video demo of BLAST.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. BLAST
2. Significance of BLAST
3. BLAST workflow
4. BLAST Metrics
5. Types of Blast Program
6. Video demo of BLAST

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: What does the E-value mean in BLAST?
5 marks: Explain the BLAST workflow.
12 marks: Describe BLAST, its significance, workflow, output metrics and types of BLAST programs.

Complete notes

BLAST

- BLAST stands for **Basic Local Alignment Search Tool**. It is a sequence similarity search program developed by Stephen Altschul and colleagues at the National Center for Biotechnology Information (NCBI) in 1990.
- The primary purpose of BLAST is to compare an unknown biological sequence with sequences stored in large public databases. Instead of comparing entire genomes base by base, BLAST searches for **local regions of similarity**, making the search much faster and computationally efficient.

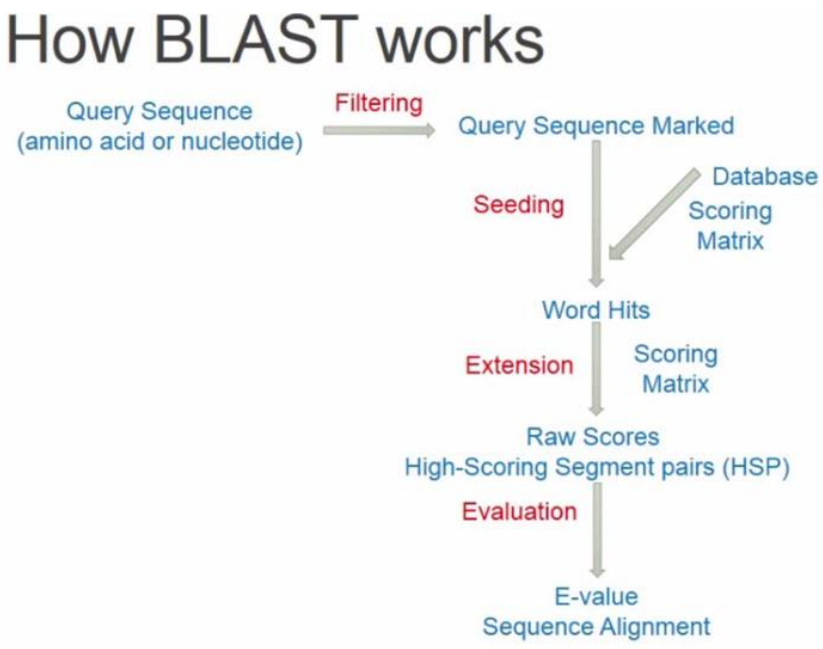
Significance of BLAST

It enables researchers to:

- Identify unknown genes.
- Predict protein functions.
- Compare genomes from different organisms.
- Study evolutionary relationships.
- Detect disease-causing mutations.
- Verify cloning and sequencing experiments.

Without BLAST, many modern genomic studies would require enormous computational effort and time.

Figure



BLAST workflow

- Although BLAST is a sophisticated algorithm, its basic workflow can be understood through a few simple steps.
- Suppose a scientist has sequenced a fragment of DNA from an unknown microorganism.
- **Step 1: Input Sequence**
- The user enters a DNA, RNA, or protein sequence into the BLAST program.
- For example,
- ATGCGTACCTGATCGGATC...
- This sequence is known as the **query sequence**.
- **Step 2: Database Search**
- BLAST compares the query sequence against millions of sequences stored in biological databases such as **GenBank**.
- Instead of comparing every nucleotide individually, BLAST first identifies **short matching words (seeds)** and then extends these matches to produce longer alignments.
- This strategy greatly reduces computational time.

Step 3: Sequence Alignment

- BLAST aligns the query sequence with similar database sequences.
- The alignment identifies:
- Matching nucleotides or amino acids
- Mismatches
- Insertions
- Deletions
- The greater the similarity, the higher the alignment score.
- **Step 4: Ranking Results**
- BLAST ranks all matching sequences according to statistical significance.
- The most similar sequences appear at the top of the results.
- Each result includes:
- Sequence identity
- Query coverage
- Alignment score
- E-value

- Organism name

BLAST Metrics

Query Coverage

- Query coverage indicates how much of the query sequence aligns with the database sequence.
- Higher query coverage generally indicates a more complete match

Percentage Identity

- Identity measures the percentage of identical nucleotides or amino acids between two aligned sequences.
- For example, 98% identity means 98 out of every 100 positions are identical.

Alignment Score/ Max Score

- The alignment score reflects the quality of the sequence match.
- Higher scores indicate stronger similarity.

Total Score

- The sum of the scores from all aligned regions (HSPs) between the query and the same subject sequence.

E-value (Expectation Value)

- The **E-value** is one of the most important statistics in BLAST.
- It estimates the number of matches that could occur **by chance** in the database.
- A lower E-value indicates a more reliable match.

Acc. Len (Accession Length)

- The total length of the database (subject) sequence.

Types of Blast Program

BLAST Program	Query	Database
BLASTN	DNA	DNA
BLASTP	Protein	Protein
BLASTX	DNA	Protein
TBLASTN	Protein	DNA
TBLASTX	DNA	DNA (translated)

Video demo of BLAST

- How to Use BLAST for Finding and Aligning DNA or Protein Sequences - YouTube

S6 Human Genome Project as big data genomics

2 MARKS The genome is the complete collection of genetic material present in an organism.
4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: What is the Human Genome?; What is the Human Genome Project?; Objectives of the Human Genome Project; Applications of the Human Genome Project.
12 TO 13 MARKS: ANSWER IN THIS ORDER 1. What is the Human Genome? 2. What is the Human Genome Project? 3. Objectives of the Human Genome Project 4. Applications of the Human Genome Project *Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.*
PRACTICE QUESTIONS 2 marks: What is the Human Genome Project? 5 marks: List the objectives of the Human Genome Project. 12 marks: Explain the Human Genome Project, its objectives and applications, and its role in big data genomics.

Complete notes

What is the Human Genome?

- The genome is the complete collection of genetic material present in an organism. In humans, the genome contains approximately **3.2 billion DNA base pairs** organized into **23 pairs of chromosomes**.
- These chromosomes contain nearly **20,000 protein-coding genes**, along with many regulatory DNA sequences that control when and where genes are expressed.
- Although every cell in the human body contains essentially the same genome, different cell types activate different sets of genes.
- For example, liver cells express genes involved in metabolism, whereas nerve cells express genes required for signal transmission.

What is the Human Genome Project?

- The **Human Genome Project (HGP)** was an international scientific research initiative launched in **1990** with the objective of determining the complete DNA sequence of the human genome and identifying all human genes.
- Scientists from the United States, the United Kingdom, Japan, France, Germany, China, and several other countries collaborated on this ambitious project. After thirteen years of research, the project was successfully completed in **2003**, two years ahead of schedule.
- The Human Genome Project is regarded as one of the greatest scientific achievements of the twentieth century because it provided the first complete reference map of the human genome.
- Its was the driving force behind the science of Genomics

The Human Genome Project



Objectives of the Human Genome Project

Technogies that enabled the genome Project

Database Management

Genome sequences are stored in specialized databases that allow researchers to retrieve genes, compare sequences, and analyze mutations efficiently.

Examples include:GenBank ,Ensembl

Algorithms

Algorithms are essential for:

DNA sequence assembly ,Pattern matching

Many of these algorithms are similar to string-matching techniques studied in computer science.

High-Performance Computing

Genome assembly requires billions of DNA fragments to be compared and arranged correctly. These computations are performed using parallel processing and high-performance computing systems.

Cloud Computing

Modern genome projects generate petabytes of data that are stored and analyzed using cloud platforms, enabling researchers worldwide to collaborate in real time.

- Determine the complete sequence of the human genome.
- Identify all protein-coding genes.
- Develop technologies for rapid DNA sequencing.
- Store genomic information in publicly accessible databases.
- Improve computational tools for genome analysis.
- Address ethical, legal, and social issues associated with genetic information.

Applications of the Human Genome Project

Medicine

- Early diagnosis of inherited diseases
- Cancer genomics
- Prenatal genetic testing
- Personalized medicine

Drug Discovery

- Scientists can identify genes associated with diseases and develop drugs that specifically target those genes.

Agriculture

- Genome sequencing has improved crop productivity, disease resistance, and nutritional quality.

Forensic Science

- DNA fingerprinting enables accurate identification of individuals in criminal investigations and disaster victim identification.

Evolutionary Biology

- Comparing genomes from different species helps scientists understand evolutionary relationships and trace the origin of organisms.

Unit 3

Protein Structures and Prediction Algorithms

How a sequence becomes a shape, and how computers predict that shape.

S1 Protein synthesis and primary structure

2 MARKS Transcription is the process of producing an RNA molecule using DNA as a template.
4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Protein Synthesis: From Code to Protein; Two major processes:; Transcription: DNA → mRNA; What is mRNA?; Pre-mRNA Formation; Pre-mRNA Processing.
12 TO 13 MARKS: ANSWER IN THIS ORDER 1. Protein Synthesis: From Code to Protein 2. Two major processes: 3. Transcription: DNA → mRNA 4. What is mRNA? 5. Pre-mRNA Formation 6. Pre-mRNA Processing 7. RNA Splicing: Introns and Exons 8. Mature mRNA 9. Translation: mRNA → Protein 10. Ribosome Structure and Function 11. Ribosomal A, P and E Sites 12. Codons: Reading the Genetic Code 13. Role of tRNA 14. Translation Process : Initiation *Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.*
PRACTICE QUESTIONS 2 marks: Define transcription. 5 marks: Explain pre-mRNA processing. 12 marks: Describe protein synthesis (transcription and translation) and primary structure.

Complete notes

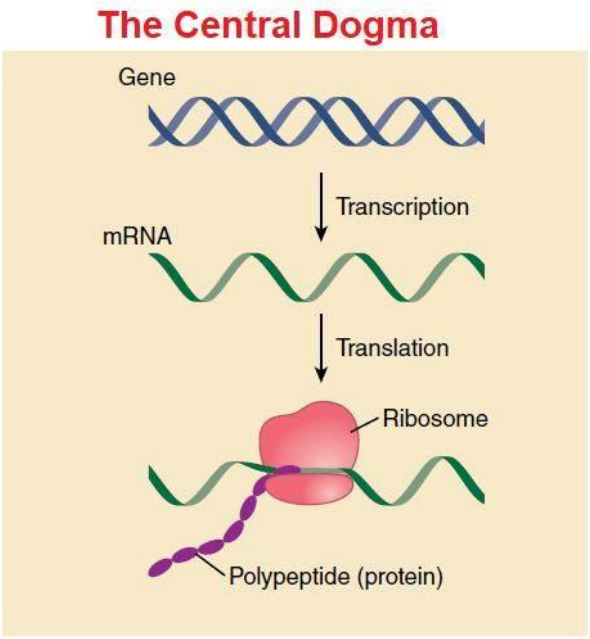
Protein Synthesis: From Code to Protein

How does a cell convert genetic information into a functional protein?

Central idea:

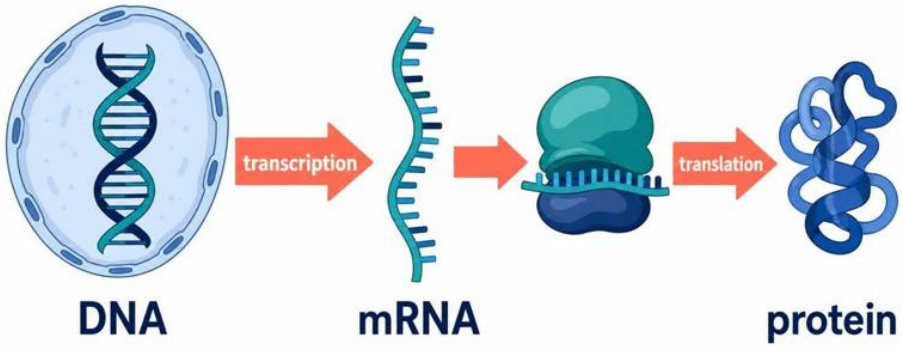
DNA → RNA → Protein

- DNA stores the instructions.
- RNA carries the required information.
- Ribosomes read the information.
- Amino acids are assembled into a protein.
- The amino acid sequence determines the protein's primary structure.



Two major processes:

- **Transcription:**
  DNA → RNA
- **Translation:**
  RNA → Protein
- This provides a simple framework for understanding how information stored in DNA ultimately produces a functional protein.



Transcription: DNA → mRNA

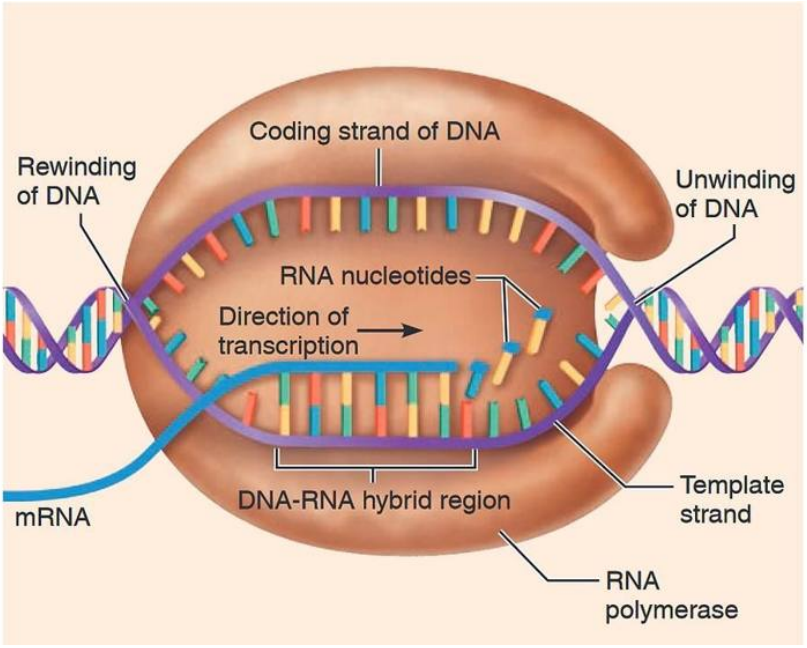
- **Transcription** is the process of producing an RNA molecule using DNA as a template.
- The enzyme **RNA polymerase** plays a central role.

Basic process:

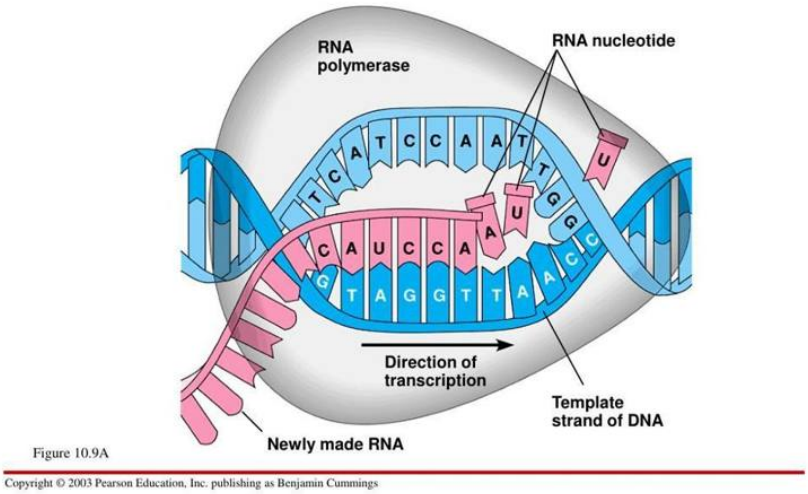
- **DNA gene → RNA polymerase → mRNA**
- The resulting **messenger RNA (mRNA)** carries the genetic information needed for protein synthesis.
- In eukaryotes, transcription occurs in the nucleus.

Why mRNA?

- DNA is the stable storage molecule, while mRNA acts as a temporary working copy of the required information.

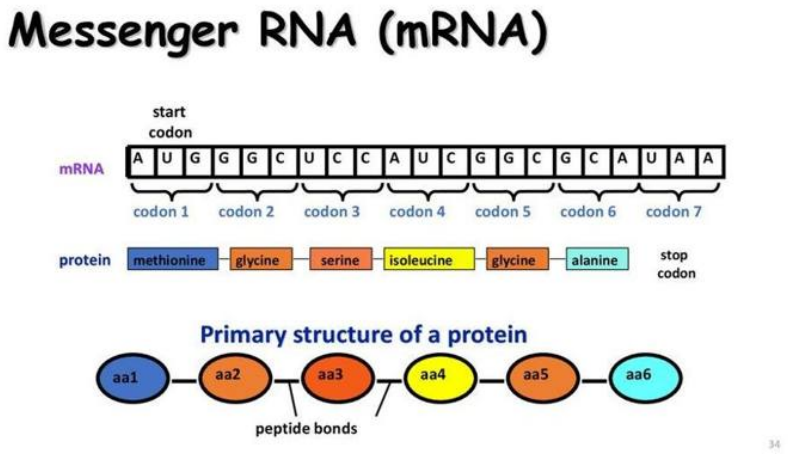


Transcription produces genetic messages in the form of mRNA



What is mRNA?

- **mRNA = messenger RNA**
- It carries genetic information from DNA to the ribosome.
- RNA contains four bases:
- **A – Adenine**
  U – Uracil
  G – Guanine
  C – Cytosine
- **DNA → T (Thymine) instead of RNA → U (Uracil)**
- The sequence of bases in mRNA determines the order in which amino acids will be incorporated into the protein.
- The initial transcript from a protein-coding gene is pre-mRNA.

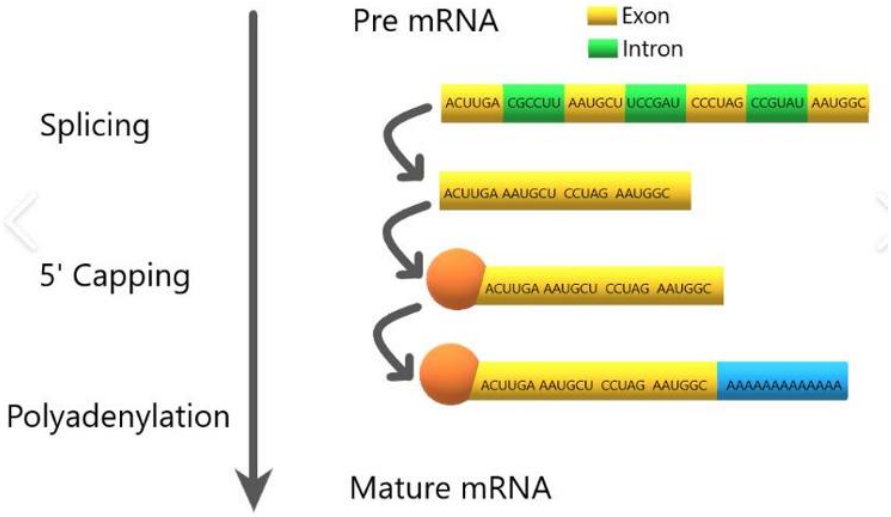


Pre-mRNA Formation

- Pre-mRNA is the primary RNA transcript produced from a protein-coding gene.
- It contains sequences that will be retained and sequences that will be removed during processing.
- Pre-mRNA undergoes processing before becoming mature mRNA.
- Major processing events are 5′ capping, splicing, and polyadenylation.

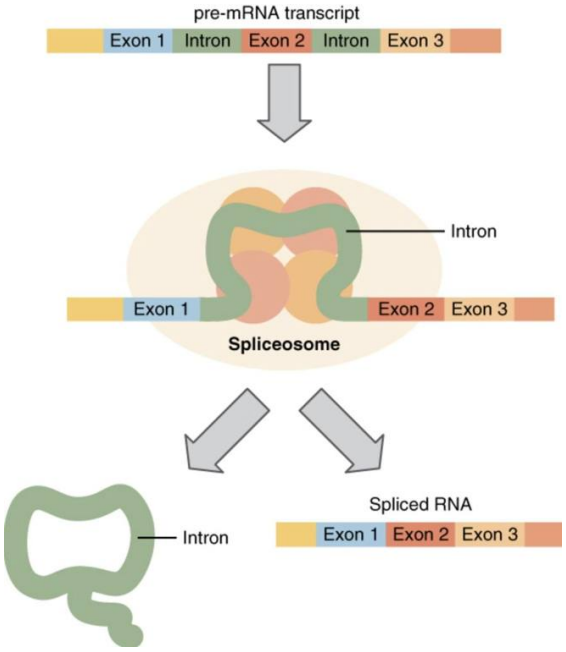
Pre-mRNA Processing

- 5′ capping adds a modified guanine nucleotide to the 5′ end.
- Splicing removes introns and joins exons.
- Polyadenylation adds a poly-A tail to the 3′ end.
- Processing produces mature mRNA suitable for export to the cytoplasm and translation



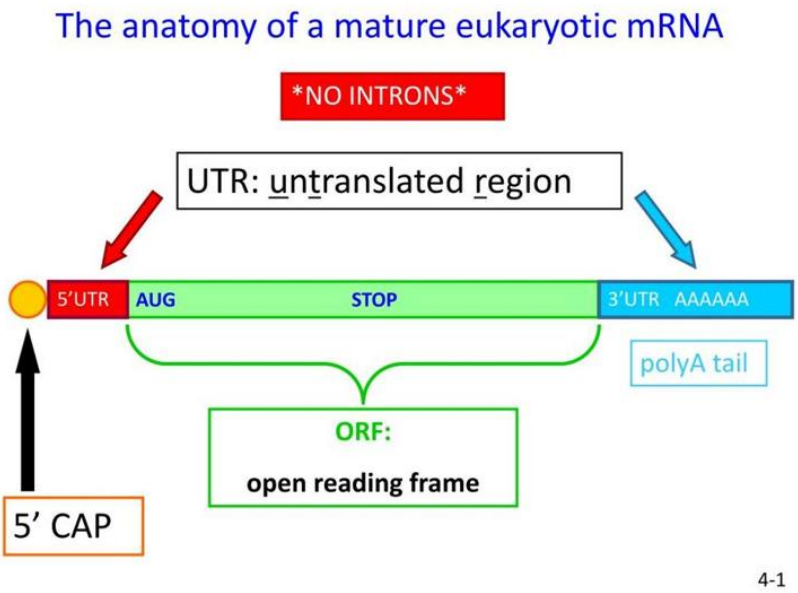
RNA Splicing: Introns and Exons

- Introns are intervening sequences removed from pre-mRNA.
- Exons are sequences retained in mature RNA.
- The spliceosome protein carries out the major steps of pre-mRNA splicing.
- Exons are joined to form the mature mRNA sequence.



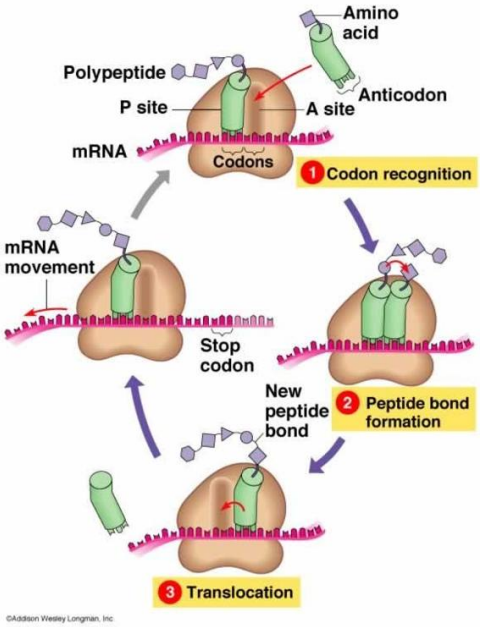
Mature mRNA

- Mature mRNA contains a 5′ cap, joined exons, and a 3′ poly-A tail.
- It is transported from the nucleus to the cytoplasm in eukaryotic cells.
- Mature mRNA provides the template for protein synthesis at the ribosome.



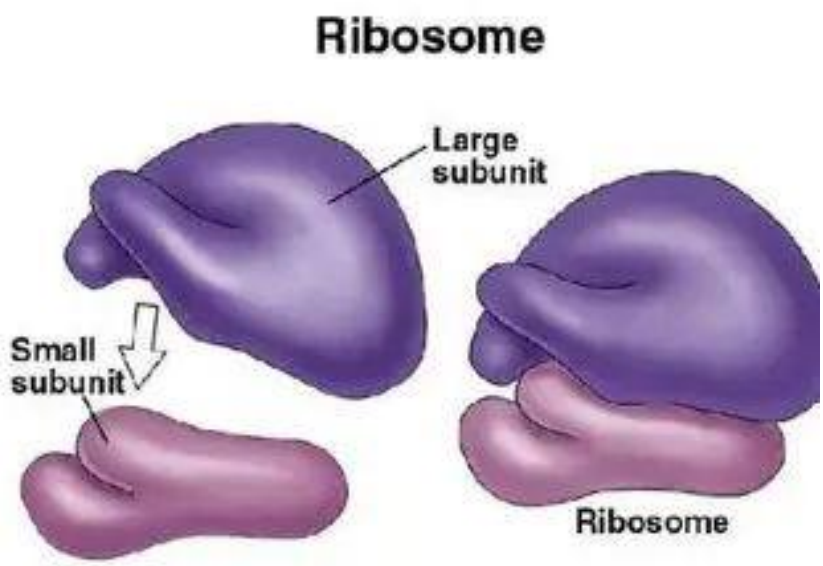
Translation: mRNA → Protein

- **Translation** is the process by which the information in mRNA is converted into an amino acid sequence.
- Translation occurs at the **ribosome**.
- The ribosome: binds to mRNA.
  - Reads the mRNA sequence.
  - Identifies codons.
  - Brings together the appropriate amino acids.
  - Links the amino acids using peptide bonds.
  - Produces a polypeptide chain.



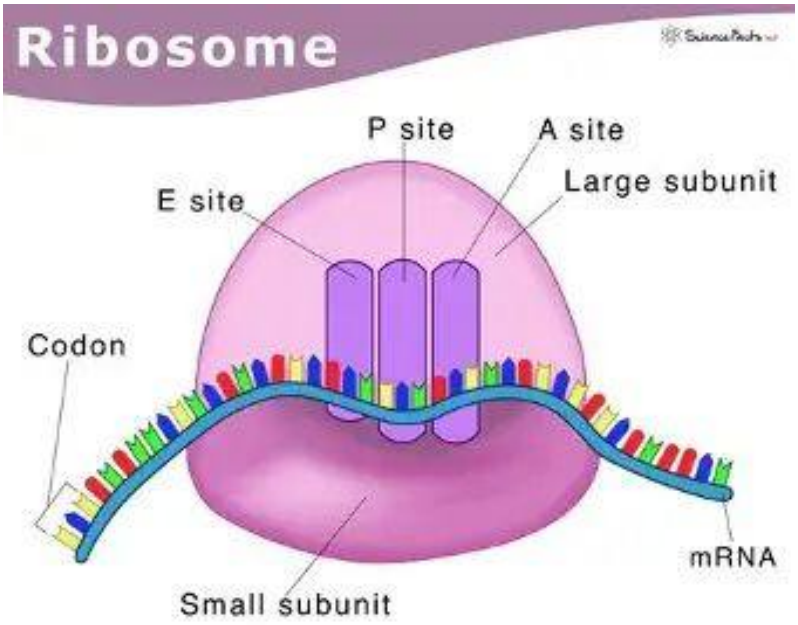
Ribosome Structure and Function

- Ribosomes are molecular machines responsible for translation.
- They consist of a small and a large subunit.
- The small subunit participates in mRNA recognition and decoding.
- The large subunit contains the catalytic center involved in peptide bond formation.



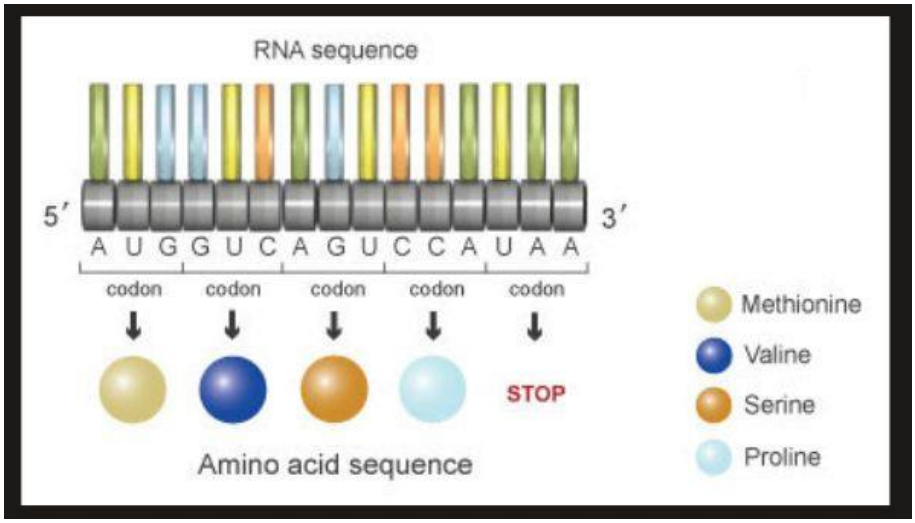
Ribosomal A, P and E Sites

- A site: accepts incoming aminoacyl-tRNA.
- P site: holds the tRNA carrying the growing polypeptide.
- E site: accommodates the exiting deacylated tRNA.
- These sites coordinate sequential addition of amino acids



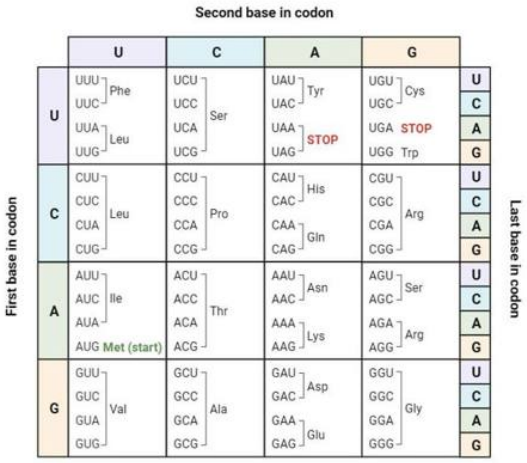
Codons: Reading the Genetic Code

- The ribosome reads mRNA in groups of three nucleotides.
- Each group of three is called a codon.
- Example:
- AUG – GCU – AAA – UGC
- Each codon specifies an amino acid or provides a start/stop signal.
- Because there are four possible RNA bases:
- 4 × 4 × 4 = 64 possible codons
- These codons specify the 20 standard amino acids, along with start and stop signals.
- Think of codons as three-character instructions in a biological programming language.



Figure

Codon Chart



Role of tRNA

- **tRNA = transfer RNA**
- tRNA helps bring the correct amino acid to the ribosome during translation.
- Each tRNA contains:
  - An **anticodon**
  - A corresponding amino acid
- Example:

mRNA codon: AUG

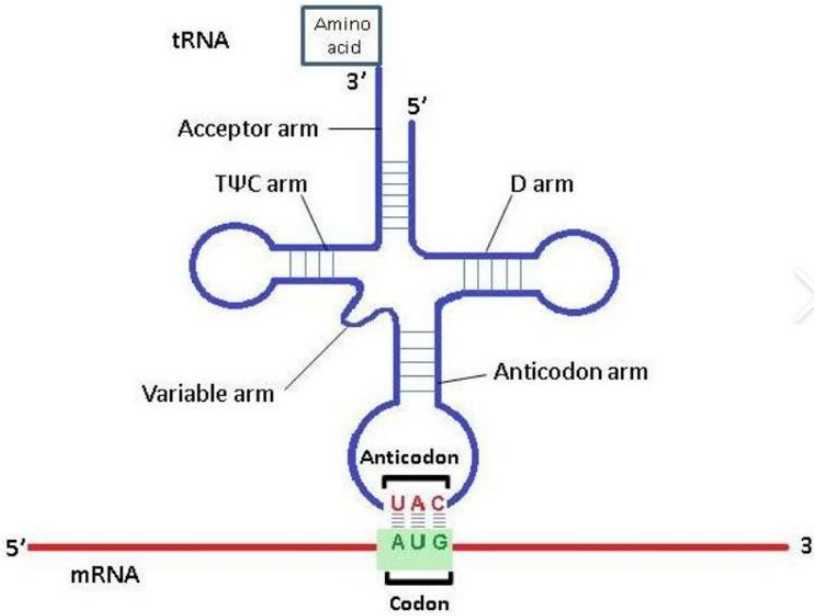
↑

complementary

↓

tRNA anticodon: UAC

- The tRNA ensures that the appropriate amino acid is incorporated into the growing protein chain.



Translation Process : Initiation

- Translation begins when the ribosome assembles on mRNA.
- The start codon, usually AUG, establishes the reading frame.
- The initiator tRNA carrying methionine pairs with the start codon.

- The complete ribosome is assembled and ready for elongation.

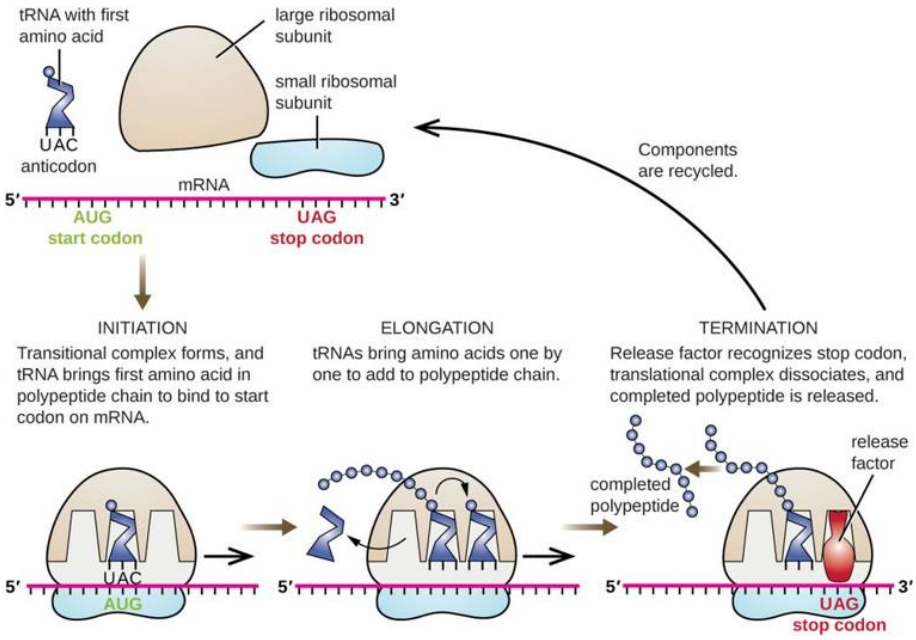
Translation: Elongation

- An aminoacyl-tRNA enters the A site and pairs with the corresponding codon.
- A peptide bond is formed between amino acids.
- The ribosome moves along mRNA in the 5′ to 3′ direction.
- tRNAs move through the A, P and E sites as the polypeptide grows.

Translation: Termination

- Translation terminates when a stop codon enters the ribosome.
- Stop codons do not encode amino acids.
- Release factors promote release of the completed polypeptide.
- Ribosomal subunits and mRNA can subsequently dissociate.

Figure



Formation of the Polypeptide Chain

- The amino acids brought to the ribosome are joined together by **peptide bonds**.

Amino acid

+

Amino acid

+

Amino acid

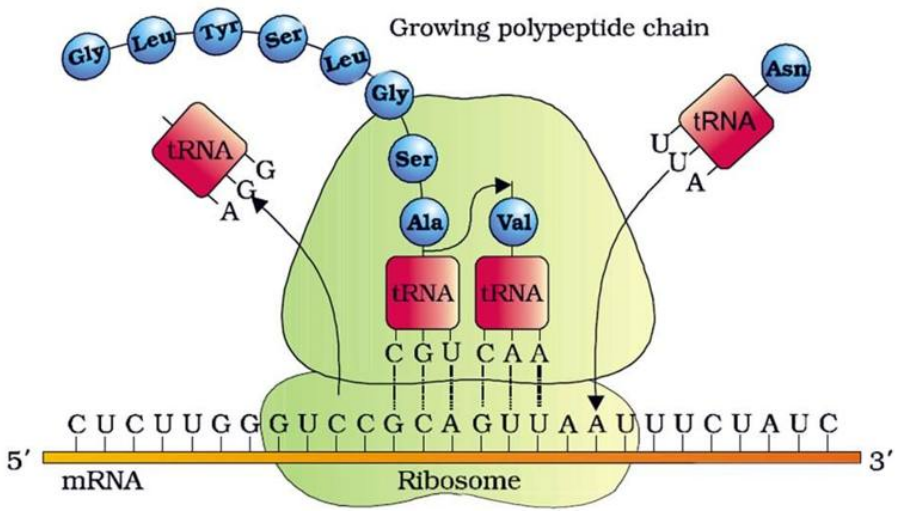
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Polypeptide chain

- The chain grows as the ribosome continues reading the mRNA.

Eventually, a **stop codon** signals the end of translation.

- The newly synthesized polypeptide then folds and may undergo further modifications to become a functional protein.



What is Primary Structure?

- The **primary structure** of a protein is the **specific linear sequence of amino acids** in its polypeptide chain.

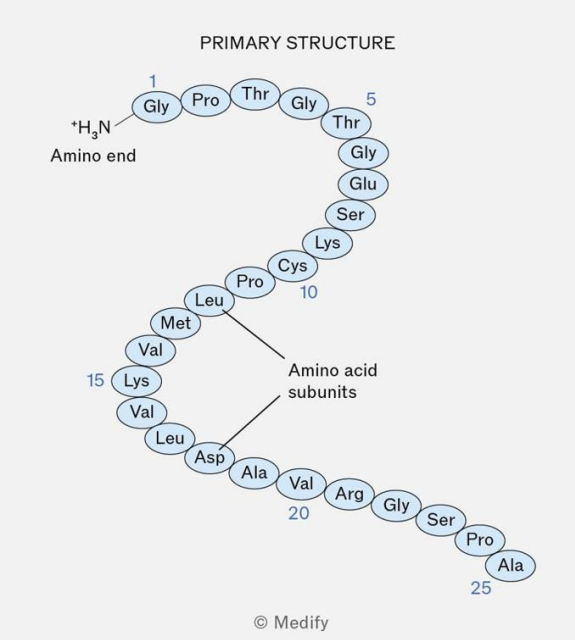
For example:

Met – Ala – Gly – Lys – Leu – Ser – Val

- The amino acids are connected by **peptide bonds**.
- The sequence is directional: **N-terminus → C-terminus**

Why is it important?

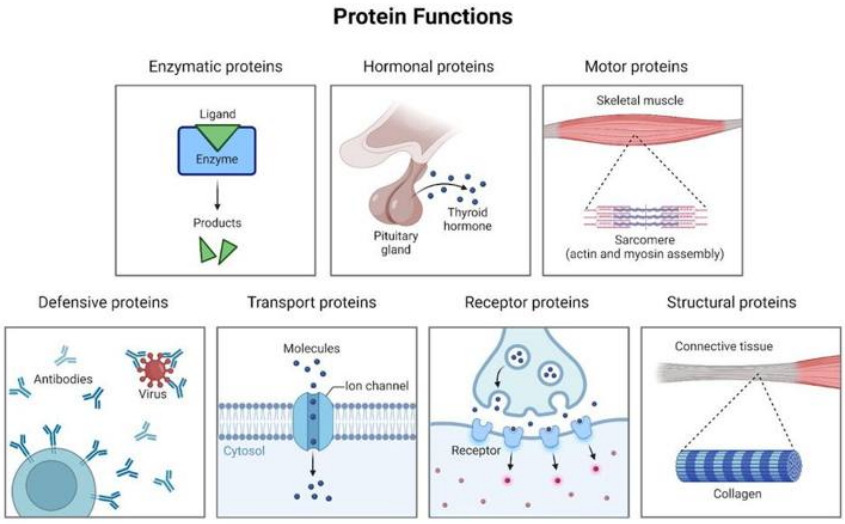
- The amino acid sequence provides the foundation for the protein's higher-order structure and ultimately influences its function.



What are Proteins?

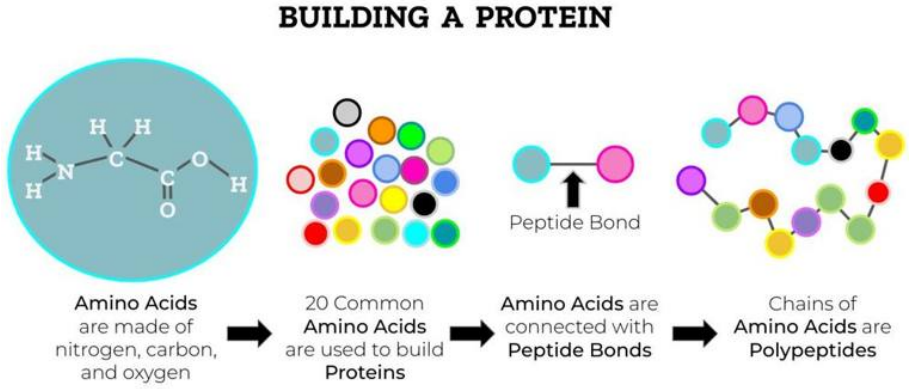
- Proteins are complex biological molecules made up of smaller units called **amino acids**.
- They perform a wide range of functions:
- Enzymes – catalyze biochemical reactions
- Structural proteins – provide support
- Transport proteins – carry molecules
- Receptors – receive signals

- Antibodies – participate in defense
- Regulatory proteins – control cellular processes

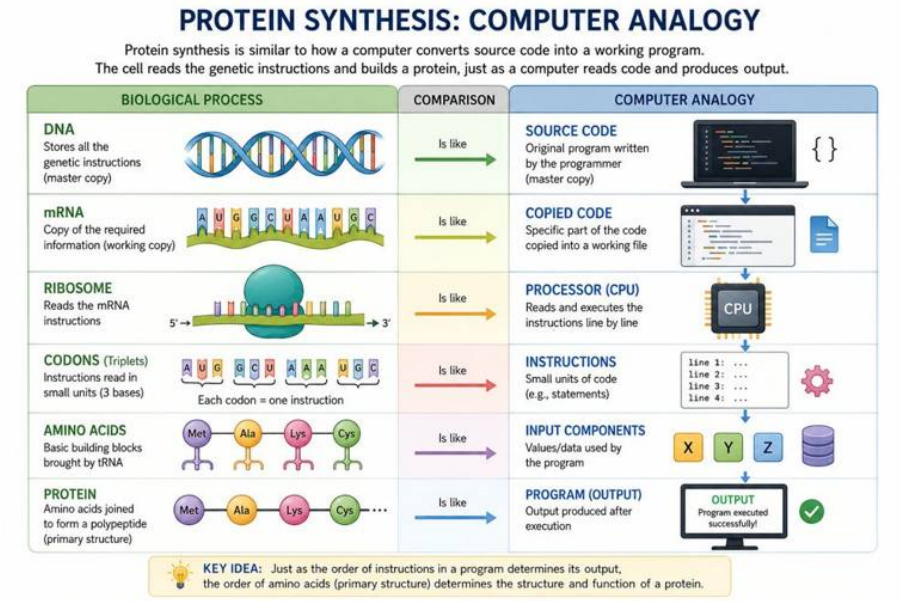


Figure

- Proteins are constructed from **20 standard amino acids**.
- Each amino acid contains:
  - Amino group ($–NH_2$)
  - Carboxyl group (–COOH)
  - Hydrogen atom
  - Variable **R group**
  - Central carbon atom
- The **R group** differs among amino acids and gives each amino acid its unique chemical properties.



Figure



More notes

2.4 PROTEIN SYNTHESIS ~ q

Protein synthesis is the process in which cells build proteins. Both prokaryotic cell and eukaryotic cell synthesize different types of proteins. It’s a multi-step process involving transcription, posttranscriptional modification, translation, and per tin, Modifications to yield the desired protein. Before understanding the i i ; , aeDNAlabd RNA te aehdh sluts of protein synthesis, a brief knowledge | DNA (Deoxyribose Nucleic Acid): DNA is a nucleic acid that contains iho soca ena used in the development and functioning of all known living organisms. The main role of DNA molecules is the long-term storage ofgenetic information, | rrr Frederick Griffith in 1928 showed that the DNA was the cell’s genetic material. Rosalind Franklin took diffraction X-ray photographs of DNA crystals. Watson and Crick in the 1950's built the first model of DNA. Structure of DNA: In 1953, Watson and Crick proposed a model to explain the arrangement of molecules in DNA. The model is characterized by the following features:

+ DNA is formed of two polynucleotide chains, spirally coiled to form a double helix. ETT

- Each chain is formed of many units called nucleotides- building blocks of DNA. agers « A nucleotide is formed of three components, namely a phosphoric acid, a deoxyribose sugar and a nitrogenous base. The bases are Adenine (A), Guanine (G), Cytosine (C) and Thymine (T). “A STEERS
- Deoxyribose sugar and nitrogenous base comprise to form a Nucleoside, with addition of a phosphoric acid is called as Nucleotide. —
- Adenine and guanine belong to a group of compounds called Purines, Thymine and cytosine belong to another group called as Pyrimidines. Purines are linked or bonded with pyrimidines.
- In a DNA molecule, adenine always pairs with thymine with two hydrogen bonds while guanine pairs with cytosine with three hydrogen | bonds, the latter being more stronger interaction. The amount of adenifie is equal to the amount of thymine and the amount of guanine 1s equal to the amount of cytosine. This is called as Chargaff’s Rule.
- The two chains of a DNA are complementary to each other. Oneend of the chain is called 3’ end and the other end is called 5’ end. The strands are antiparallel in nature. i
- The two complementary chains are twisted around each other to form a double helix. One turn of the helix is about 34A and contains ten paired nucleotides with a distance of 3.4A between each pair of bases. The width of the DNA molecule is 20A.

DNA Replication: DNA replication is a biological process that OCCurs in all living organisms and copies their DNA; it is the basis for biologica inheritance. The process starts with one double-stranded DNA molecule ang produces two identical copies of the molecule. There are number differences between

the mode of prokaryotic and eukaryotic DNA replications. In a cel), DNA replication generally begins at specific locations in.the genome, calleq Steps involved:

- Unwinding of DNA at the origin. + Break of hydrogen bonds by unzipping of the, strands by various enzymes and proteins. Th
- Addition of bases and synthesis of new strand from the template strand F by DNApolymerase. seihaie H Ribonucleic acid (RNA): RNA is a nucleic acid containing ribose sugar, at It is found in large amounts in the cytoplasm and fewer amounts in the nucleus. RNA is single stranded, sometimes may be folded back upon itself to-form double stranded coiled form. The purines and pyrimidines amount are not equal, and thymine is replaced by another base called uracil (U), \ Three types of RNA are present:

“ (a) Messenger RNA (mRNA) constitutes 3-5% of total RNA content. (b) Transfer RNA (tRNA) constitutes 15-17% of total RNA content. (c), Ribosomal RNA (rRNA) constitutes 80% of total RNA content. (a) Messenger RNA (mRNA) Messenger RNA is a ribonucleic acid which carries the genetic information for “protein synthesis from DNA to the cytoplasm. The mRNA constitutes of 3-5% of the total cellular RNA. The mRNA is synthesised as a complementary strand upon the chromosomal or genomic DNA. The mRNA carries the message in the form of triplet codes. It carries the information for a specific protein and is made up of several hundreds to few thousands nucleotides One end of mRNA is called the 5’end and the other is the 3’end. At the 5’end, acap region is present which helps the mRNA to bind to with ribosomes adenylate (A) nucleotides. These are called post-transcriptional modifications which can also provide stability to the mRNA. AUG generally represents translation initiation codon (methionine) and UAA represents translation termination codon in the mRNA.

} - The Molecular and Biochemical Basis ofan Organism 55 1) AUG UAA é f | Coding region | oes Poly (A tail Non-Coding region Non-Coding region sae AUG UAA Poly(A) tail . | Non coding Region = ;

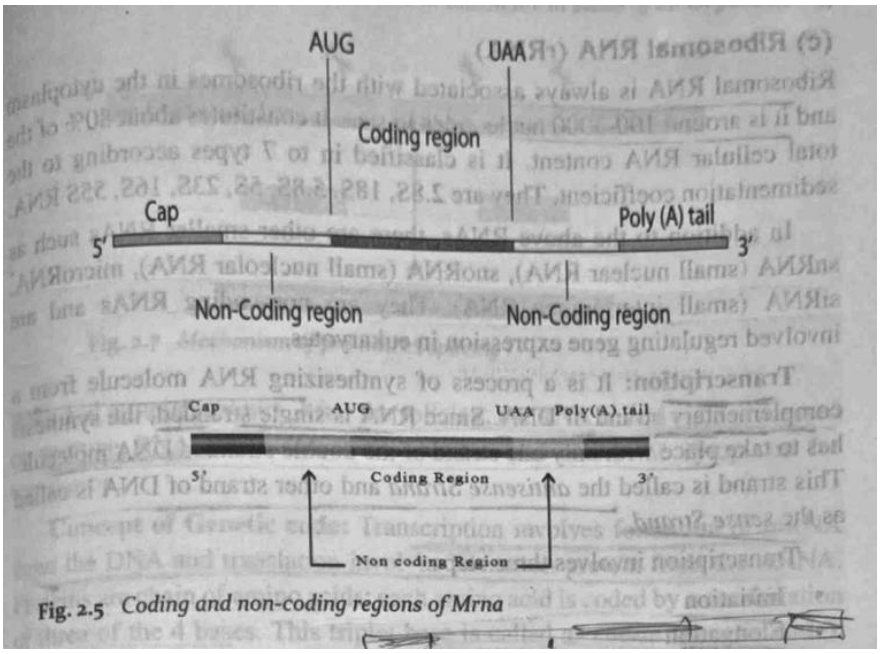


Fig. 2.5 *Coding and non-coding regions of Mrna*

Transfer RNA is a type of RNA which transports amino acids to the site of protein synthesis. It is single stranded and constitutes about 15-17% of the total cellular RNA. Each tRNA can transport only one amino acid. There are 60 types of tRNA. As there are 20 different amino acids, certain amino acids are carried by more than one tRNA.

Itcontains of 75-80 nucleotides.

The tRNA assigns a clover leaf atest | like structure (Fig. 2.6). It contains four arms namely: Tarm (a) Acceptor arm or amino acid- Vests binding site (b) D arm or enzyme site Anticodon arm (c) Ty*C arm or Ribosome site anederkig (d) — Anticodon arm Or_ Fig. 2.6 Schematic representation of a Recognition site tRNA :

The anticodon loop contains codon complementary to the mRNA sequence. The respective anticodon | Makes the respective amino acid to bind at the acceptor arm. The D and T arm Possess unusual

bases like pseudouridine and dihydrouridine.

(c) Ribosomal RNA (rRNA) \ Ribosomal RNA is always associated with the ribosomes in the cytoplasm and it is around 100-3000 nucleotides in size. It constitutes about 80% of the totaf cellular RNA content. It is classified in to 7 types according to the In addition to the above RNAs, there are other smaller RNAs such as snRNA (small nuclear RNA), snoRNA (small nucleolar RNA), microRNA, siRNA (small interference RNA). They are non-coding RNAs and are involved regulating gene expression in eukaryotes. Transcription: It is a process of synthesizing RNA molecule from a complementary strand of DNA. Since RNA is single stranded, the synthesis has to take place with only one strand of the double stranded DNA molecule, i This strand is called the antisense Strand and other strand of DNA is called j as the sense Strand, % Transcription involves three steps:

+ Elongatior gation ’ ' Initiation: The RNA polymerase binds to the double stranded DNA by recognizing _a_sequence on it for attachment; this is called as promoter Sega TATA Box After ihe binding of RNA polymerase, the DNA mofeetle unzips liself by 20 base pairs. The region which has to be transcribed following the promoter sequence is called_as_downstream_region and sequence before the promoter is called as upst -sequence.-The site (nucleotide) where transcription starts is led ee site and it is denoted as +1. 5 Elongation: As the unwinding of DNA takes place the RNA polymerase adds complementary bases and synthesizes RNA in the 5’ —» 3° direction. It continues until a gene is completely transcribed. =——— Termination: When the RNA polymerase ‘reaches a stop signal in the DNA sequence, the RNA polymerase stops its work and unbinds itself from the DNA and THRNA falls off from the complex ‘Teading to the stop in transcription. After the_synthesis of RNA from DNA by.transeription, the resultant RNA is called as precursor mRNA (pre-mRNA). This contains both coding sequences (exons) and: non-coding sequénces (introns). Thus, the introns have to be removed or spliced from the pre-mRNA (Fig. 2.7). Splicing is a mechanism of removing of introns and joining of exons that occurs in spliceosome. A spliceosome contains a complex of small nuclear

A, Parise zaman

- | Tat Sree eel f Fig. 2-7 Mechanism of pre-mRNA splicing s } RNA and protein subunits. After the splicing, the RNA is said to be processed i or mere BNA and it leaves the nucleus and enters the cytoplasm for | Concept of Genetic code: Transcription involves formation of mRNA from the DNA and translation involves formation protein from the mRNA. i Proteins are chain of amino acids; each amino acid is coded by a combination of three of the 4 bases. This triplet base is called as codon-(genetic code). | There are 4 bases so total of 4 x 4 x 4= 64 codons. Totally only twenty amino i | ‘ cers meee eee f acids are present but there are 64 codons and obviously, there are extra . codons. Fi =e Properties of genetic code:
- Genetic code is universal among living organisms.
- Each amino acid is coded by more than one codon.- Redundancy
- No codon codes for more than one amino acid. — Ambiguous Four codons among the 64 codons have the function to determine the start and end of the franslation process (Table 2.1). Tu ad Table 2.1 List of codons coding for the respective amino acids. ; First | Second Base Third ey uuu | ucu UAU UGU ie per Phenylalanine | Serine ‘Tyrosine __| Cysteine eee UUC UCC UAC UGC asa Phenylalanine Serine Tyrosine Cysteine peeeeey UUA | UCA | UAA S20 | UGA ie aes Py) Leucine Serine Seip eta iE: op. aie eae] UUG UCG UAG =| UGG 3 Leucine Serine Stop —_| Tryptophan }

Oe ae

Leucine Proline Histidine Arginine Leucine Proline Histidine Arginine

Cua cca caa cga

Leucine Proline Glutamine Arginine | CUG CCG CAG cGG | Leucine Proline Glutamine Arginine | Isoleucine Threonine Asparagine Serine

Auc acc aac agc

f Isoleucine Threonine Asparagine Serine

Aua aca aaa aga

Tsoleucine Threonine Lysine Arginine Methionine ‘| Threonine Lysine Arginine jet a GUU GCU GAU GGU Seay Valine Alanine Aspartate Glycine sey GUC GCC GAC GGC Valine Alanine Aspartate Glycine

Gua gca

Valine Alanine | Glutamate Glycine es GUG GCG GAG GGG Sie Valine Alanine | Glutamate Glycine a) Start codon: AUG _ b) Stop codons: UAG, UGA, UAA Translation involves the transfer of the genetic code in an mRNA molecule into the amino acid sequence of a polypeptide (Fig. 2.8). It occurs in the cytoplasm. This involves four components: — ribosomes (with large and small subunits) — amino acids.

Translation mechanism comprises of three steps involving (a) Initiation, (b) Elongation, and (c) Termination. fa eee Sts nas:

(A) Initiation of Translation : The_mRNA binds to the smaller unit of the ribosome (30S). The attachment is made at the first codon of mRNA which is AUG coding c Sa eee”

Initiation step

charged tRNA Rac ELONGATION STEP empty tRNA q L i es i aa 8 Translocation (movement of the ribosome Any > x relative to the mRNA) allows the cycle of fs Ge BAY protein synthesis to continue.As the complex pais ey re reach the stop codon the Protein synthesis stops e ee ae aE

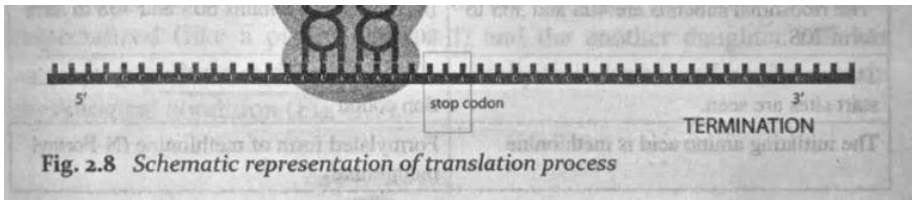


Fig. 2.8 *Schematic representation of translation process*

+ ThetRNA having the anticodon UAC complementary to AUG in mRNA attaches itself to the initiation codon and transports methionine to the 30S subunit. This complex of mRNA, tRNA and 30S is called preinitiation complex. Later, 50S subunit joins to forms the complex, Initiation Complex.

- The complex has two slots for the entry of tRNA. They are P site (peptidyl site) and A site (aminoacy] site). The first tRNA attaches to the P site.

(B) Elongation and (C) Termination _* Itrefers to the addition of amino acids one by one to the first aming acid | nethionine, as per the sequence of the codons in mRNA.

- The second codon in the mRNA is recognized and the corresponding tRNA enters the A site and the respective amino acid is brought along with it,’ The tRNA with respective amino acid is called as chargeq
- The formation of peptide bond occurs. between the first amino acid and the second amino acid... - i
- As the complex moves and reads the next codon in the mRNA the first tRNA leaves the complex through the P site and falls off, followed the second tRNA occupying the P site and entry of new tRNA at the A site and corresponding amino acid is added to the chain and peptide bond is
- This process continues as the stop codon UAA/UGA/UAG is reached, then the chain terminates and the process stops leaving behind the synthesized protein. The protein formed is a sequence of amino acids linked by peptide bond. This is called as termination.

+ Proteins can further_undergo post-translational modifications such glycosylation, phosphorylation, acetylation, sumoylation which may be required for their intracellular localization, activity, or other purposes,

Difference between Eukaryotic and prokaryotic protein synthesis:

Prokaryotes } Eukaryotes

The ribosomal subunits are 40S and 30S to | In contrast, it contains 60S and 40S to form

The initiation codon varies and multiple In eukaryotes only AUG serves as the initia-

Start sites are seen. tioncodon

The initiating amino acid is methionine Formylated form of methionine (N-Formyl

- Nano-bioelectronics is a multi-disciplinary research area comprising biotechnology, nanofabrication, and imaging. It examines — the properties of biomolecules by nanotechnology and uses biomolecules to add novel functions to nanoelectronic devices. The ultimate goal is to incorporate biomaterials like DNA as structural framework in nanoelectronic devices. The ability of DNA’s self-assembly is combined with electronic properties of nanotubes to overcome the

- problem of organizing nanotubes into nanoscale electronic circuits.

+ Role of Technology in Human Genome Project: Human “genome project’s goal was to sequence the human DNA. The final sets of at scientific papers were published in 2006 even though the w é "ae draft was released in 2000, The Department of Energy, USA was able

S2 Secondary/tertiary/quaternary organisation

2 MARKS Secondary structure describes the local folding patterns of the polypeptide backbone.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: α-helix; β-sheet; Turns and Loops; Tertiary Structure; Factors Stabilising Tertiary Structure; Protein Domain.

12 TO 13 MARKS: ANSWER IN THIS ORDER

1. α-helix
2. β-sheet
3. Turns and Loops
4. Tertiary Structure
5. Factors Stabilising Tertiary Structure
6. Protein Domain
7. Quaternary Structure
8. Example: Hemoglobin

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS

2 marks: What is a protein domain?
5 marks: Differentiate the alpha-helix and the beta-sheet.
12 marks: Explain secondary, tertiary and quaternary structure with examples and stabilising forces.

Complete notes

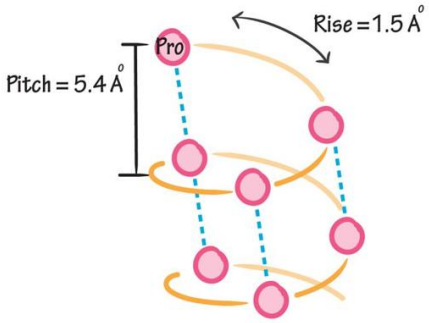
Figure

- Secondary structure describes the local folding patterns of the polypeptide backbone.
- The major types are:
- α-helix-A coiled, spiral arrangement of the polypeptide backbone.
- β-sheet-Extended sections of the polypeptide chain arranged next to one another.
- Both are stabilized mainly by hydrogen bonds involving the peptide backbone.

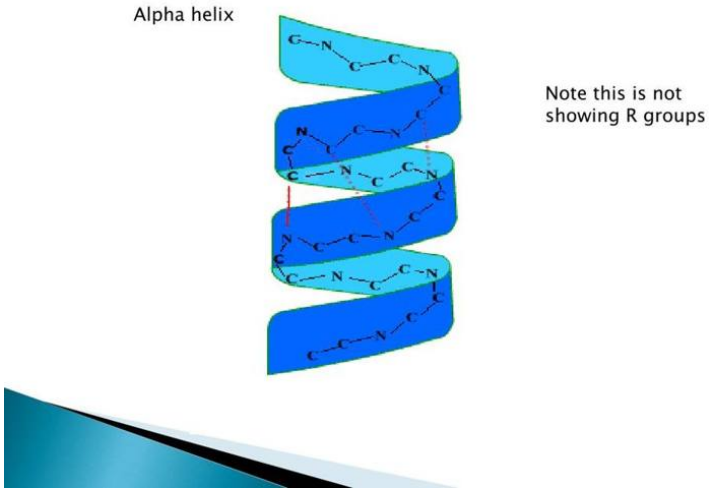
α-helix

- An α-helix is a regular helical structure formed by twisting of the polypeptide backbone.
- Stabilized mainly by hydrogen bonds.
- Hydrogen bonds occur between groups in the peptide backbone.
- Amino acid side chains project outward from the helix.
- A protein can contain several α-helical regions.

Alpha Helix

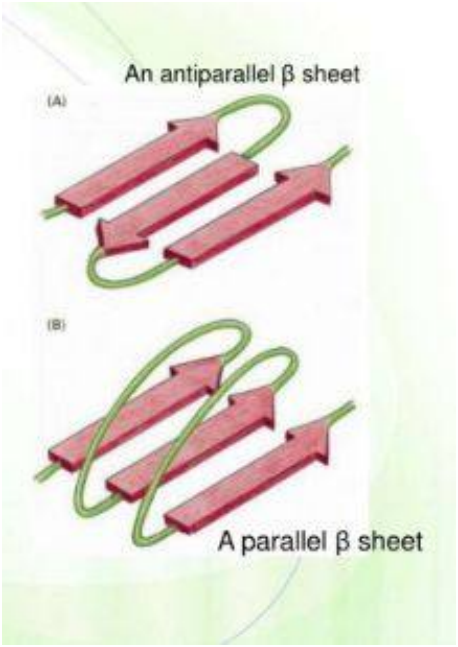


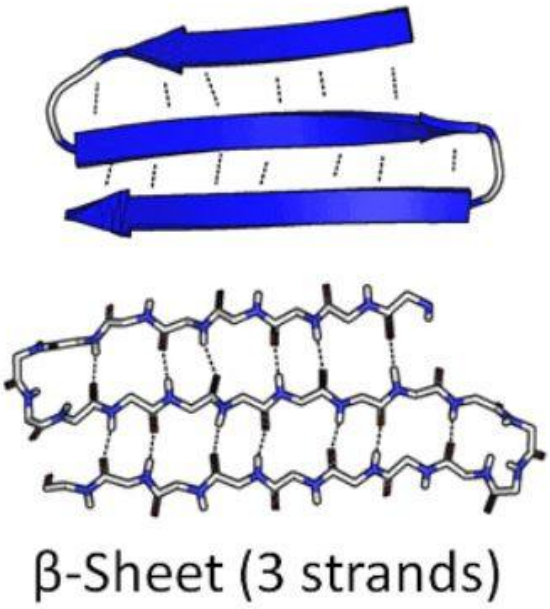
Secondary Structure Alpha Helix



β-sheet

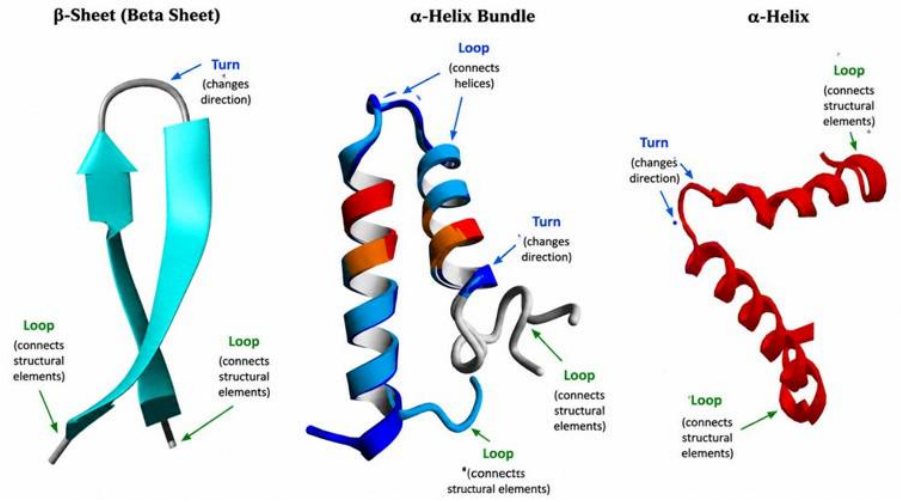
- A β-sheet is formed when extended sections of a polypeptide chain, called β-strands, are positioned next to one another.
- Stabilized by hydrogen bonds between peptide backbones.
- β-strands can be arranged as:
  - Parallel
  - Antiparallel





Turns and Loops

- Proteins also contain:
  - **Turns**
  - **Loops**
- Irregular structural regions
- These regions connect different secondary structural elements and allow the polypeptide chain to change direction.

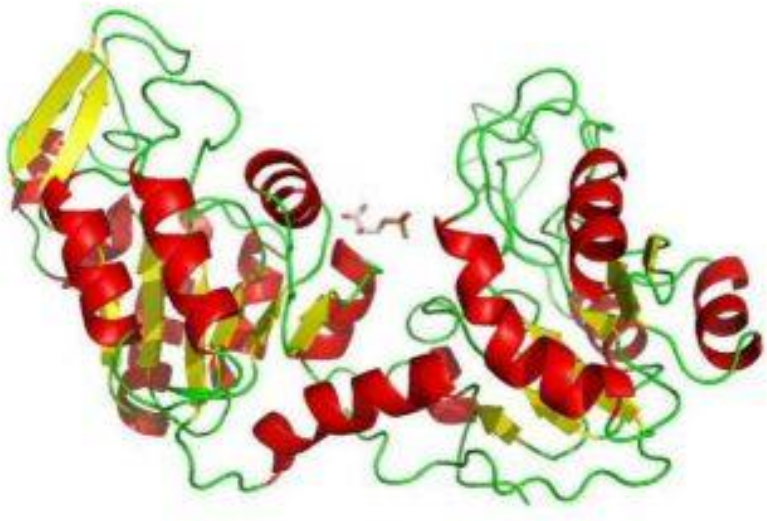


Tertiary Structure

- **Tertiary structure** is the overall three-dimensional arrangement of a **single polypeptide chain**.
- It describes how different secondary structural elements are positioned relative to one another.

For example:

- **α-helix + β-sheet + loops + turns → Complete 3D structure**
- Tertiary structure creates functional regions such as:
  - Active sites
  - Binding sites
  - Interaction surfaces

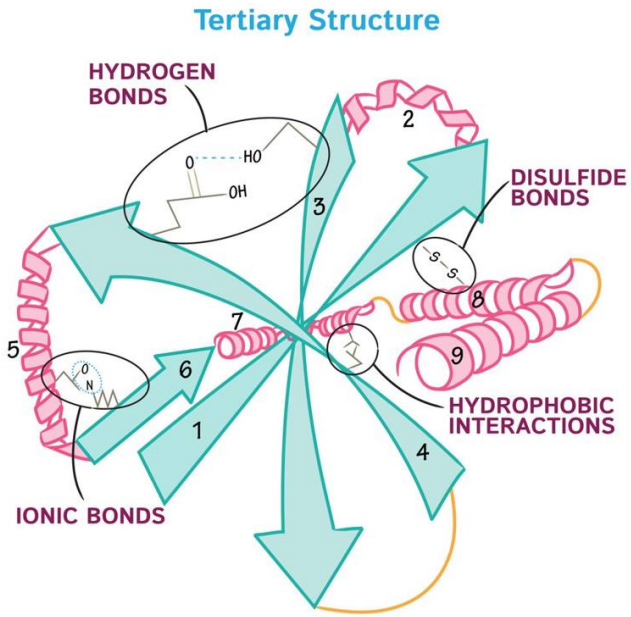


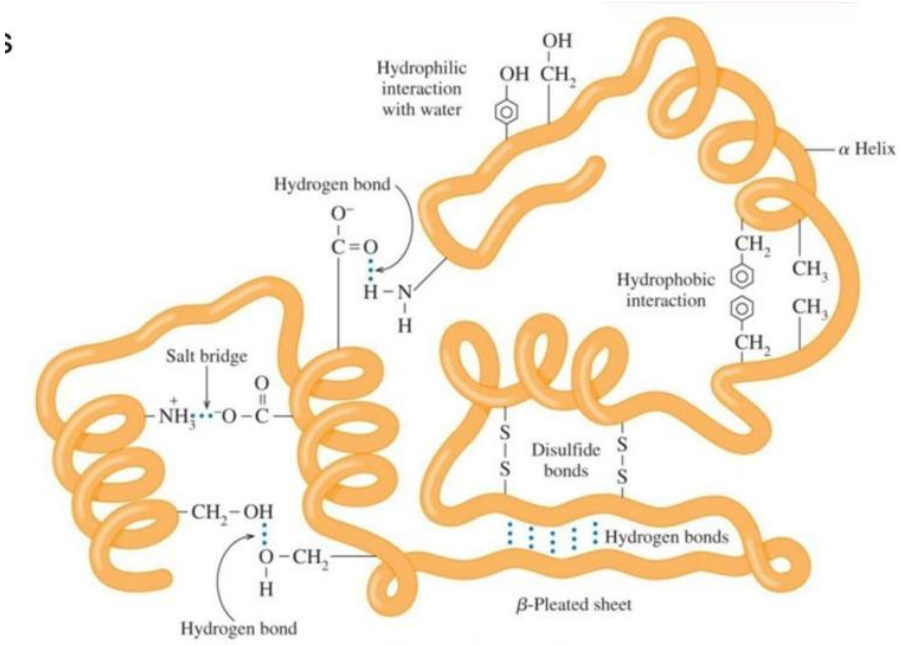
Factors Stabilising Tertiary Structure

- **Hydrophobic interactions**
  Non-polar groups tend to become buried within the protein.
- **Hydrogen bonds**
  Form between suitable polar groups.
- **Ionic interactions**
  Occur between oppositely charged groups.
- **Disulfide bonds**
  Can form between cysteine residues.

- **Van der Waals interactions**
  Contribute through close contact between atoms.

Figure



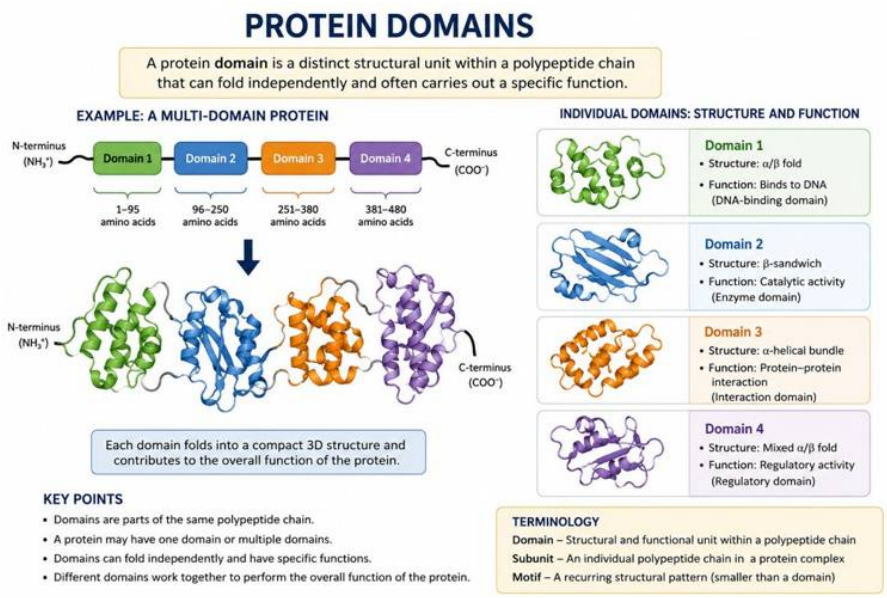


Protein Domain

- A **protein domain** is a distinct structural and often functional region within a protein.
- A protein may contain: **One domain or multiple domains**
- Different domains may perform different functions.

For example:

- **Domain 1 → Molecular binding**
- **Domain 2 → Catalytic activity**
- Domains are important for understanding the relationship between **protein structure and function**.



Quaternary Structure

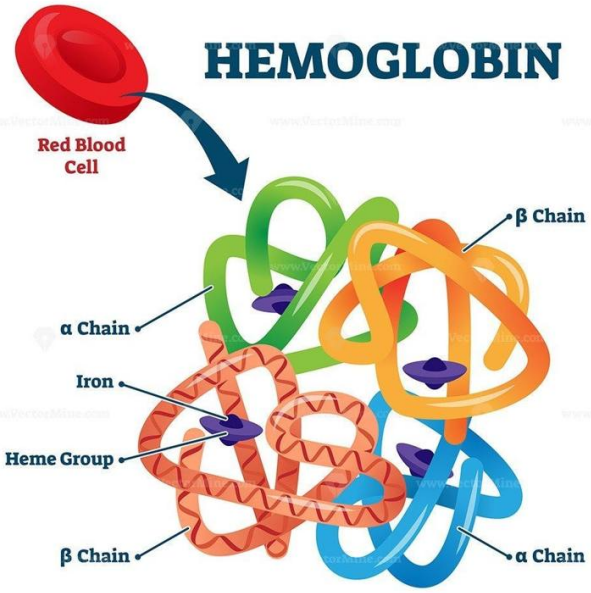
- Some proteins contain more than one polypeptide chain.
- The arrangement of these individual chains into a functional protein is called **quaternary structure**.
- The individual chains are called **subunits**.

For example:

- **Subunit A + Subunit B + Subunit C → Functional protein**
- Each subunit may have its own tertiary structure.

Example: Hemoglobin

- **Hemoglobin** is an important example of a protein with quaternary structure.
- It consists of four polypeptide subunits:
- **2 α subunits + 2 β subunits**
- Each subunit contains a **heme group**.
- The four subunits work together in the functional hemoglobin molecule, which is involved in **oxygen transport**.



A subunit is an individual polypeptide chain within a protein complex. A domain is a distinct structural and often functional region within a polypeptide chain. Hemoglobin has four polypeptide subunits—two α and two β. Each globin subunit has a characteristic globin structural fold and binds one heme group.

S3 PyMOL for 3D protein visualisation

2 MARKS PyMOL is a molecular visualization software used to view biological molecules in three dimensions.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Why Visualize Proteins?; From Protein Data to 3D Structure; SOURCE OF 3D structures; Protein Structure in PyMOL; Cartoon – Overall Protein Structure; Sticks – Atoms and Bonds.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. Why Visualize Proteins?
2. From Protein Data to 3D Structure
3. SOURCE OF 3D structures
4. Protein Structure in PyMOL
5. Cartoon – Overall Protein Structure
6. Sticks – Atoms and Bonds
7. Spheres – Individual Atoms
8. Surface – Outer Molecular Shape
9. Other tools for molecular visualizations
10. DIY- Explore a protein in a molecular visualization tool

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: What is PyMOL?
5 marks: Describe the display styles in PyMOL.
12 marks: Explain how protein structures are obtained and visualised in PyMOL.

Complete notes

Figure

- **PyMOL** is a molecular visualization software used to view biological molecules in **three dimensions**.
- It can be used to visualize:
  - Proteins
  - DNA and RNA
  - Small molecules
  - Protein–ligand complexes

Why Visualize Proteins?

- Protein function is closely related to its **three-dimensional structure**.
- A protein sequence alone does not easily show:
  - How the protein is folded
  - Where α-helices and β-sheets are located
  - Where ligands bind
  - How different chains are arranged
- 3D visualization helps us **see and understand protein structure**.

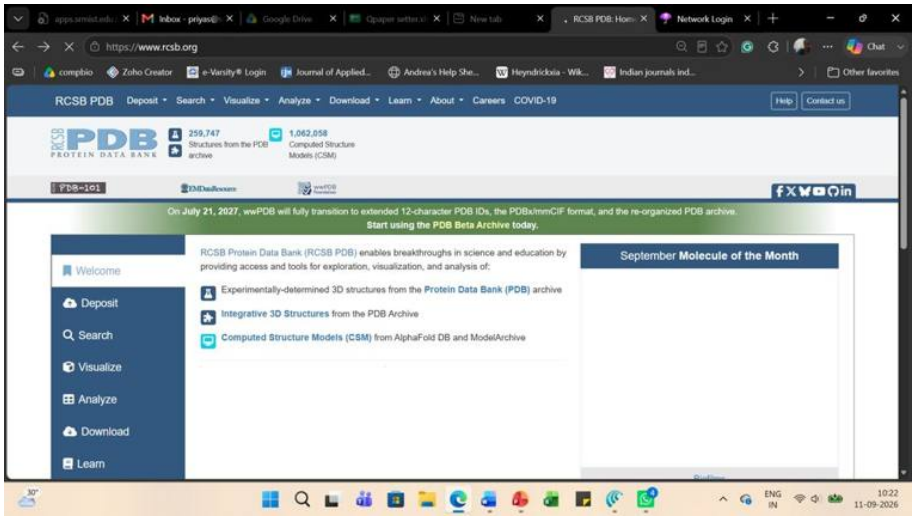
From Protein Data to 3D Structure

- A protein structure contains information about the **position of atoms in three-dimensional space**.
- Each atom has coordinates: **X – Y – Z**
- **The source of the coordinates for Protein Structure is the RCSB PDB**

- The **Protein Data Bank (PDB)** is a freely accessible database that stores the **three-dimensional structures of biological macromolecules**, mainly proteins, nucleic acids and their complexes.
- PyMOL uses these coordinates to display the molecule.

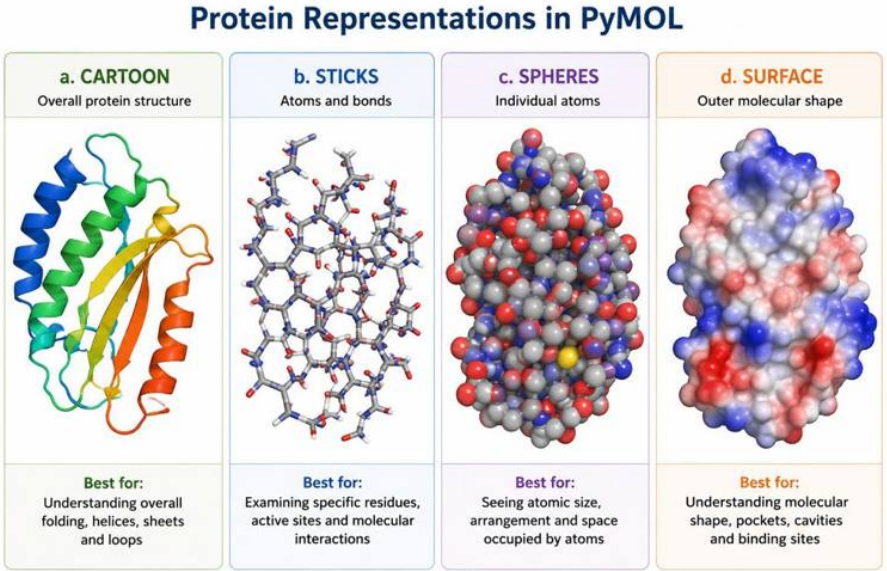
SOURCE OF 3D structures

- PDB-RCSB PDB: Homepage
- Search PDB → Select structure → Check structure information → Download PDB/mmCIF file → Visualize or analyse in Pymol
- Try the PDB code 1HHO- retrive the PDB file of Hemoglobin



Protein Structure in PyMOL

- PyMOL can display a protein using different representations.
- **Common representations:**
  - **Cartoon** – overall protein structure
  - **Sticks** – atoms and bonds
  - **Spheres** – individual atoms
  - **Surface** – outer molecular shape
- Different representations highlight different features of the same protein.
- Other than the said representations there are many more types of representations of proteins available in PyMol and other tools



Cartoon – Overall Protein Structure

- The **cartoon representation** is a simplified way of displaying the **overall folding pattern of a protein**.
- Instead of showing every atom, PyMOL represents the protein backbone using ribbons and arrows.
- **α-helices** appear as coiled or spiral structures.
- **β-sheets** appear as broad arrows.
- **Loops and turns** appear as flexible connecting lines or ribbons.
- Individual atoms are generally **not shown**.

Why is it useful?

- It allows us to quickly understand the **secondary structure and overall 3D organisation** of a protein without being distracted by thousands of atoms.

Sticks – Atoms and Bonds

- The **stick representation** shows individual atoms and the bonds connecting them.
- Atoms are represented as small shapes, while bonds appear as **cylindrical sticks** between atoms.
- For example, if we want to examine a small region of a protein:
- **Amino acid → atoms → bonds → molecular arrangement**
- Sticks provide much more detail than the cartoon representation.

Why is it useful?

- It is particularly useful when looking closely at:
- Specific amino acids
- Active-site residues
- Ligands
- Protein–ligand interactions
- Hydrogen-bonding or other close-contact regions

Spheres – Individual Atoms

- In the **sphere representation**, atoms are displayed as **spherical objects**.
- The size of the spheres can represent the approximate size or spatial occupation of atoms, depending on the display setting.
- When many atoms are shown as spheres, the protein can appear like a collection of tightly packed balls.

Why is it useful?

- It helps us appreciate:
- The physical space occupied by atoms
- Atomic arrangement
- How closely atoms are packed
- The approximate size and shape of individual atoms
- However, showing an entire large protein as spheres can make the structure look **crowded and difficult to interpret**.

Surface – Outer Molecular Shape

- The **surface representation** displays the approximate **outer boundary of the molecule**.
- Instead of seeing individual atoms and bonds, we see the molecular surface surrounding them.
- Imagine placing a flexible sheet around the outside of the protein—the resulting shape represents its molecular surface.
- It allows us to see features such as:
  - Protein shape

- Grooves
- Cavities
- Pockets
- Surface-exposed regions

Why is it useful?

- Surface representation is particularly helpful for understanding **where another molecule might interact with the protein**.
- For example, a drug molecule may fit into a pocket on the protein surface.

Other tools for molecular visualizations

- **UCSF ChimeraX** – advanced 3D visualization and structural analysis
- **Jmol** – interactive molecular visualization, commonly used for teaching
- **RasMol** – one of the older molecular visualization programs
- **Avogadro** – molecular structure building and visualization
- **iCn3D** – web-based 3D structure viewer from NCBI

DIY- Explore a protein in a molecular visualization tool.

- Go to https://www.ncbi.nlm.nih.gov/Structure/icn3d,
- Next add **1ZNI** as the code – its for insulin protein-press enter
- Goto Style□Protein□ribbon/strand/ schematic
- Try each one of those options to understand visualization of proteins. Explore other tabs to play with the coloring of the protein. Also using the middle key of the mouse zoom in and out of the protein

S4 Secondary structure prediction methods

2 MARKS Secondary structure prediction estimates, from the amino acid sequence alone, whether each residue forms an alpha-helix, a beta-sheet or a coil.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Secondary Structure Prediction; What Does a Prediction Tell Us?; Basic Principle of Prediction; What are Sequence-Based Methods?; Examples of Sequence-Based Methods; What are Homology-Based Methods?.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. Secondary Structure Prediction
2. What Does a Prediction Tell Us?
3. Basic Principle of Prediction
4. What are Sequence-Based Methods?
5. Examples of Sequence-Based Methods
6. What are Homology-Based Methods?
7. Examples of Homology-Based Approaches
8. What are Hybrid Methods?
9. Examples of Hybrid Predictors
10. DIY- Predict secondary structure of the proteins using NETSURFP
11. How Do We Know if a Prediction is Good?

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: What is secondary structure prediction?
5 marks: Explain sequence-based, homology-based and hybrid methods.
12 marks: Describe the methods of secondary structure prediction and how accuracy is judged.

Complete notes

Secondary Structure Prediction

- Proteins are long chains of amino acids that fold into specific three-dimensional structures.
- **Secondary structure prediction** is the computational prediction of local structural elements from an amino acid sequence.
- **Major structural elements**
  - **α-helix**
  - **β-strand / β-sheet**
  - **Loops and turns**

Classification of Prediction Methods

- Secondary structure prediction methods can be broadly classified according to the **type of information used for prediction**.
- **Three major approaches**
  - **Sequence-based methods**
    → Use information from the target amino acid sequence
  - **Homology-based methods**
    → Use information from related proteins with known structures
  - **Hybrid methods**
    → Combine sequence, evolutionary/homology information and computational learning

What Does a Prediction Tell Us?

Symbol	Structure
H	α-helix

Symbol	Structure
E	β-strand
C	Coil/loop

Sequence: M A L K E L G V A A T K L

Prediction: H H H H H H C E E E C C C

Basic Principle of Prediction

- Different amino acids have different tendencies to occur in different structural environments.
- Prediction methods examine:
  - Amino acid composition
  - Sequence patterns
  - Neighbouring residues
  - Evolutionary information
  - Conserved regions
- These features are used to estimate the most likely secondary structure at each position.

What are Sequence-Based Methods?

- Sequence-based methods predict secondary structure **directly from the amino acid sequence**.
- They examine properties and patterns within the sequence to predict whether a region is likely to form:
- **α-helix | β-strand | Coil**
- **Basic workflow**

Amino acid sequence

↓

Sequence properties / patterns

↓

Prediction algorithm

↓

Secondary structure

- These methods do not require a structurally characterized homolog.

Examples of Sequence-Based Methods

Classical methods

- **Chou–Fasman** – uses amino acid structural propensities
- **GOR (Garnier–Osguthorpe–Robson)** – considers neighbouring residues and statistical information

Machine-learning approaches

- SVM-based predictors
- Neural-network-based predictors
- **Main advantage**
- They can make predictions **even when a suitable homologous structure is unavailable**.
- **Limitation**
- A single sequence may not contain enough information to accurately predict every structural feature.

What are Homology-Based Methods?

- Homology-based methods use information from **related proteins whose structures are already known**.

- The underlying principle is:
- **Similar protein sequences often have similar structural features.**
- **Basic workflow**

Target protein sequence

↓

Search for related proteins

↓

Identify homolog with known structure

↓

Transfer structural information

↓

Predict secondary structure

Examples of Homology-Based Approaches

Examples of tools/resources that use homologous or evolutionary information include:

- **PHDsec / PHD**
- **PredictProtein**
- **HHpred**
- **Major advantage**
- When a **close structural homolog** is available, prediction can be highly reliable.
- **Major limitation**
- Performance decreases when there is **no suitable homolog** or when sequence similarity is very low.

What are Hybrid Methods?

- Hybrid methods combine information from **multiple sources** rather than relying on a single type of information.
- They may combine:
  - Amino acid sequence
  - Local sequence patterns
  - Evolutionary information
  - Multiple sequence alignments
  - Sequence profiles
  - Machine learning/deep learning
- **More information → better representation of the protein → improved prediction**

Examples of Hybrid Predictors

Examples include:

- **PSIPRED**
- **JPred**
- **NetSurfP**
- **DeepCNF**
- **SPOT-1D**
- These approaches use combinations of sequence-derived, evolutionary and machine-learning information.
- **Advantages**
  - Combines complementary information
  - Generally more powerful than relying on one information source

- Well suited to modern ML/deep-learning approaches

- **Limitations**
  - More computationally complex
  - Dependent on the quality of evolutionary/profile information

DIY- Predict secondary structure of the proteins using NETSURFP

- Go to NetSurfP 3.0 - DTU Health Tech - Bioinformatic Services
- Paste this sequence

>2WNS.A.1 TRANSFERASE

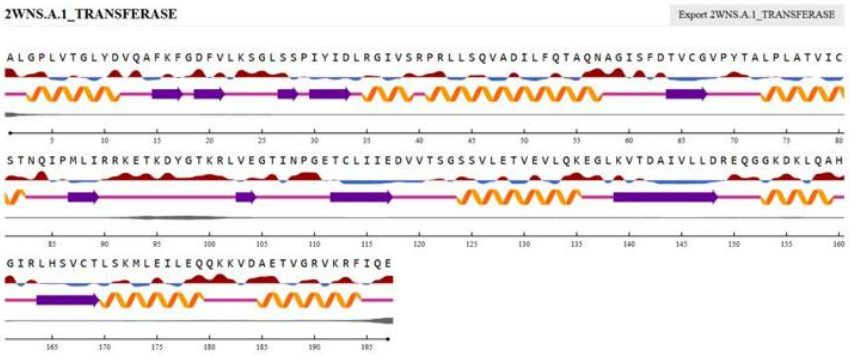
ALGPLVTGLYDVQAFKFGDFVLKSGLSSPIYIDLRGIVSRPRLLSQVADILFQTAQNAGIS

FDTVCGVPYTALPLATVICSTNQIPMLIRRKETKDYGTKRLVEGTINPGETCLIIEDVVTS

GSSVLETVEVLQKEGLKVTDAIVLLDREQGGKDKLQAHGIRLHSVCTLSKMLEILEQQKKV

DAETVGRVKRFIQE

3. Press submit and visualize the result



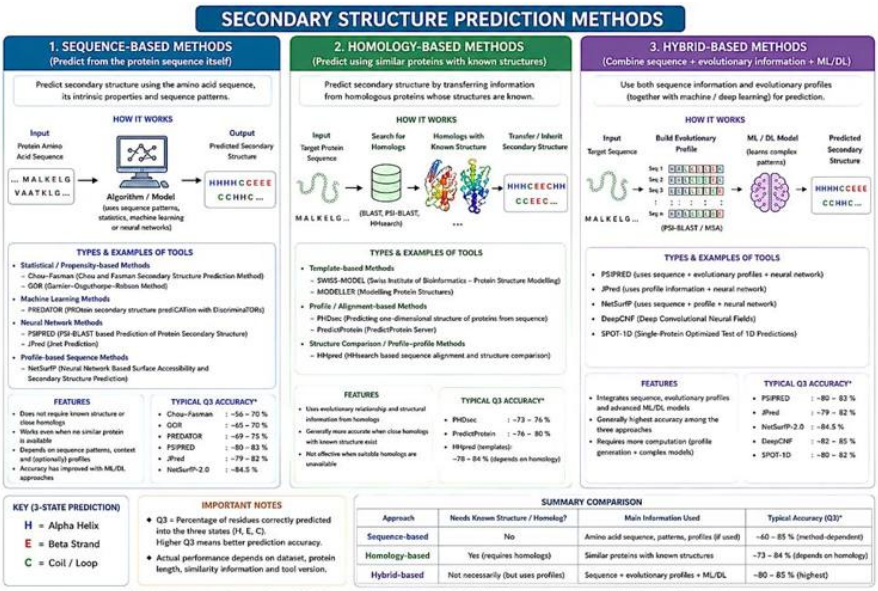
How Do We Know if a Prediction is Good?

- Predictions are evaluated by comparing them with experimentally determined structures.
- One commonly used measure is **Q3 accuracy**.
- **Q3 accuracy** Measures how many residues are correctly classified into: **H → α-helix E → β-strand C → coil**

For example:

- **85 correctly predicted residues / 100 residues**
- **→ Q3 = 85%**

Figure


SECONDARY STRUCTURE PREDICTION METHODS
1. SEQUENCE-BASED METHODS
(Predict from the protein sequence itself)
Predict secondary structure using the amino acid sequence, its intrinsic properties and sequence patterns.
HOW IT WORKS
Input
Protein Amino Acid Sequence
... MALKELG
VAATKLG ...
Algorithm / Model
(uses sequence patterns, statistics, machine learning or neural networks)
Output
Predicted Secondary Structure
HHHHCCEEEE
CCHHC ...
TYPES & EXAMPLES OF TOOLS
Statistical / Propensity-based Methods
– Chou–Fasman (Chou and Fasman Secondary Structure Prediction Method)
– GOR (Garnier–Osguthorpe–Robson Method)
Machine Learning Methods
– PREDATOR (PREdiction secondary structure predICAFion with DiscriminaTORs)
Neural Network Methods
– PSIPRED (PSI-BLAST based Prediction of Protein Secondary Structure)
– JPred (Jnet Prediction)
Profile-based Sequence Methods
– NetSurfP (Neural Network Based Surface Accessibility and Secondary Structure Prediction)
FEATURES
Does not require known structure or close homologs
Works even when no similar protein is available
Depends on sequence patterns, context and (optionally) profiles
Accuracy has improved with ML/DL approaches
TYPICAL Q3 ACCURACY*
Chou–Fasman : ~56 – 70 %
GOR : ~65 – 70 %
PREDATOR : ~69 – 75 %
PSIPRED : ~80 – 83 %
JPred : ~79 – 82 %
NetSurfP-2.0 : ~84.5 %
2. HOMOLOGY-BASED METHODS
(Predict using similar proteins with known structures)
Predict secondary structure by transferring information from homologous proteins whose structures are known.
HOW IT WORKS
Input
Target Protein Sequence
Search for Homologs
Homologs with Known Structure
Transfer / Inherit Secondary Structure
MALKELG ...
(BLAST, PSI-BLAST, HHsearch)
...
HHHCEECHH
CCEEC ...
TYPES & EXAMPLES OF TOOLS
Template-based Methods
– SWISS-MODEL (Swiss Institute of Bioinformatics – Protein Structure Modelling)
– MODELLER (Modelling Protein Structures)
Profile / Alignment-based Methods
– PHDsec (Predicting one-dimensional structure of proteins from sequence)
– PredictProtein (PredictProtein Server)
Structure Comparison / Profile–profile Methods
– HHpred (HHsearch based sequence alignment and structure comparison)
FEATURES
Uses evolutionary relationship and structural information from homologs
Generally more accurate when close homologs with known structure exist
Not effective when suitable homologs are unavailable
TYPICAL Q3 ACCURACY*
PHDsec : ~73 – 76 %
PredictProtein : ~76 – 80 %
HHpred (templates): ~78 – 84 % (depends on homology)
3. HYBRID-BASED METHODS
(Combine sequence + evolutionary information + ML/DL)
Use both sequence information and evolutionary profiles (together with machine / deep learning) for prediction.
HOW IT WORKS
Input
Target Sequence
Build Evolutionary Profile
ML / DL Model
(learns complex patterns)
Predicted Secondary Structure
MALKELG ...
Seq 1
Seq 2
Seq 3
Seq n
(PSI-BLAST / MSA)
HHHHCCEEEE
CCHHC ...
TYPES & EXAMPLES OF TOOLS
PSIPRED (uses sequence + evolutionary profiles + neural network)
JPred (uses profile information + neural network)
NetSurfP (uses sequence + profile + neural network)
DeepCNF (Deep Convolutional Neural Fields)
SPOT-1D (Single-Protein Optimized Test of 1D Predictions)
FEATURES
Integrates sequence, evolutionary profiles and advanced ML/DL models
Generally highest accuracy among the three approaches
Requires more computation (profile generation + complex models)
TYPICAL Q3 ACCURACY*
PSIPRED : ~80 – 83 %
JPred : ~79 – 82 %
NetSurfP-2.0 : ~84.5 %
DeepCNF : ~82 – 85 %
SPOT-1D : ~80 – 82 %
KEY (3-STATE PREDICTION)
H = Alpha Helix
E = Beta Strand
C = Coil / Loop
IMPORTANT NOTES
Q3 = Percentage of residues correctly predicted into the three states (H, E, C). Higher Q3 means better prediction accuracy.
Actual performance depends on dataset, protein length, similarity information and tool version.
SUMMARY COMPARISON
Approach | Needs Known Structure / Homolog? | Main Information Used | Typical Accuracy (Q3)*
Sequence-based | No | Amino acid sequence, patterns, profiles (if used) | ~60 – 85 % (method-dependent)
Homology-based | Yes (requires homologs) | Similar proteins with known structures | ~73 – 84 % (depends on homology)
Hybrid-based | Not necessarily (but uses profiles) | Sequence + evolutionary profiles + ML/DL | ~80 – 85 % (highest)
*Q3 values are representative ranges from published reports and benchmarks (e.g., CB513, CASP). They may vary with dataset, tool version and evaluation criteria.

S5 Python programming for structure analysis

2 MARKS Biopython is an open-source Python library designed for working with biological data.
4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Important Biopython Modules for Structural Analysis; Biopython IDE- example program; Protein; Number of chains; Chain IDs.
12 TO 13 MARKS: ANSWER IN THIS ORDER 1. Important Biopython Modules for Structural Analysis 2. Biopython IDE- example program 3. Protein 4. Number of chains 5. Chain IDs *Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.*
PRACTICE QUESTIONS 2 marks: What is Biopython? 5 marks: List important Biopython modules and their uses. 12 marks: Explain Python programming for structure analysis with an example program.

Complete notes

Figure

- **Biopython** is an open-source Python library designed for working with biological data. It provides modules for handling **DNA, RNA, protein sequences and protein structures**.
- For structural bioinformatics, the **Bio.PDB** module is particularly useful. It allows a PDB structure downloaded from the Protein Data Bank to be read and analysed using Python.
- With Biopython, we can:
  - Read **PDB/mmCIF structure files**
  - Access protein **models, chains, residues and atoms**
  - Obtain amino acid and atom information
  - Work with **3D coordinates**
  - Calculate distances between atoms
  - Identify residues around ligands
  - Perform basic structural analysis

Important Biopython Modules for Structural Analysis

Biopython Module	Main Use
Bio.PDB	Reading, analysing and manipulating protein 3D structures from PDB/mmCIF files
Bio.PDB.DSSP	Obtaining secondary-structure information such as α-helices, β-sheets and coils from structures
Bio.PDB.Superimposer	Superimposing two protein structures and calculating structural differences
Bio.PDB.NeighborSearch	Finding atoms or residues that are spatially close to one another

Biopython IDE- example program

```
from Bio.PDB import PDBParser

from urllib.request import urlopen

from io import StringIO
```

Download the PDB structure

url = "https://files.rcsb.org/download/1LYZ.pdb"

data = urlopen(url).read().decode("utf-8")

Read the structure

parser = PDBParser(QUIET=True)

structure = parser.get_structure("Lysozyme", StringIO(data))

Get the first model

model = structure[0]

Count the chains

chains = list(model.get_chains())

print("Protein: Lysozyme")

print("Number of chains:", len(chains))

Display chain names

print("Chain IDs:")

for chain in chains:

print(chain.id)

Expected output

Protein: Lysozyme

Number of chains: 1

Chain IDs: A

try the code here:

Online Python environment-python-fiddle.com/new?checkpoint=1789107121-

Integrated Development Environment (IDE)

S6 Basics for drug design

2 MARKS Drug design is the process of discovering and developing molecules that can interact with a specific biological target to produce a desired therapeutic effect.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Introduction to Drug Design; Drug Targets and Drug–Target Interactions; Approaches and Computational Methods; From Drug Candidate to Drug.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. Introduction to Drug Design
2. Drug Targets and Drug–Target Interactions
3. Approaches and Computational Methods
4. From Drug Candidate to Drug

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: Define a drug target.
5 marks: Explain the approaches of computational drug design.
12 marks: Describe the basics of drug design, from target to approved drug.

Complete notes

Introduction to Drug Design

- Drug design is the process of discovering and developing molecules that can interact with a specific biological target to produce a desired therapeutic effect.
- Most drug targets are biological molecules such as proteins, enzymes, receptors and ion channels.
- The basic concept is:
- Disease → Identify biological target → Find/design drug molecule → Drug–target interaction → Desired therapeutic effect
- A drug must not only bind to its target but should ideally show good potency, selectivity, safety and drug-like properties.

Drug Targets and Drug–Target Interactions

- A drug target is a biological molecule whose activity can be modified by a drug to produce a beneficial effect.
- Common targets(mostly proteins) include
  - Enzymes – drugs may inhibit or activate their activity
  - Receptors – drugs can activate or block cellular signalling
  - Ion channels – drugs can influence the movement of ions
  - Transporters – drugs can alter the movement of molecules across membranes
- Drugs interact with their targets through interactions such as hydrogen bonds, hydrophobic interactions, electrostatic interactions and van der Waals forces.
- A binding site is the region of the target where the drug interacts. Its shape and chemical properties strongly influence drug binding.

Approaches and Computational Methods

Two important approaches are used in drug design.

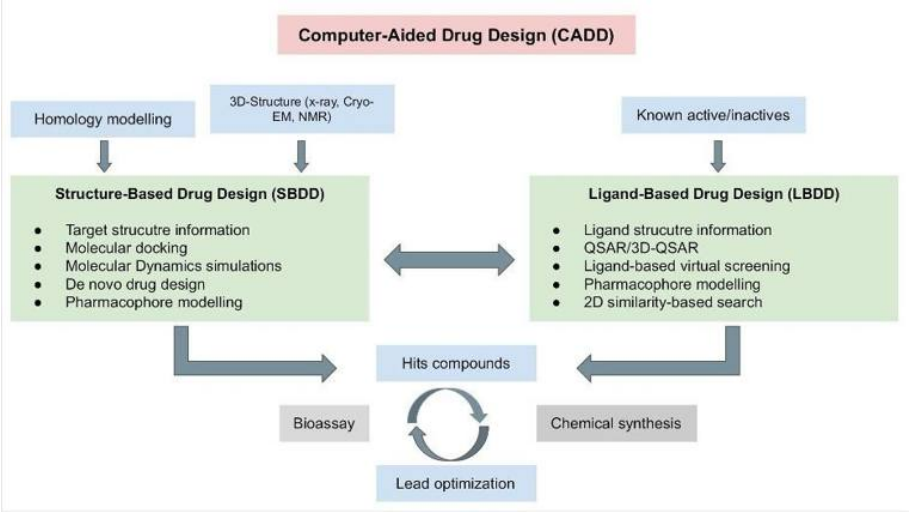
- **Structure-Based Drug Design (SBDD)**
- Uses the 3D structure of the target protein to identify binding pockets and design or select molecules that can interact with them.
- Target structure → Binding site → Compound screening/design → Docking → Candidate molecules
- **Ligand-Based Drug Design (LBDD)**

- Used when the target structure is unavailable but compounds with known biological activity are available.
- Known active compounds → Identify common features → Build model → Design/search for new compounds

Computational techniques

- Molecular docking – predicts possible drug–target binding poses
- Virtual screening – computationally screens large compound libraries
- Molecular dynamics – studies molecular movement and stability
- QSAR – relates chemical properties to biological activity
- AI/ML – predicts activity and assists in compound discovery

Figure



From Drug Candidate to Drug

Drug design is part of a much larger **drug discovery pipeline**.

Target identification
↓
Target validation
↓
Hit identification
↓
Lead identification
↓
Lead optimization
↓
Preclinical testing
↓
Clinical trials
↓
Regulatory approval

Unit 4

AI/ML Foundations in Biology

Learn the machine learning basics needed to work with biological data.

S1 Overview of AI for biological big data

2 MARKS Artificial Intelligence (AI) refers to computer systems capable of performing tasks that involve learning, pattern recognition, prediction and decision-making.
4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Introduction to AI in Biology; Why Does Biology Need AI?; What is Biological Big Data?; AI, ML and Deep Learning; AI is used across multiple biological disciplines.
12 TO 13 MARKS: ANSWER IN THIS ORDER 1. Introduction to AI in Biology 2. Why Does Biology Need AI? 3. What is Biological Big Data? 4. AI, ML and Deep Learning 5. AI is used across multiple biological disciplines *Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.*
PRACTICE QUESTIONS 2 marks: Define biological big data. 5 marks: Differentiate AI, ML and deep learning. 12 marks: Explain the role of AI in biological big data with examples.

Complete notes

Introduction to AI in Biology

- **Artificial Intelligence (AI)** refers to computer systems capable of performing tasks that involve learning, pattern recognition, prediction and decision-making.
- Biology is increasingly becoming a **data-intensive science**.
- Examples of biological data include:
  - DNA and RNA sequences
  - Protein sequences and structures
  - Gene-expression data
  - Medical images
  - Clinical records
  - Molecular structures
- **AI provides computational methods to extract useful information from these large datasets.**

Why Does Biology Need AI?

- Modern biological experiments generate data at a scale that is difficult to analyse manually.
- For example:
  - **Genome sequencing → Millions/Billions of DNA bases**
  - **Gene-expression experiment → Thousands of genes**
  - **Medical imaging → Millions of pixels**
  - **Drug screening → Thousands/Millions of compounds**
- AI can help:
  - Process large datasets
  - Identify hidden patterns

- Classify biological samples
- Make predictions
- Automate repetitive analysis

What is Biological Big Data?

Biological big data refers to **large, complex and diverse datasets generated from biological and healthcare studies**.

The data may differ in:

- **Volume** – very large amounts of data
- **Variety** – sequences, images, numbers, clinical records
- **Velocity** – data may be generated continuously
- **Veracity** – data may contain errors or uncertainty

Examples

- Genome databases
- Protein databases
- Gene-expression datasets
- Electronic health records
- Medical imaging repositories

AI, ML and Deep Learning

These terms are related but not identical.

Artificial Intelligence

Broad field concerned with intelligent computational systems.

↓

Machine Learning

AI systems that learn patterns from data.

↓

Deep Learning

Machine learning based on multilayer neural networks.

Relationship

AI ⊃ Machine Learning ⊃ Deep Learning

Deep learning is therefore **one approach within machine learning**, and machine learning is one approach within AI.

AI is used across multiple biological disciplines.

Area	Example AI application
Genomics	Variant prediction
Proteomics	Protein-function prediction
Structural biology	Structure prediction
Drug discovery	Virtual screening
Medical imaging	Image classification
Systems biology	Network analysis
Healthcare	Risk prediction

S2 Supervised and unsupervised learning basics

2 MARKS For biological applications, supervised and unsupervised learning are especially important.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: What is Machine Learning?; MACHINE LEARNING FOUNDATIONS; Supervised Learning; Classification and Regression; What is Classification?; Multiclass classification.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. What is Machine Learning?
2. MACHINE LEARNING FOUNDATIONS
3. Supervised Learning
4. Classification and Regression
5. What is Classification?
6. Multiclass classification
7. Common Classification Algorithms
8. What is Regression?
9. Suppose we want to predict a person's blood glucose level
10. Classification vs Regression
11. Unsupervised Learning
12. Applications
13. Clustering in Biology
14. Basic Idea of Clustering

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: Define supervised learning.
5 marks: Differentiate classification and regression.
12 marks: Explain supervised and unsupervised learning, with k-means and hierarchical clustering.

Complete notes

What is Machine Learning?

Machine Learning (ML) allows a computer to learn patterns from data rather than requiring every rule to be explicitly programmed.

Traditional programming:

Rules + Data → Output

Machine learning:

Data + Known examples → Learning algorithm → Model

Then:

New data → Trained model → Prediction

This makes ML particularly useful for biological systems where the underlying relationships can be extremely complex.

MACHINE LEARNING FOUNDATIONS

Major machine-learning approaches include:

- **Supervised Learning**
- Learns from labelled examples.
- **Unsupervised Learning**
- Discovers patterns in unlabelled data.
- **Reinforcement Learning**
- Learns actions through rewards and penalties.

- For biological applications, **supervised and unsupervised learning** are especially important.

Supervised Learning

Input	Label
Gene expression	Cancer
Gene expression	Normal
Gene expression	Cancer
Gene expression	Normal

Supervised learning uses **labelled training data**.

Example:

The model learns the relationship between the input features and known labels.

It can then classify a new sample.

Applications

- Disease classification
- Protein-function prediction
- Drug activity prediction
- Medical image classification

Classification and Regression

Two common supervised-learning tasks are:

Classification

Predicts a category.

Examples:

- Tumour → Benign / Malignant
- Patient → High risk / Low risk

Regression

Predicts a numerical value.

Examples:

- Predicting drug activity
- Predicting protein expression
- Predicting disease-risk score

What is Classification?

Classification is a supervised learning problem in which the model predicts which **predefined category** an input belongs to.

The training data contain **inputs + known class labels**.

Binary classification

Only **two possible categories**.

- Examples:
- Cancer / Normal
- Diseased / Healthy
- Toxic / Non-toxic
- Active / Inactive

Multiclass classification

More than two categories.

For example, classifying a cell into:

Neuron / Muscle cell / Liver cell / Immune cell-The model selects one of the predefined classes.

Multilabel classification-An observation can belong to **multiple categories simultaneously**.

For example, a protein could be associated with:

DNA binding + Nuclear localization + Enzyme activity

Common Classification Algorithms

Some commonly used algorithms are:

Logistic Regression

Decision Trees

Random Forest

Support Vector Machines (SVM)

k-Nearest Neighbours (k-NN)

Neural Networks

Deep-learning models

What is Regression?

Regression is a supervised learning problem in which the model predicts a **continuous numerical value**.

Instead of asking:

"Which category does this sample belong to?"

we ask:

"What numerical value should be predicted?“

Suppose we want to predict a person's blood glucose level.

We give the model information such as:

Age

Weight

Previous glucose level

Activity level

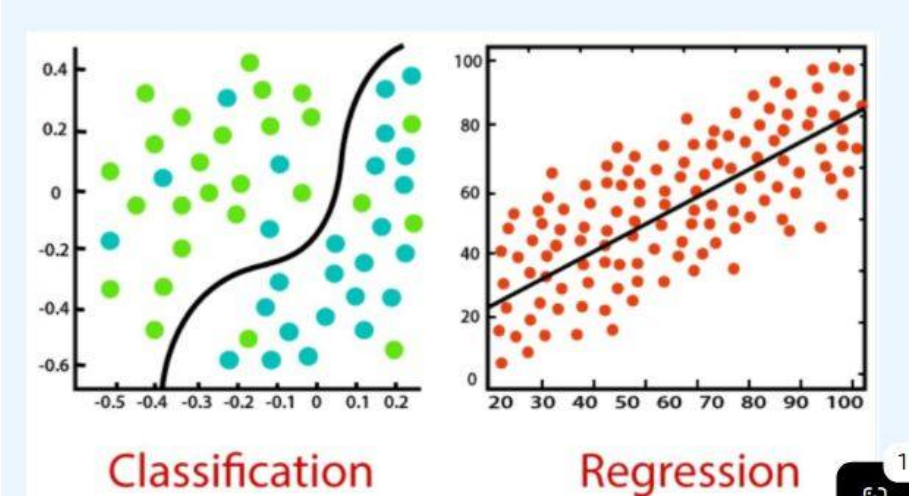
The model may predict:

Blood glucose = 125 mg/dL

The answer is a **number**, so this is regression.

Classification vs Regression

Feature	Classification	Regression
Output	Category	Numerical value
Type	Discrete	Continuous
Example	Cancer / Normal	Blood glucose = 125 mg/dL
Output example	Cancer	78.5
Common use	Diagnosis	Prediction of measurement



Unsupervised Learning

Unsupervised learning works with **unlabelled data**.

The algorithm attempts to discover natural patterns.

Example:

Gene-expression data

↓

Clustering

↓

Patient Group 1

Patient Group 2

Patient Group 3

The researcher may then investigate whether these groups correspond to different biological subtypes.

Applications

Patient clustering

Gene-expression analysis

Molecular subtype discovery

Biological pattern discovery

Clustering in Biology

Clustering groups similar data points together.

Suppose researchers have gene-expression data from 500 patients.

The algorithm may identify:

Cluster 1 → Similar expression pattern

Cluster 2 → Different expression pattern

Cluster 3 → Another expression pattern

Researchers can then investigate the biological significance of these groups.

Common approaches include:

K-means clustering

Hierarchical clustering

Basic Idea of Clustering

Gene B

|

90| ● P5

| ● P4 ● P6

|

|

20| ● P2

| ● P1 ● P3

|

+---------- Gene A

- The objective of clustering is to allow the **computer to discover these groups automatically**.

We can visually see two groups.

What Does "Similarity" Mean?

Clustering requires some way to measure how similar or different two observations are.

A common measure is **distance**.

Small distance → More similar

Large distance → Less similar

In biological datasets, the definition of similarity depends on the type of data.

What is K-Means Clustering?

K-Means is one of the simplest and most widely used clustering algorithms.

The **K** represents the number of clusters we want to create.

For example:

K = 2 → Two clusters

K = 3 → Three clusters

K = 5 → Five clusters

The algorithm attempts to divide the data into **K groups** such that points within the same group are relatively similar.

K-Means Algorithm – Basic Steps

Step 1 – Choose K

Decide how many clusters you want.

For example:

K = 2

S**tep 2 – Select Initial Centroids(average position of all the points belonging to that cluster**.)

The algorithm initially chooses two points as cluster centres.

● = Data point

X = Initial centroid

● ●

X X

● ●

Step 3 – Assign Points to the Nearest Centroid

For every data point:

Calculate distance to each centroid

↓

Choose the closest centroid

↓

Assign the point to that cluster

For example:

Cluster 1 Cluster 2

● ● ● ● ● ●

● ●

Step 4 – Recalculate the Centroids

Once points have been assigned, calculate the new centre of each cluster.

The centroid moves toward the centre of the points.

Then:

Reassign points → Recalculate centroids →

This continues until the clusters become stable.

Advantages of K-Means

Simple

Easy to understand and implement.

Fast

Generally computationally efficient for large datasets.

Scalable

Can be applied to relatively large datasets.

Easy to visualize

Clusters can be plotted when the data have two or three dimensions.

Useful for exploration

Can help researchers discover previously unknown groups.

Limitations of K-Means

1. K must be chosen

The algorithm requires the number of clusters to be specified.

2. Sensitive to initialization

Different starting centroids can sometimes produce different results.

3. Sensitive to outliers

Extreme data points can influence the centroid.

4. Assumes certain cluster shapes

K-Means works particularly well when clusters are relatively compact and separated.

5. Scaling matters

Features measured on very different scales can dominate the distance calculation.

For biological data, **normalization or appropriate scaling** may therefore be important.

Hierarchical clustering

Hierarchical clustering is an unsupervised learning method that groups similar data points and creates a **tree-like structure of clusters**.

Unlike K-means, we **do not need to specify the number of clusters at the beginning**.

The result is represented as a **dendrogram**.

Basic process:

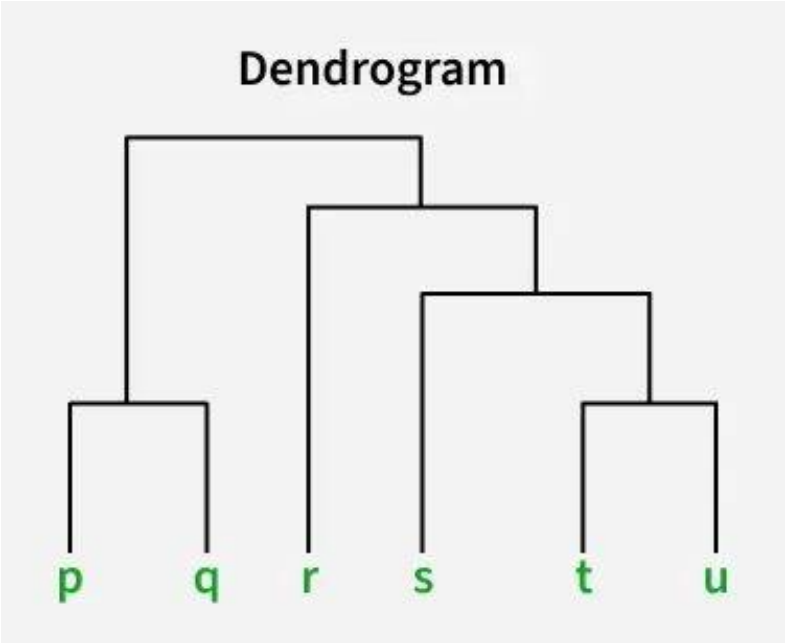
Start with each data point as a separate cluster.

Find the two most similar clusters.

Merge them into one cluster.

Continue merging until all points form one large cluster.

Cut the dendrogram at a chosen level to obtain the desired groups.



Two approaches

- **Agglomerative clustering**
  - Starts with individual data points.
  - Gradually **merges** similar clusters.
  - Most commonly used.
- **Divisive clustering**
  - Starts with one large cluster.
  - Gradually **splits** it into smaller clusters.

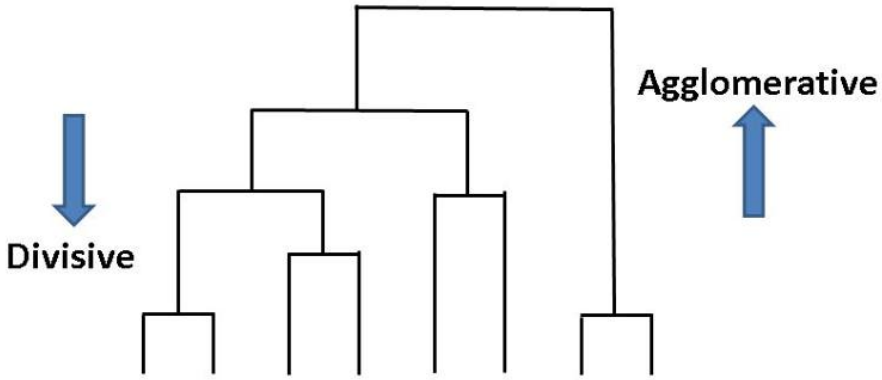
Biological applications

Grouping genes with similar expression

Clustering patients based on molecular profiles

Grouping similar species or organisms

Analysing patterns in omics datasets



Figure

Feature	Supervised	Unsupervised
Labels	Required	Not required
Goal	Prediction	Pattern discovery
Example	Cancer classification	Patient clustering
Output	Known categories/values	Groups/patterns

S3 Feature representations for sequences and omics

2 MARKS Feature representation means converting biological data into a form that a machine-learning algorithm can understand and process.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Feature Representation; Representing DNA and RNA Sequences; B. k-mer representation; C. Representing Protein Sequences; What is Omics Data?.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. Feature Representation
2. Representing DNA and RNA Sequences
3. B. k-mer representation
4. C. Representing Protein Sequences
5. What is Omics Data?

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: What is one-hot encoding?
5 marks: Explain k-mer representation.
12 marks: Explain feature representation for DNA, RNA, proteins and omics data.

Complete notes

Feature Representation

Feature representation means converting biological data into a form that a machine-learning algorithm can understand and process.

Biological data such as DNA sequences, protein sequences, gene-expression values, or metabolomics data are usually not directly suitable as ML inputs. They must first be represented as **numerical features**.

Feature representation is the process of converting biological information into numerical form so that an AI or machine-learning algorithm can process it.

Representing DNA and RNA Sequences

A. One-hot encoding

Each nucleotide is represented using a binary vector.

For example:

ATGC becomes:

This representation does not assume that one nucleotide is numerically greater or smaller than another.

Applications:

DNA classification

Promoter prediction

Enhancer prediction

Variant analysis

RNA sequence analysis

Base	Representation
A	1 0 0 0
T	0 1 0 0
G	0 0 1 0

Base	Representation
C	0 0 0 1

B. k-mer representation

A **k-mer** is a sequence of **k consecutive nucleotides**.

For example, for: ATGCGT

the 2-mers are:

AT, TG, GC, CG, GT

The algorithm can count how frequently different k-mers occur.

For DNA, if **k = 2**, there are:

4^2 = 16 possible 2-mers

Thus, a sequence can be represented by a vector containing the frequencies of its k-mers.

Applications:

- Genome classification
- Species identification
- Promoter prediction
- Regulatory sequence analysis
- Microbial classification

C. Representing Protein Sequences

Protein sequences contain **20 standard amino acids**.

Example:

MKTLLV...

Each amino acid can be represented using different features.

Simple representation

Each amino acid can be converted into numerical properties such as:

Molecular weight, Hydrophobicity, Charge, Polarity, amino acid composition

For example, a protein can be represented by the **percentage of each amino acid**.

This produces a **20-dimensional feature vector**:

[A%, R%, N%, D%, C%, ..., Y%, V%]

This is called **amino-acid composition**.

What is Omics Data?

Omics refers to large-scale biological datasets that measure many biological molecules simultaneously.

The major challenge is that omics datasets can contain **thousands or millions of features**.

For example, a gene-expression dataset may contain:

Samples × Genes

500 patients × 20,000 genes

This means there are 20,000 potential features for each patient.

The ML model can use them to predict something such as:

- Disease vs normal
- Cancer subtype
- Drug response
- Patient outcome

S4 AI data pipeline: preprocess, train, evaluate

2 MARKS Preprocessing is the process of cleaning and preparing raw biological data before using it for machine learning.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Data Pipeline – Overview; Data Preprocessing; Feature Representation and Data Splitting; Step 3 – Model Training; Step 4 – Model Evaluation; Step 5: Prediction.

12 TO 13 MARKS: ANSWER IN THIS ORDER
1. Data Pipeline – Overview
2. Data Preprocessing
3. Feature Representation and Data Splitting
4. Step 3 – Model Training
5. Step 4 – Model Evaluation
6. Step 5: Prediction

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS
2 marks: What is data preprocessing?
5 marks: Write the steps in an AI data pipeline.
12 marks: Explain the AI data pipeline with the diabetes-prediction example.

Complete notes

Data Pipeline – Overview

An AI model does not directly learn from raw biological data. The data must pass through a series of steps before the model can make reliable predictions.

Basic AI pipeline:

- **Preprocess:** Clean and prepare the data.
- **Feature representation:** Convert biological information into suitable numerical features.
- **Train:** Use training data to learn patterns.
- **Evaluate:** Test how well the trained model performs on unseen data.
- **Prediction:** Apply the trained model to new biological samples.

The quality of each step affects the final performance of the AI model.

Raw Data →

Preprocess →

Feature Representation →

Train →

Evaluate →

Prediction

Data Preprocessing

Preprocessing is the process of cleaning and preparing raw biological data before using it for machine learning.

Biological datasets may contain:

- Missing values
- Duplicate observations
- Measurement errors
- Noise
- Different scales between variables

- Inconsistent formats

Common preprocessing steps include:

- Removing or handling missing values
- Removing duplicate or incorrect data
- Normalizing or standardizing numerical values
- Encoding categorical information
- Checking for outliers
- Ensuring that all samples are represented consistently

Feature Representation and Data Splitting

After preprocessing, biological information is converted into **features** that can be used by the AI model.

Examples:

DNA → k-mer frequencies or one-hot encoding

Protein → amino-acid features or embeddings

Gene expression → numerical expression values

Clinical data → age, laboratory values, treatment history, etc.

The dataset is then divided into different subsets:

Training set → Used to teach the model

Validation set → Used to tune and select the model

Test set → Used for final evaluation

A common approach is to use approximately **70–80% for training** and the remaining data for validation/testing, although the exact proportions depend on the dataset and study design.

Step 3 – Model Training

During **training**, the AI model learns patterns from the training data.

The model receives:

Input features → Model → Prediction

The prediction is compared with the known answer, called the **target or label**.

The model calculates an **error (loss)** and adjusts its parameters to reduce this error.

This process is repeated many times until the model learns useful patterns.

Example:

Step 4 – Model Evaluation

After training, the model must be tested using **unseen data**.

This determines whether the model has learned general patterns rather than simply memorizing the training data.

For **classification**, commonly used measures include:

Accuracy – overall proportion of correct predictions

Precision – how many predicted positives were actually positive

Recall/Sensitivity – how many actual positives were correctly identified

Specificity – how many actual negatives were correctly identified

F1-score – balance between precision and recall

For **regression**, common measures include:

MAE – Mean Absolute Error

MSE – Mean Squared Error

RMSE – Root Mean Squared Error

R^2 – coefficient of determination

Step 5: Prediction

Once the AI model has been **trained and evaluated**, it can be used to make predictions on **new, unseen biological data**.

Example:

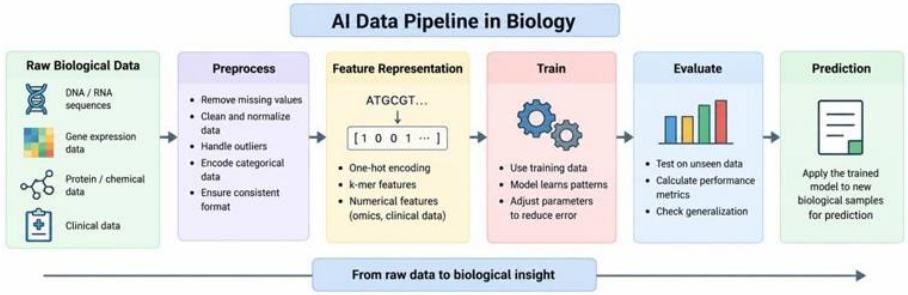
A model is trained using patient data where the disease status is already known.

New patient data → Trained AI model → Prediction

For example:

Clinical + genomic data → AI model → Disease: High Risk / Low Risk

Figure



S5 Applications in genomics and systems biology

2 MARKS In genomics AI identifies genes and variants and analyses gene expression; in systems biology it models how many components of a cell interact.
4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: AI in Genomics; AI for Gene and Variant Analysis; AI for Gene Expression Analysis; Systems Biology.
12 TO 13 MARKS: ANSWER IN THIS ORDER 1. AI in Genomics 2. AI for Gene and Variant Analysis 3. AI for Gene Expression Analysis 4. Systems Biology *Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.*
PRACTICE QUESTIONS 2 marks: Name two AI applications in genomics. 5 marks: Explain AI in gene expression analysis. 12 marks: Explain AI applications in genomics and systems biology.

Complete notes

AI in Genomics

Genomics is the study of an organism's complete DNA.

AI can analyse large genomic datasets and identify patterns that may be difficult to detect manually.

Applications:

Genome sequence analysis

Identification of genetic variants

Disease-associated mutation prediction

Gene function prediction

Genome classification

Prediction of regulatory regions

Example:

DNA sequence → AI model → Predict whether a genetic variant may be associated with disease

AI for Gene and Variant Analysis

A genome can contain millions of genetic variations between individuals.

AI can help identify which variations may be biologically important.

Workflow:

DNA sequence → Identify variants → AI analysis → Predict possible effect

For example, an AI model can analyse a mutation and predict whether it is likely to have:

Little or no effect

A possible functional effect

A potentially harmful effect

This can help researchers prioritize variants for further experimental investigation.

AI for Gene Expression Analysis

Gene expression tells us how actively genes are being expressed in a cell or tissue.

AI can analyse gene-expression data to identify patterns associated with different biological conditions.

Example:

Gene-expression data from:

Healthy cells + Cancer cells

↓

AI/ML analysis

↓

Identify genes or patterns associated with cancer

Applications include:

Disease classification

Cancer subtype identification

Prediction of treatment response

Systems Biology

Systems biology studies how different biological components work together as a system.

Instead of studying only one gene or protein, it considers interactions between:

Genes ↔ Proteins ↔ Metabolites ↔ Cells

AI can analyse these complex relationships and identify patterns within biological systems.

Examples include:

Gene regulatory networks

Protein–protein interaction networks

Cell signalling pathways

S6 Benefits and limitations of AI in biology

2 MARKS AI in biology offers speed and pattern discovery, but it is limited by data bias, overfitting, poor explainability and the need for lab validation.
4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Benefits of AI in Biology; Limitations of AI in Biology; Overfitting; Common causes OF Overfitting.
12 TO 13 MARKS: ANSWER IN THIS ORDER 1. Benefits of AI in Biology 2. Limitations of AI in Biology 3. Overfitting 4. Common causes OF Overfitting *Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.*
PRACTICE QUESTIONS 2 marks: What is overfitting? 5 marks: List the benefits and limitations of AI in biology. 12 marks: Discuss the benefits, limitations and overfitting problem of AI in biology.

Complete notes

Benefits of AI in Biology

AI can process large amounts of biological data and identify patterns that may be difficult to detect manually.

Major benefits:

Handles large datasets – Can analyse genomic, clinical and molecular data at large scale.

Fast analysis – Performs computational analysis much faster than manual approaches.

Pattern recognition – Identifies relationships and patterns in complex biological data.

Automation – Automates repetitive analysis tasks.

Prediction – Can predict disease risk, drug properties, gene functions and other biological outcomes.

Personalization – Can support personalized medicine by analysing individual patient data.

Limitations of AI in Biology

AI models also have important limitations.

Data quality: Poor-quality data can produce poor predictions.

Limited training data: Some biological problems have relatively small datasets.

Bias: If the training data are biased, the model can produce biased results.

Overfitting: A model may perform well on training data but poorly on new data.

Generalization: A model trained using one population, laboratory or dataset may not work equally well in another.

Interpretability: Some complex AI models are difficult to understand.

Overfitting

Overfitting means the AI model learns the training data too well — including its noise and small details — instead of learning the general pattern.

So:

Training data → Very good performance
New/unseen data → Poor performance

Simple example:

Suppose we train an AI to predict diabetes using patient data.

The model is given 1,000 previous patient records.

If it memorizes the specific patterns of those 1,000 patients rather than learning general relationships between age, weight, glucose level, etc., it may perform like this:

This is **overfitting**.

Data	Accuracy
Training patients	98%
New patients	65%

Common causes OF Overfitting

Too complex a model

Too little training data

Too many features

Noisy data

Training the model for too long in some settings

Some ways to reduce overfitting?

Use more training data

Remove irrelevant features

Use a simpler model

Use **regularization**

Use **cross-validation**

Use techniques such as **dropout** in neural networks

Evaluate the model on **unseen data**

Unit 5

Deep Learning, Healthcare and Ethics

Neural networks in medicine, and the responsibilities that come with them.

S1 Deep learning for images, signals and sequences

2 MARKS Deep learning is a branch of machine learning that uses neural networks with multiple layers to learn increasingly complex representations from data.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Core Idea; Neurons and layers; Deep Learning for Biological Images; Convolutional Neural Networks; Deep Learning for Biological Signals; Transformers and Attention in Biology.

12 TO 13 MARKS: ANSWER IN THIS ORDER

1. Core Idea
2. Neurons and layers
3. Deep Learning for Biological Images
4. Convolutional Neural Networks
5. Deep Learning for Biological Signals
6. Transformers and Attention in Biology
7. Training Deep Models
8. Strengths and Challenges

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

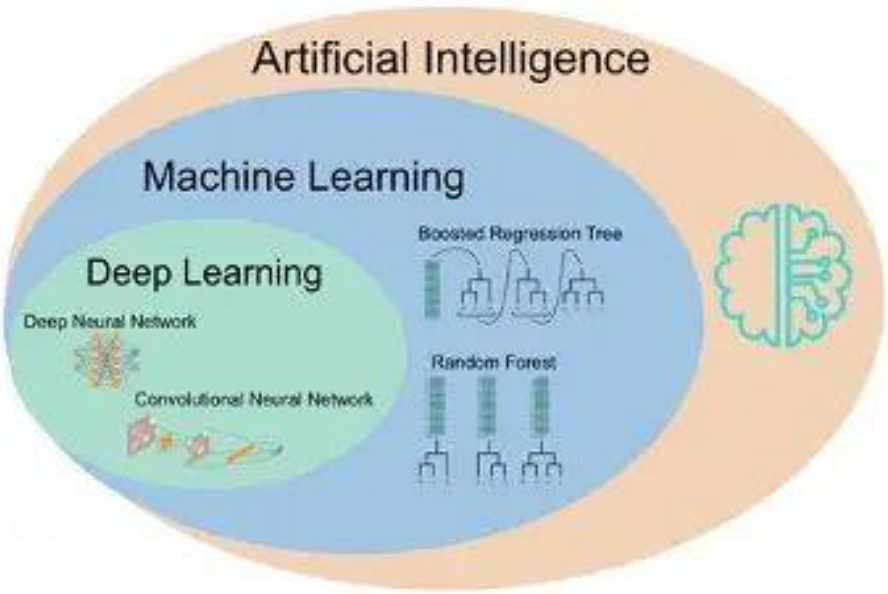
PRACTICE QUESTIONS

2 marks: What is deep learning?
5 marks: Explain convolutional neural networks.
12 marks: Explain deep learning for images, signals and sequences, including training, strengths and challenges.

Complete notes

Core Idea

- Definition
- Deep learning is a branch of machine learning that uses neural networks with multiple layers to learn increasingly complex representations from data.
- Why biology?
- Biological data are high-dimensional and contain spatial, temporal or sequential patterns that are difficult to capture with a few manually designed rules.
- Workflow
- Biological data → preprocessing → numerical representation → deep neural network → learned representation → prediction.
- Key distinction
- Deep learning can learn useful internal features from data, whereas conventional machine learning often depends more strongly on manually engineered features.



Neurons and layers

A neural network contains interconnected computational units. Each unit combines inputs using weights and a bias and applies an activation function.

Layered learning

Early layers can learn simpler patterns, and deeper layers can combine them into more complex representations.

Representation learning

The model learns a numerical representation that is useful for the task instead of requiring every feature to be specified manually.

Training

Parameters are adjusted by minimizing a loss function using backpropagation and optimization methods such as gradient descent.

Neural Networks and Representation Learning

Deep Learning for Biological Images

- Image data
- Examples include microscopy, histopathology, X-ray, CT, MRI and fluorescence images.
- Challenge
- Images contain many pixels and spatial relationships, with variation in scale, orientation, illumination, sample preparation and biological morphology.
- Tasks
- Classification assigns categories; detection identifies relevant regions; segmentation assigns labels to pixels or regions.
- Applications
- Deep learning can classify cells, detect abnormalities, quantify structures and identify tissue patterns.

Convolutional Neural Networks

- Hierarchical features
- Earlier layers may respond to edges or textures, while later layers can represent more complex shapes or structures.
- Downsampling

- Pooling or related operations can reduce spatial dimensions and create compact representations.
- Outputs
- CNN representations can support image classification, object detection, segmentation and quantitative image analysis.

Deep Learning for Biological Signals

- Signals
- Examples include ECG, EEG, heart rate, blood pressure and continuous glucose measurements.
- Challenges
- Signals may contain noise, missing observations, baseline drift, individual variation and patterns at different time scales.
- Models
- CNNs can learn local temporal patterns; recurrent models can capture sequential dependencies; transformers can model long-range relationships.
- Example
- An ECG can be represented as a numerical time series and processed to classify defined rhythm patterns.

Transformers and Attention in Biology

- Attention
- Attention lets a model assign different importance to different elements of an input when forming a representation or prediction.
- Long-range relationships
- This is useful when distant sequence positions may influence one another.
- Biological language models
- Transformer-based models can learn representations of DNA, RNA and protein sequences from large sequence collections.
- Embeddings
- The learned numerical embeddings can be reused for downstream classification, prediction and similarity tasks.

Training Deep Models

- Training data
- The model learns from examples containing inputs and, in supervised learning, target labels or numerical outcomes.
- Loss
- The loss function measures disagreement between model output and the desired output.
- Optimization
- Backpropagation calculates parameter gradients and an optimizer updates the model parameters.
- Generalization
- The objective is not just low training error; the model should perform well on unseen data.

Strengths and Challenges

- Strengths
- Deep learning can model nonlinear relationships and learn complex representations from high-dimensional images, signals and sequences.
- Data needs
- Large, representative datasets are often valuable, especially for high-capacity models.
- Computational needs
- Training may require substantial compute, storage and careful hyperparameter selection.

- Interpretation
- Complex models can be difficult to interpret, so predictions require appropriate validation and uncertainty assessment.

Figure

- Images
- CNN-based approaches learn spatial patterns and support classification, detection and segmentation.
- Signals
- Deep models can learn temporal patterns from physiological measurements.
- Sequences
- DNA, RNA and proteins can be converted into numerical representations and analysed with deep networks.
- Central idea
- Deep learning creates learned representations that support biological prediction, but model output still requires biological interpretation and validation.

S2 AI in drug discovery and biotechnology

2 MARKS AI speeds up drug discovery by predicting protein structures, screening millions of compounds virtually and designing new molecules.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Definition; Key features or structure; Process or mechanism; Examples; Applications; Limitations.

12 TO 13 MARKS: ANSWER IN THIS ORDER

1. Definition
2. Key features or structure
3. Process or mechanism
4. Examples
5. Applications
6. Limitations
7. Conclusion

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS

2 marks: Name two uses of AI in drug discovery.
5 marks: Explain virtual screening.
12 marks: Discuss the role of AI in drug discovery and biotechnology.

S3 AI-assisted medical diagnostics and imaging

2 MARKS AI-assisted diagnostics uses deep learning to read medical images such as X-rays, CT and MRI, to detect disease and support the doctor's decision.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Definition; Key features or structure; Process or mechanism; Examples; Applications; Limitations.

12 TO 13 MARKS: ANSWER IN THIS ORDER

1. Definition
2. Key features or structure
3. Process or mechanism
4. Examples
5. Applications
6. Limitations
7. Conclusion

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS

2 marks: Define sensitivity and specificity.
5 marks: Explain AI in medical imaging.
12 marks: Discuss AI-assisted diagnostics and imaging, with benefits and challenges.

S4 AI for integrated bioinformatics and health data

2 MARKS Integrated health data combines records, genomes, images and wearable data so that AI can support precision medicine.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Definition; Key features or structure; Process or mechanism; Examples; Applications; Limitations.

12 TO 13 MARKS: ANSWER IN THIS ORDER

1. Definition
2. Key features or structure
3. Process or mechanism
4. Examples
5. Applications
6. Limitations
7. Conclusion

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS

2 marks: What is federated learning?
5 marks: Explain integration of health data for precision medicine.
12 marks: Discuss AI for integrated bioinformatics and health data.

S5 Real-time AI for monitoring and decision support

2 MARKS Real-time AI analyses streaming sensor data to flag dangerous changes in a patient at once and support quick clinical decisions.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Definition; Key features or structure; Process or mechanism; Examples; Applications; Limitations.

12 TO 13 MARKS: ANSWER IN THIS ORDER

1. Definition
2. Key features or structure
3. Process or mechanism
4. Examples
5. Applications
6. Limitations
7. Conclusion

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS

2 marks: What is alarm fatigue?
5 marks: Explain real-time AI monitoring.
12 marks: Discuss real-time AI for monitoring and decision support.

S6 Ethical and social issues of AI in biology

2 MARKS The main ethical issues of medical AI are patient privacy and consent, bias and fairness, accountability, transparency and equal access.

4 TO 5 MARKS Write the definition, then any 4 or 5 of these points, one line each: Definition; Key features or structure; Process or mechanism; Examples; Applications; Limitations.

12 TO 13 MARKS: ANSWER IN THIS ORDER

1. Definition
2. Key features or structure
3. Process or mechanism
4. Examples
5. Applications
6. Limitations
7. Conclusion

Finish with a labelled diagram or table, one real example, one application and a one-line conclusion.

PRACTICE QUESTIONS

2 marks: Name two ethical issues in medical AI.
5 marks: Explain bias and fairness in medical AI.
12 marks: Discuss the ethical and social issues of AI in biology.